

Exploratory Verification

in the age of

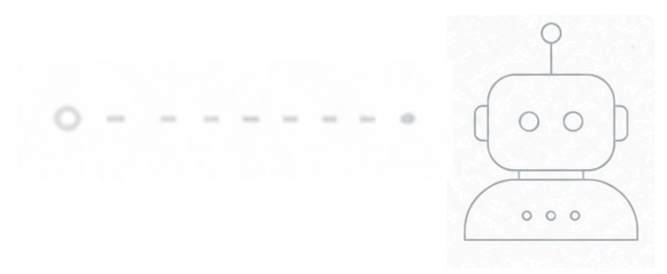
Probabilistic Code

Mourjo Sen

September 2026

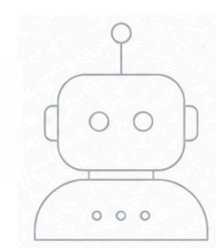






\$ mvn clean test



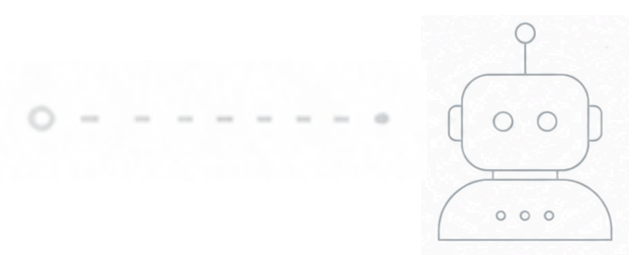


```
$ mvn clean test
```

```
...  
tries = 187 | # of calls to property  
checks = 187 | # of not rejected calls  
generation = RANDOMIZED | parameters are randomly generated  
...
```

```
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
```





```
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Invariant 'no-overlap' failed after the following actions: [

]

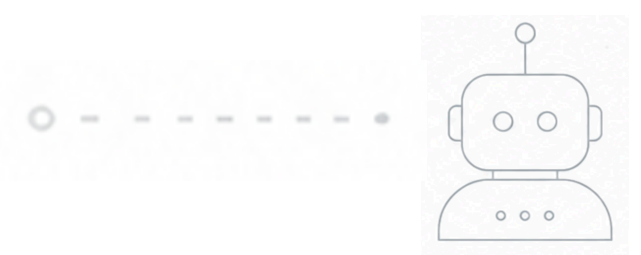
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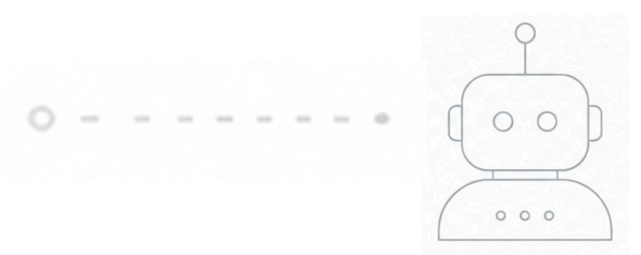
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$ mvn clean test
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Invariant 'no-overlap' failed after the following actions: [  
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
]
```

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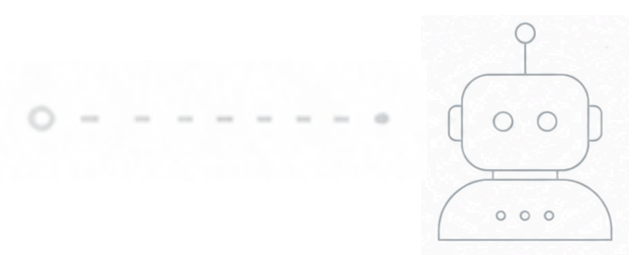
```
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```
Invariant 'no-overlap' failed after the following actions: [  
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
  bob creates Meeting-2 in calendar default (UTC) from 10:00 to 10:01  
  
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Invariant 'no-overlap' failed after the following actions: [  
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
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  bob invites alice to Meeting-2  
]
```

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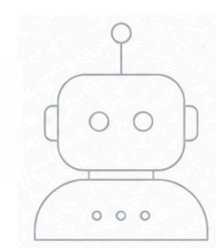
Invariant 'no-overlap' failed after the following actions: [
 alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
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 bob invites alice to Meeting-2
 alice accepts invitation to Meeting-2
]

...
tries = 187 | **# of calls to property**
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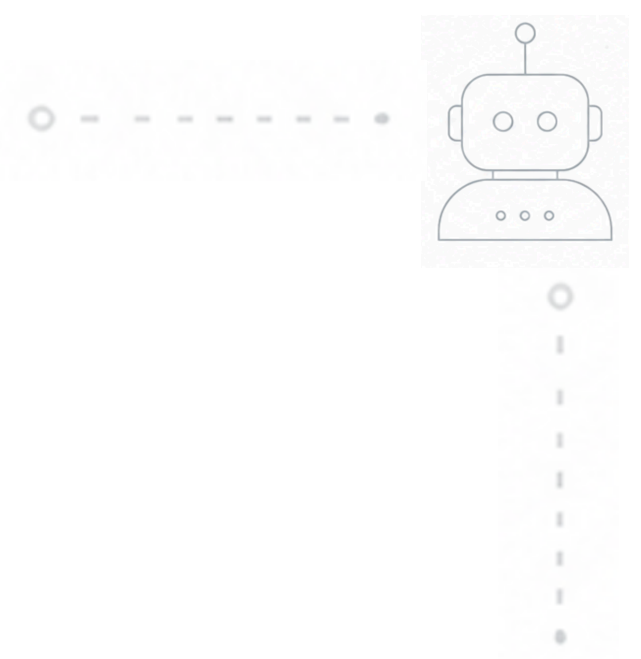
How did I miss that?

Invariant 'no-overlap' failed after the following actions: [
 alice creates **Meeting-1** in calendar default (UTC) from 10:00 to 10:01
 bob creates **Meeting-2** in calendar default (UTC) from 10:00 to 10:01
 bob invites **alice** to Meeting-2
 alice accepts invitation to Meeting-2
]

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...  
tries = 187           | # of calls to property  
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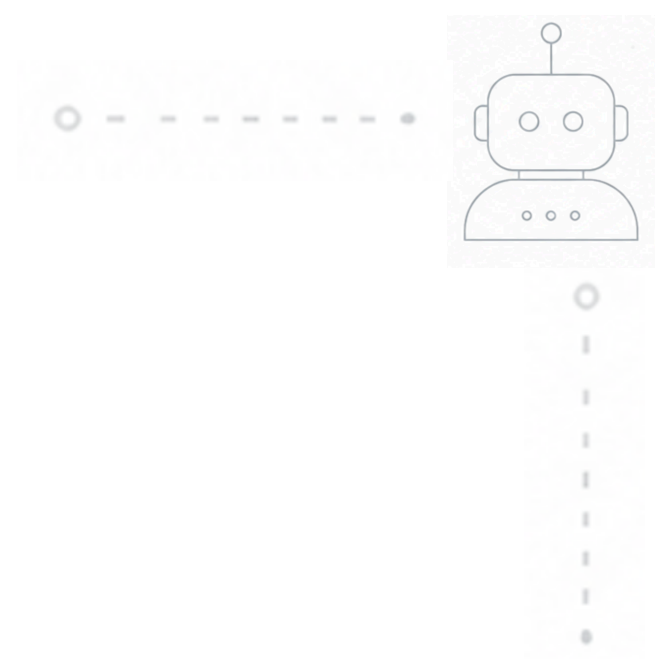




Namaste

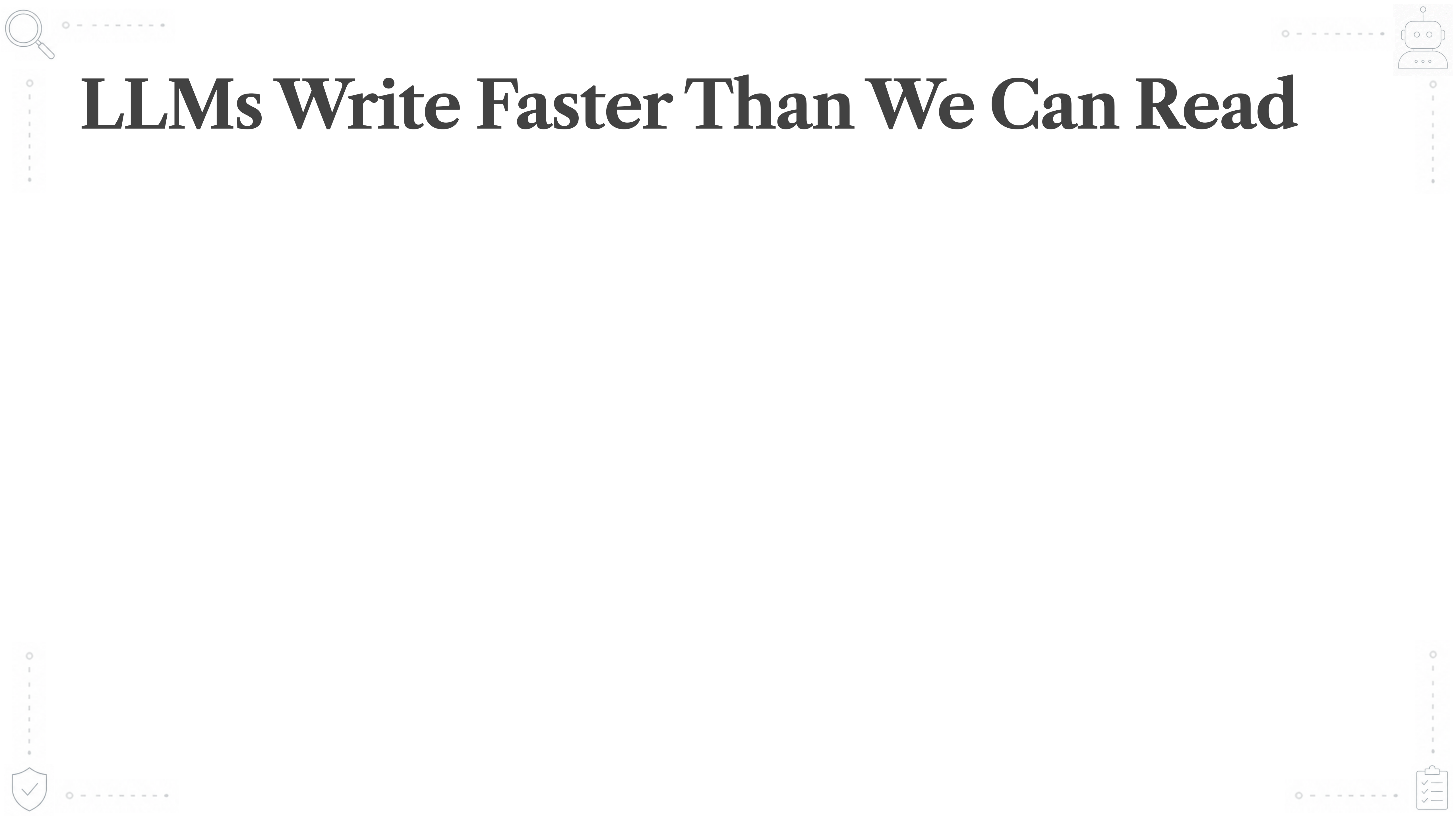
- India → Netherlands
- 10 years in engineering
- Staff Engineer @ booking.com



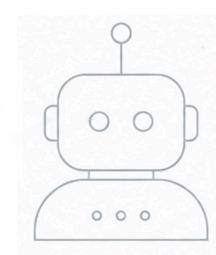


Everything's Changed





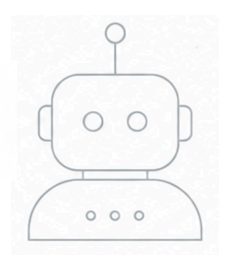
LLMs Write Faster Than We Can Read



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- Code generation is fast but probabilistic
 - Quality of software cannot be probabilistic





LLMs Write Faster Than We Can Read

- Code generation is fast but probabilistic
 - Quality of software cannot be probabilistic
- Use LLMs for testing software?
 - Because we cannot read at that speed





Are LLMs Good At Testing Software?



Are LLMs Good At Testing Software?

Source: <https://arxiv.org/abs/2507.06920>

Rethinking Verification for LLM Code Generation: From Generation to Testing

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Minnan Luo^{2,3,†}, Songyang Zhang^{1,†,‡}, Kai Chen^{1,†}

¹Shanghai AI Laboratory

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{mazihan880}@stu.xjtu.edu.cn

{zhangtaolin, zhangsongyang}@pjlab.org.cn

{minnluo}@xjtu.edu.cn

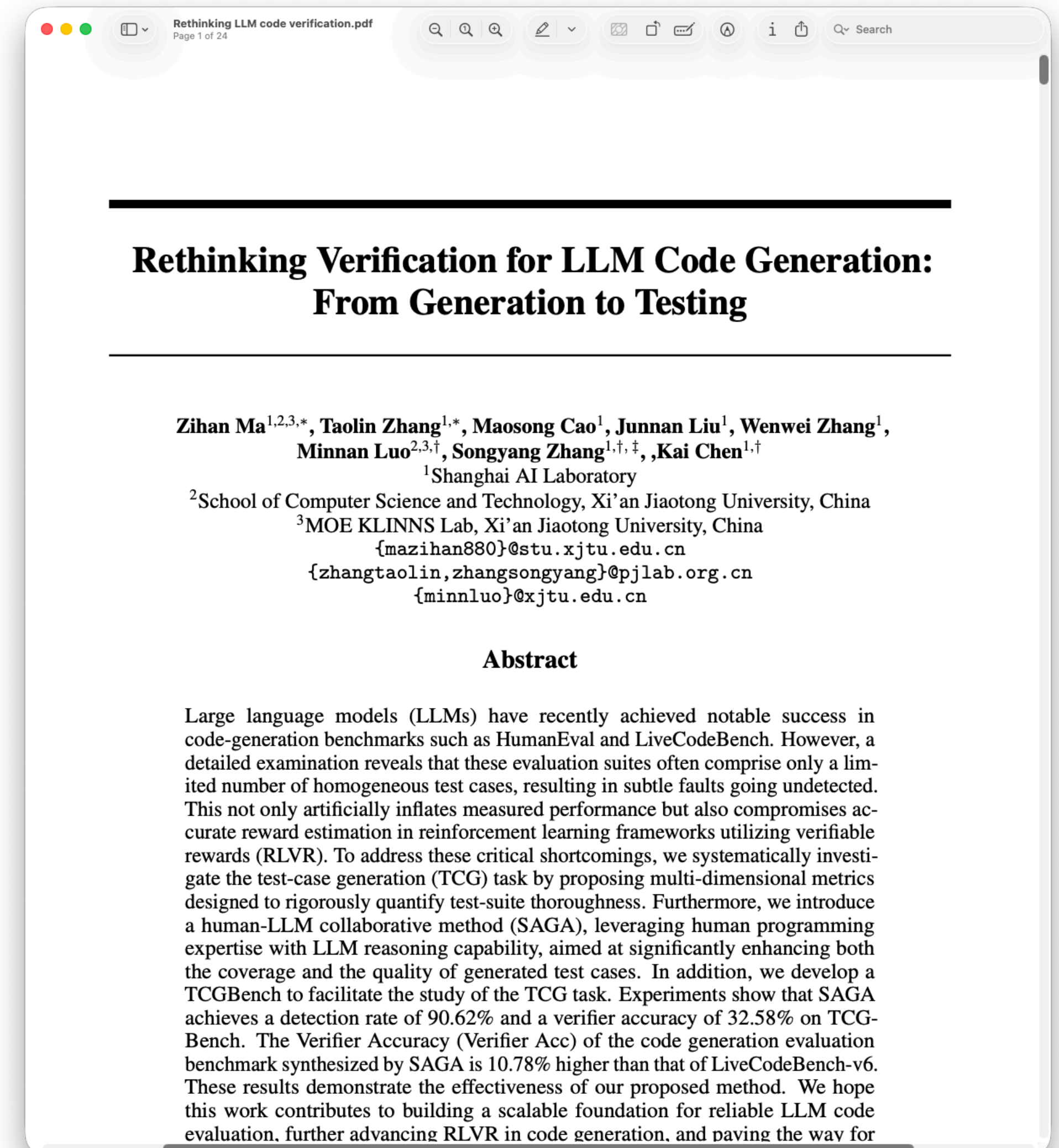
Abstract

Large language models (LLMs) have recently achieved notable success in code-generation benchmarks such as HumanEval and LiveCodeBench. However, a detailed examination reveals that these evaluation suites often comprise only a limited number of homogeneous test cases, resulting in subtle faults going undetected. This not only artificially inflates measured performance but also compromises accurate reward estimation in reinforcement learning frameworks utilizing verifiable rewards (RLVR). To address these critical shortcomings, we systematically investigate the test-case generation (TCG) task by proposing multi-dimensional metrics designed to rigorously quantify test-suite thoroughness. Furthermore, we introduce a human-LLM collaborative method (SAGA), leveraging human programming expertise with LLM reasoning capability, aimed at significantly enhancing both the coverage and the quality of generated test cases. In addition, we develop a TCGBench to facilitate the study of the TCG task. Experiments show that SAGA achieves a detection rate of 90.62% and a verifier accuracy of 32.58% on TCGBench. The Verifier Accuracy (Verifier Acc) of the code generation evaluation benchmark synthesized by SAGA is 10.78% higher than that of LiveCodeBench-v6. These results demonstrate the effectiveness of our proposed method. We hope this work contributes to building a scalable foundation for reliable LLM code evaluation, further advancing RLVR in code generation, and paving the way for

Are LLMs Good At Testing Software?

- LLMs struggle to catch bugs
 - ~60% chance of identifying a buggy solution

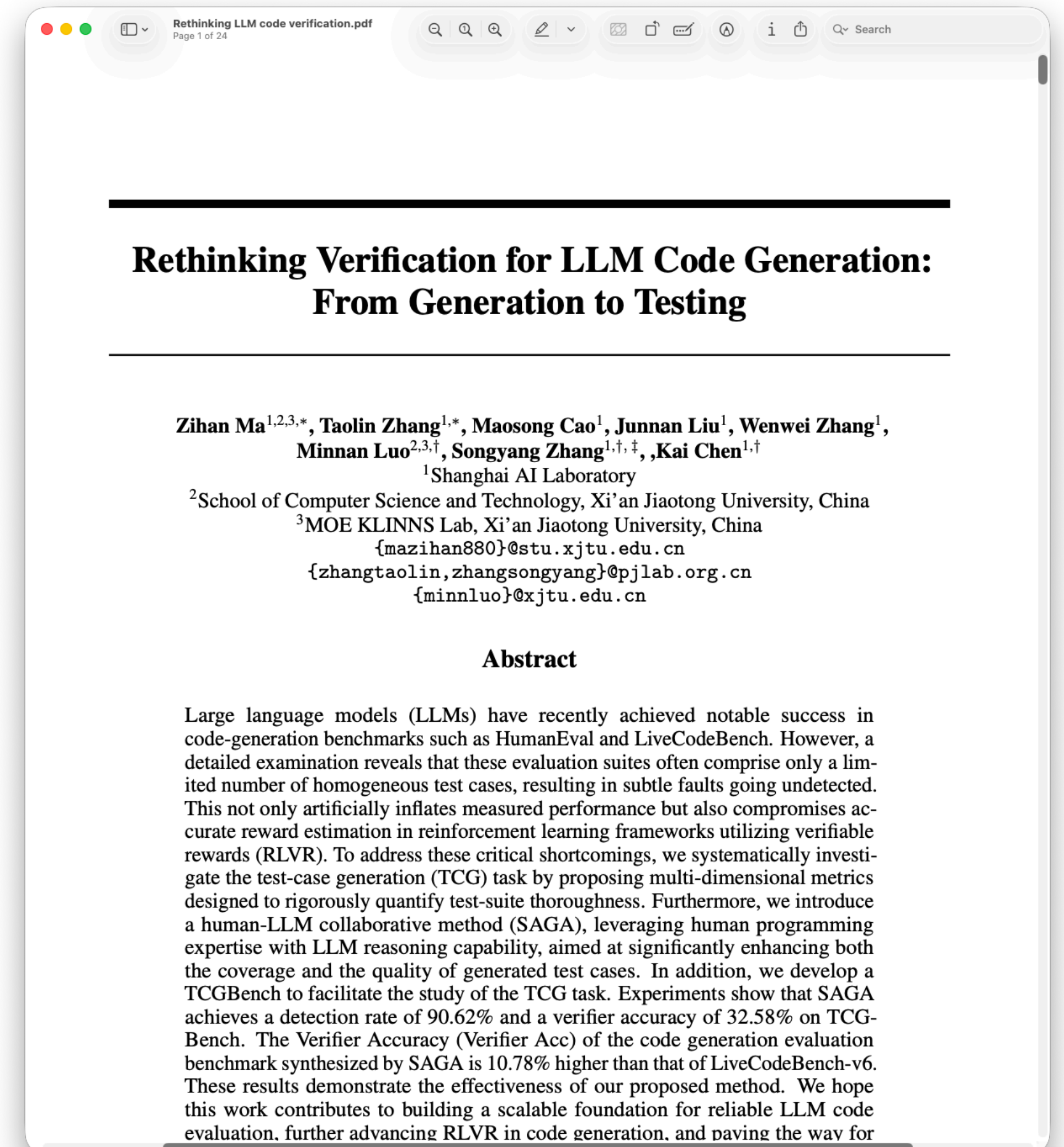
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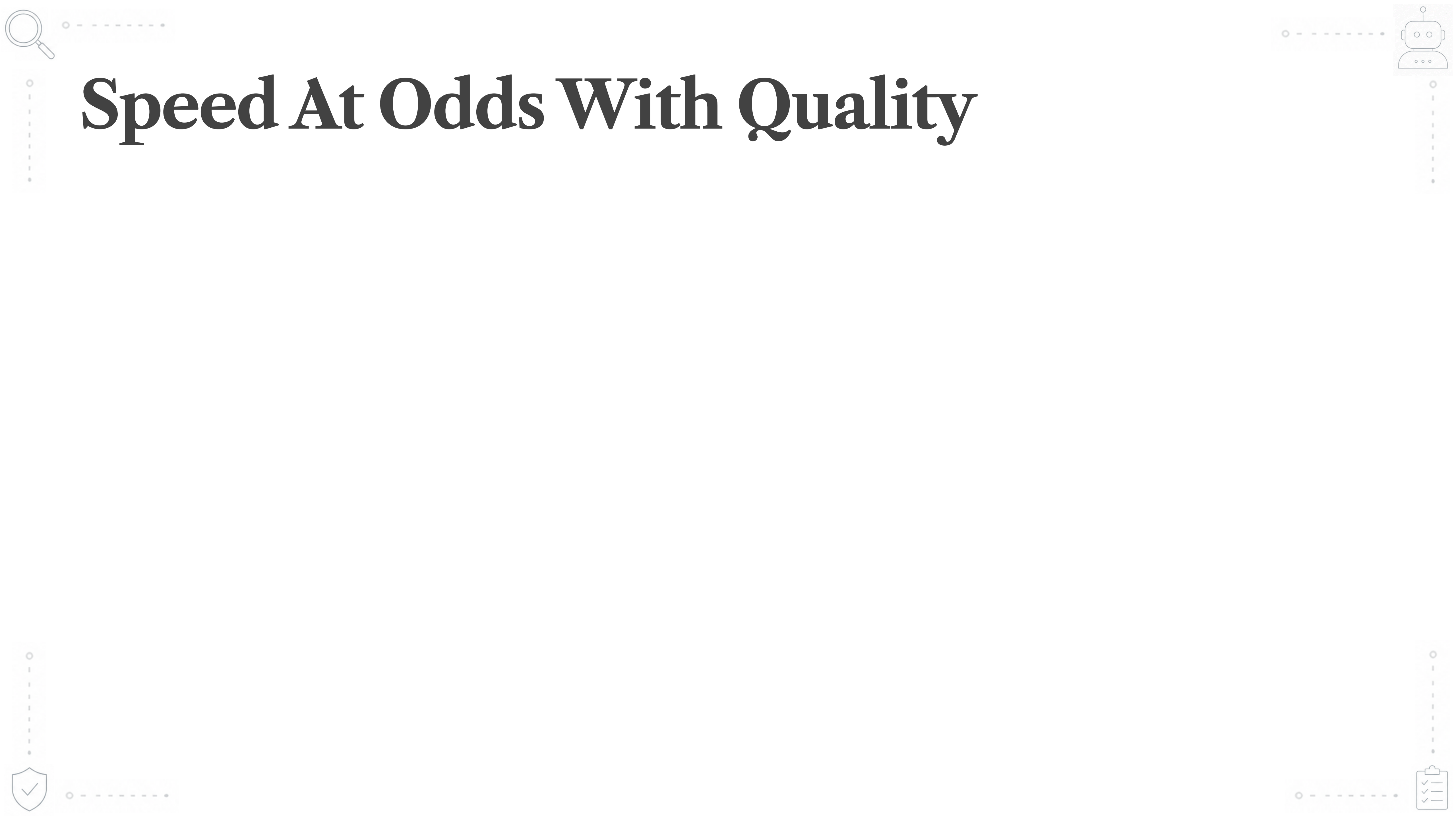


Are LLMs Good At Testing Software?

- LLMs struggle to catch bugs
 - ~60% chance of identifying a buggy solution
- Inflated self-testing success
 - LLM-written tests have a high success rate against LLM-written source

Source: <https://arxiv.org/abs/2507.06920>





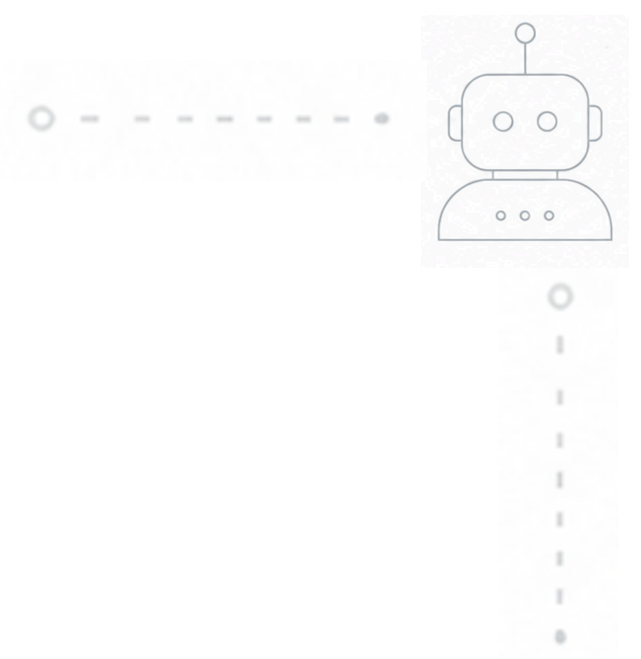
Speed At Odds With Quality



Speed At Odds With Quality

- Code generation is fast
 - AI writes code faster than we can read it

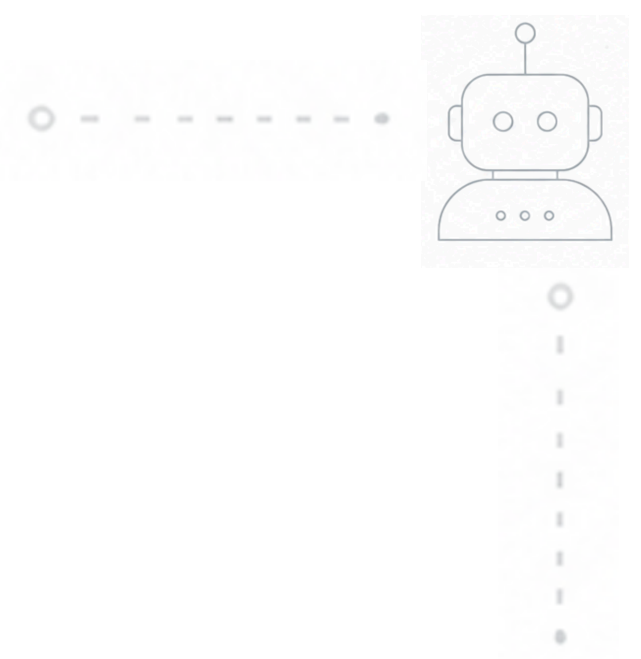




Speed At Odds With Quality

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- But it is also probabilistic
 - If we can't read it, we can't verify its correctness

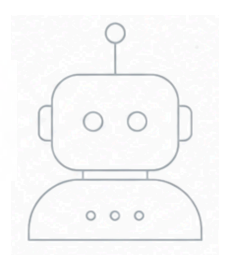




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Speed At Odds With Quality

- Code generation is fast
 - AI writes code faster than we can read it
- But it is also probabilistic
 - If we can't read it, we can't verify its correctness
 - LLMs are not good at testing software
- Human comprehension
 - Our ability to understand and make meaning from information

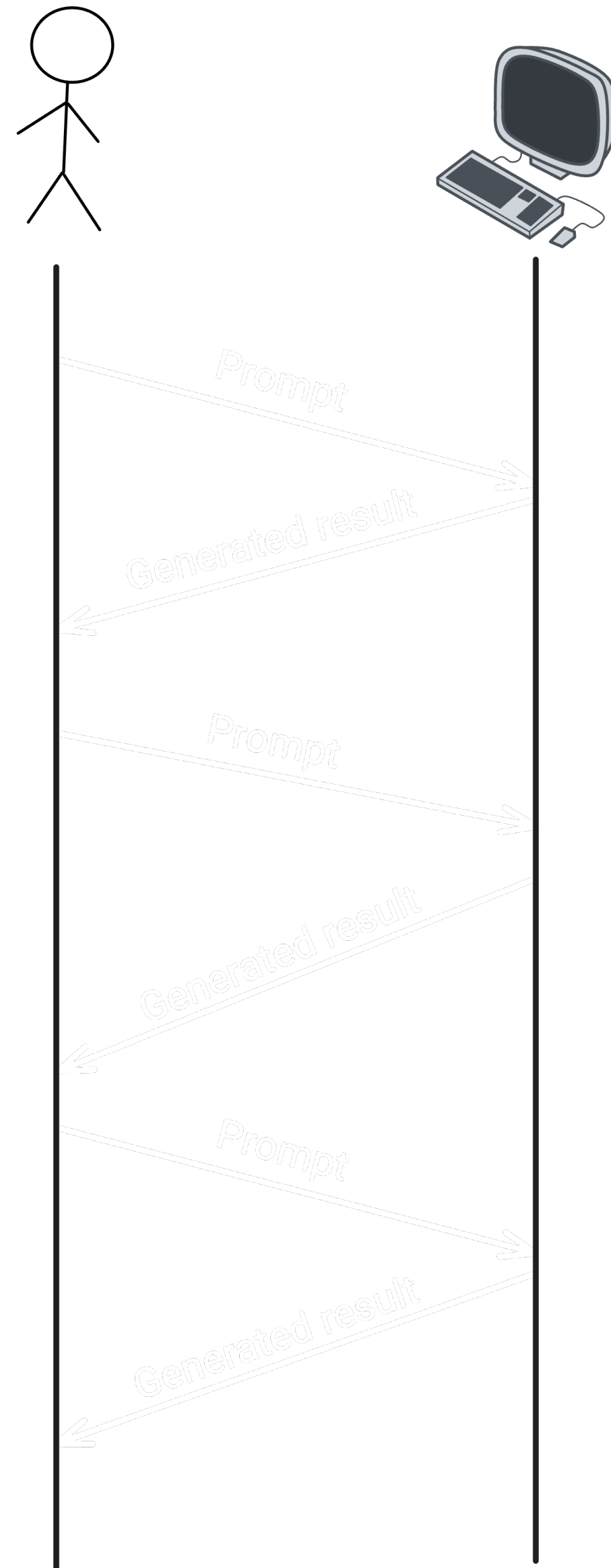




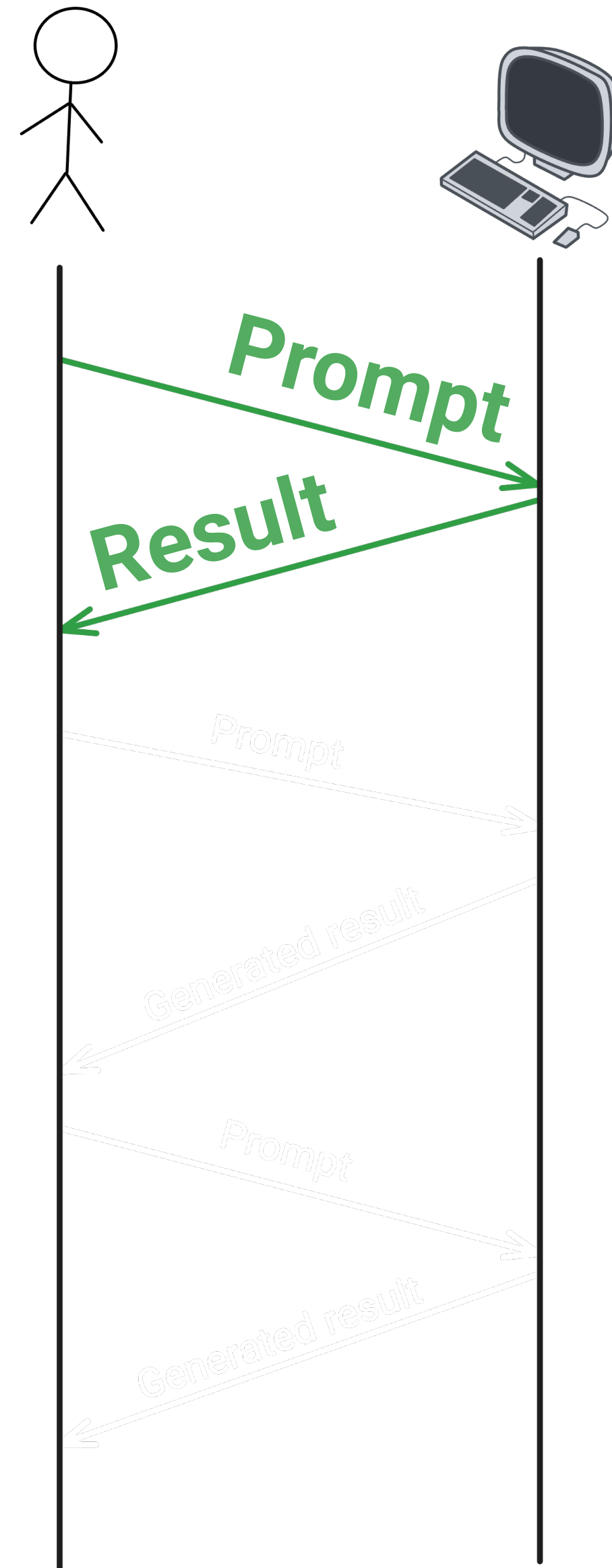
Playing To Our Strengths



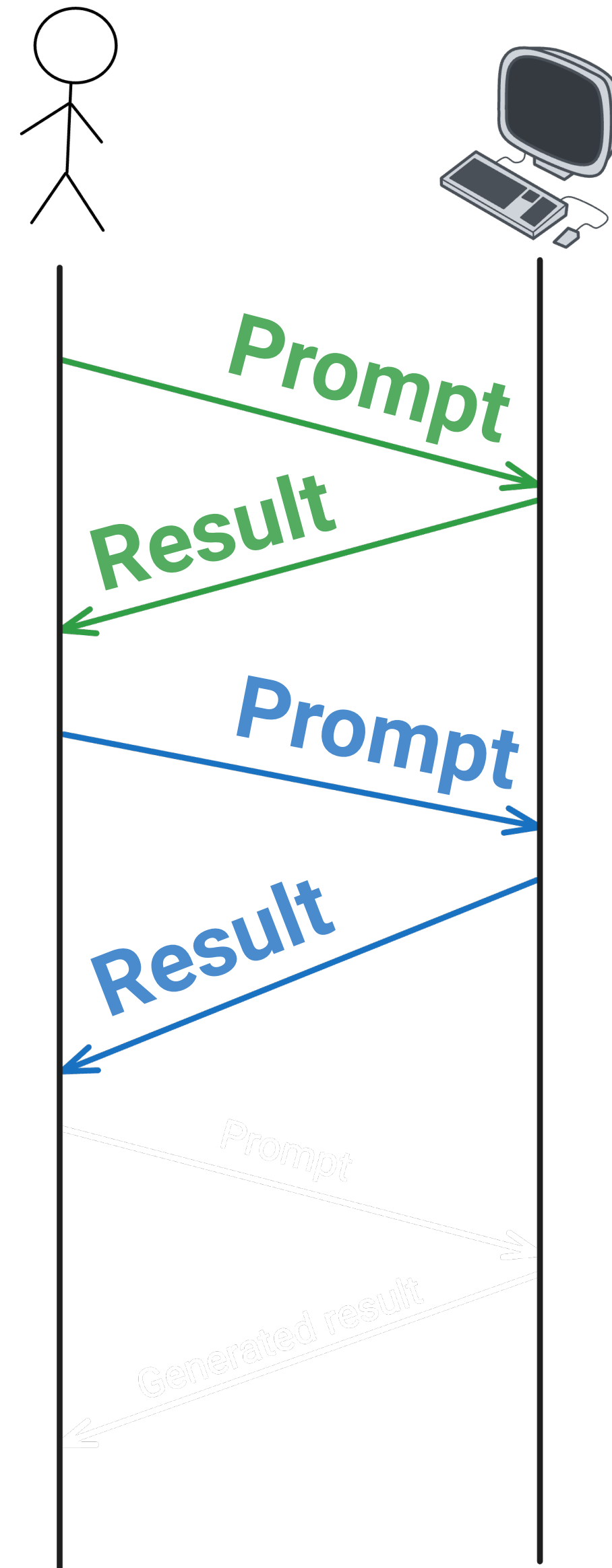
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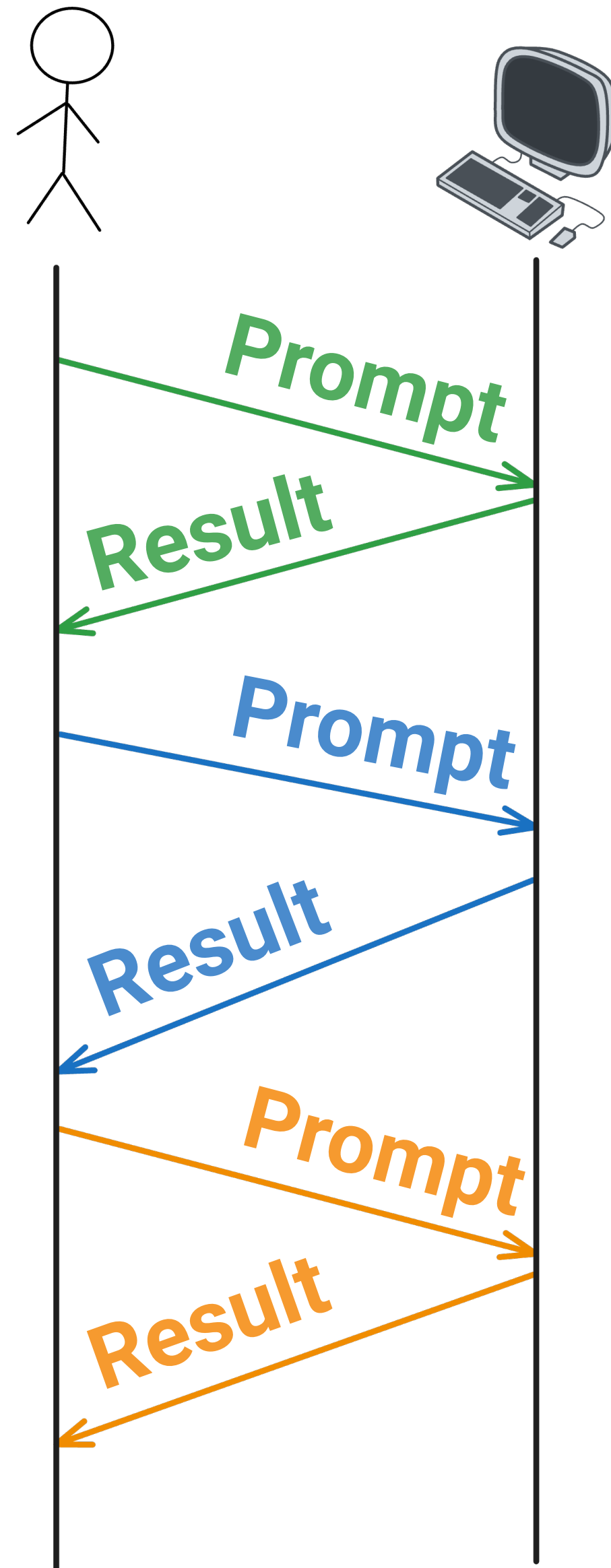
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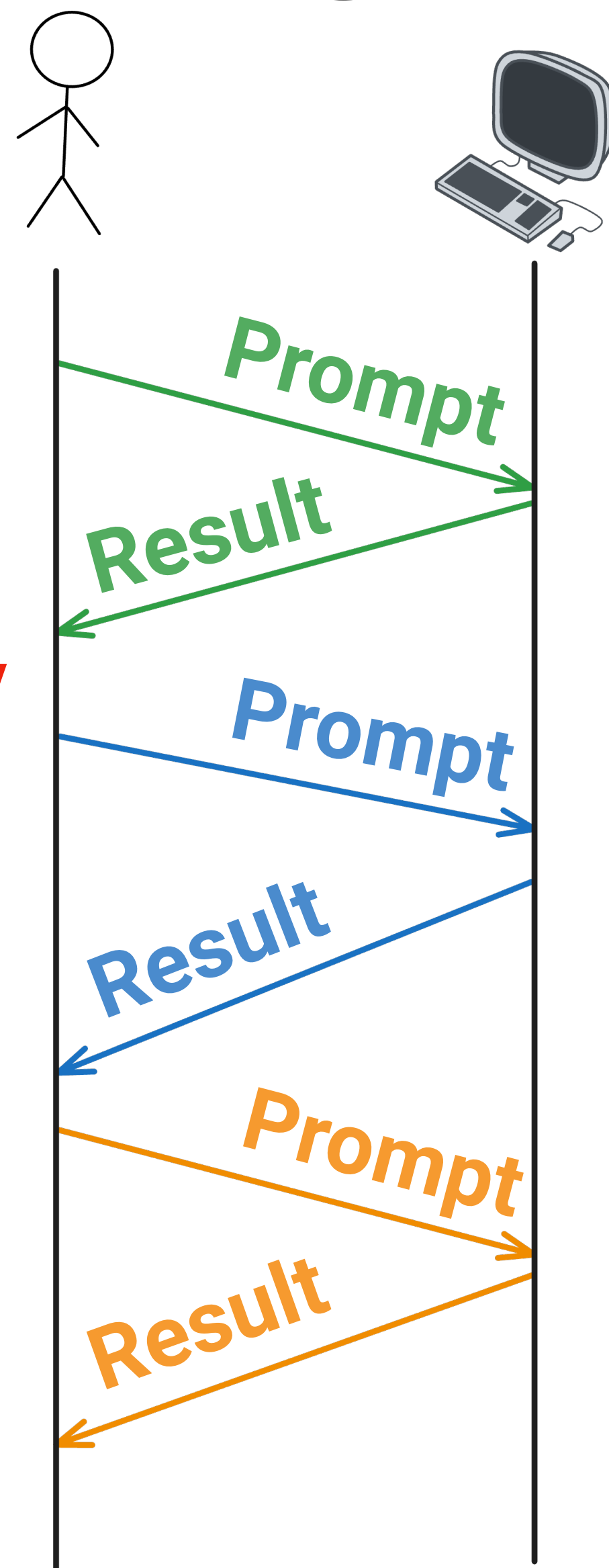


Playing To Our Strengths



Playing To Our Strengths

Happens too quickly



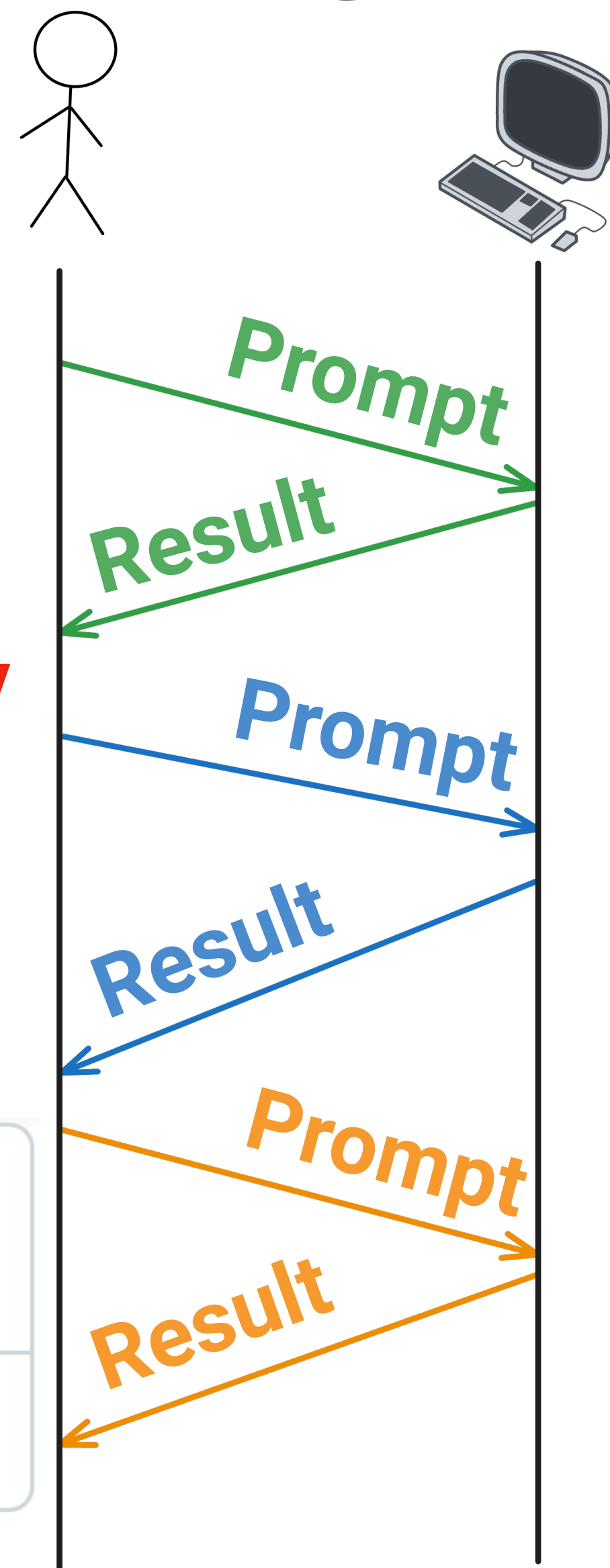
Playing To Our Strengths

Happens too quickly

feat: implement user, meeting,
invitation endpoints

25 files changed

+1,030 -32



Playing To Our Strengths

Active Participant

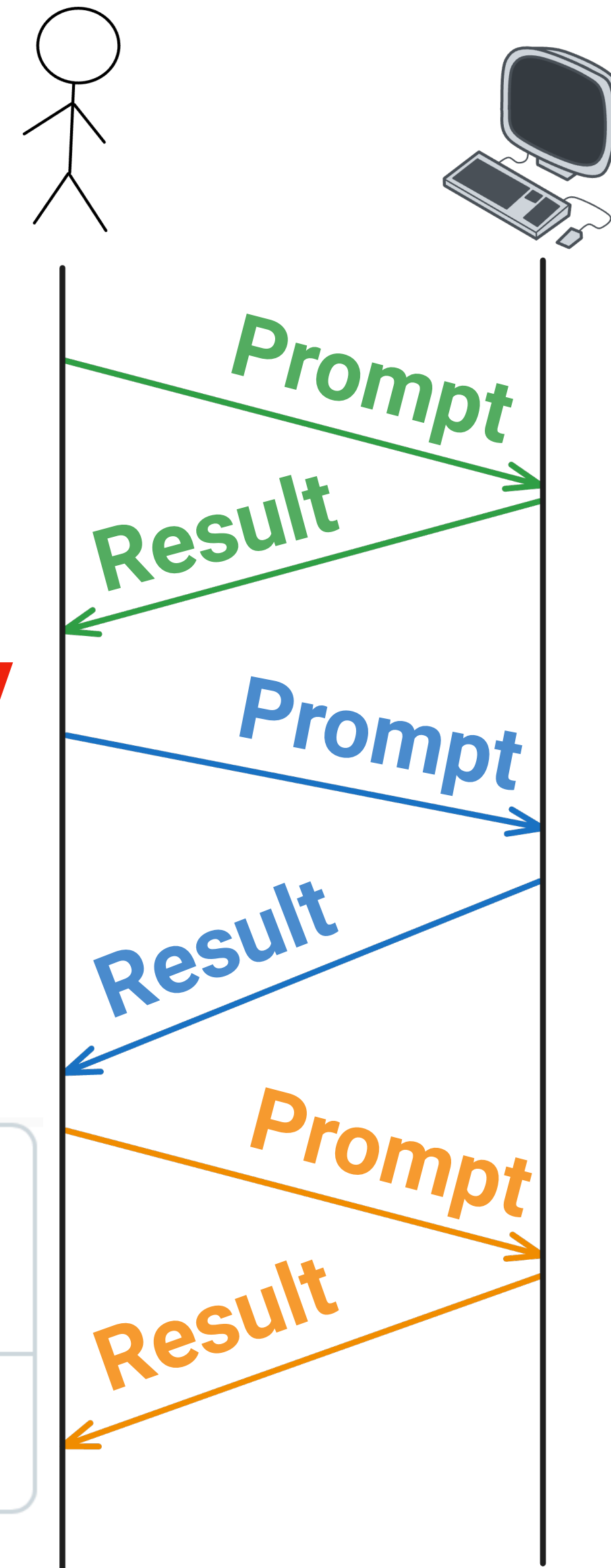
***Superficial* Overseer**

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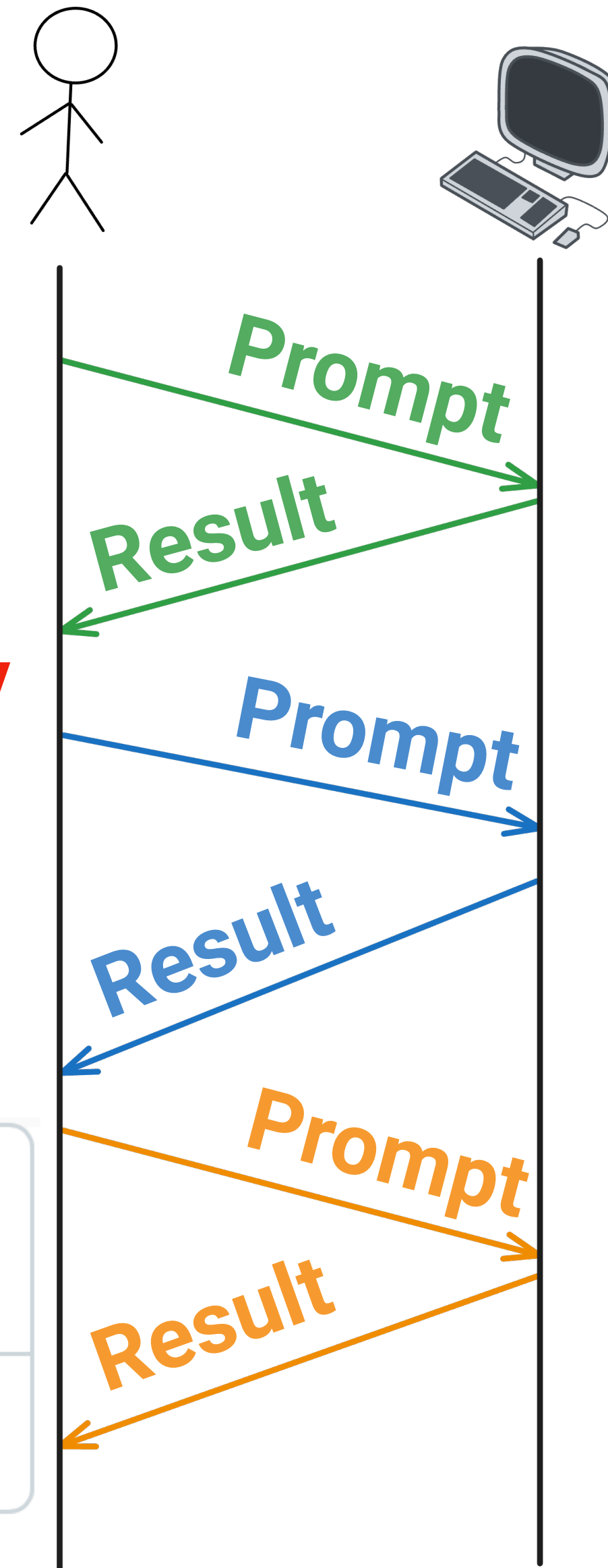
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Superficial Overseer

Overseeing does not **engage** our
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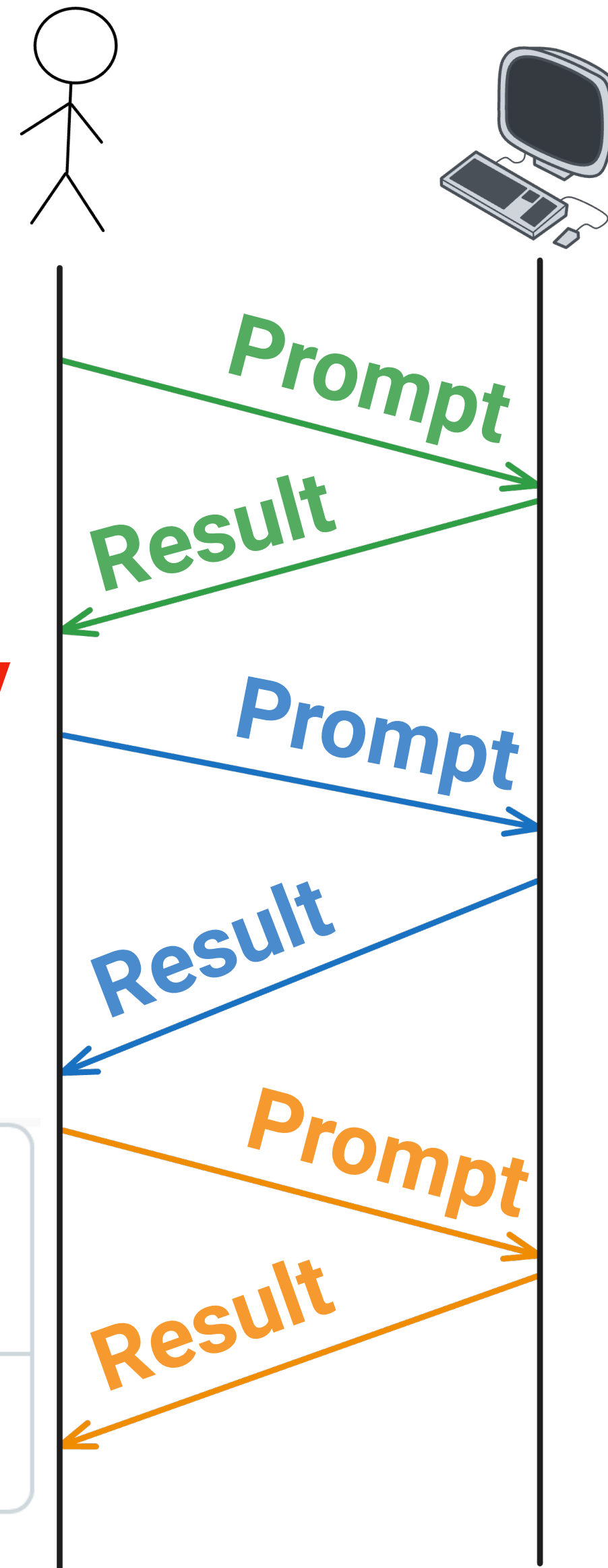
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How did I miss that?

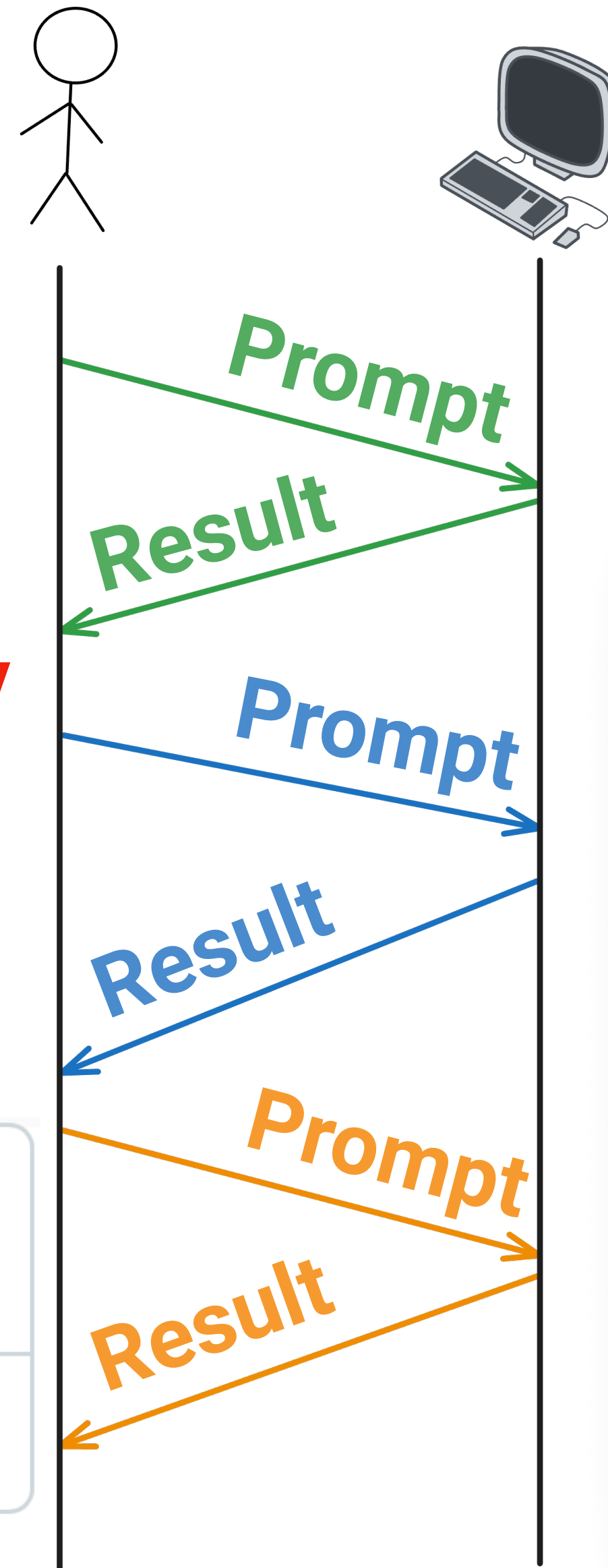
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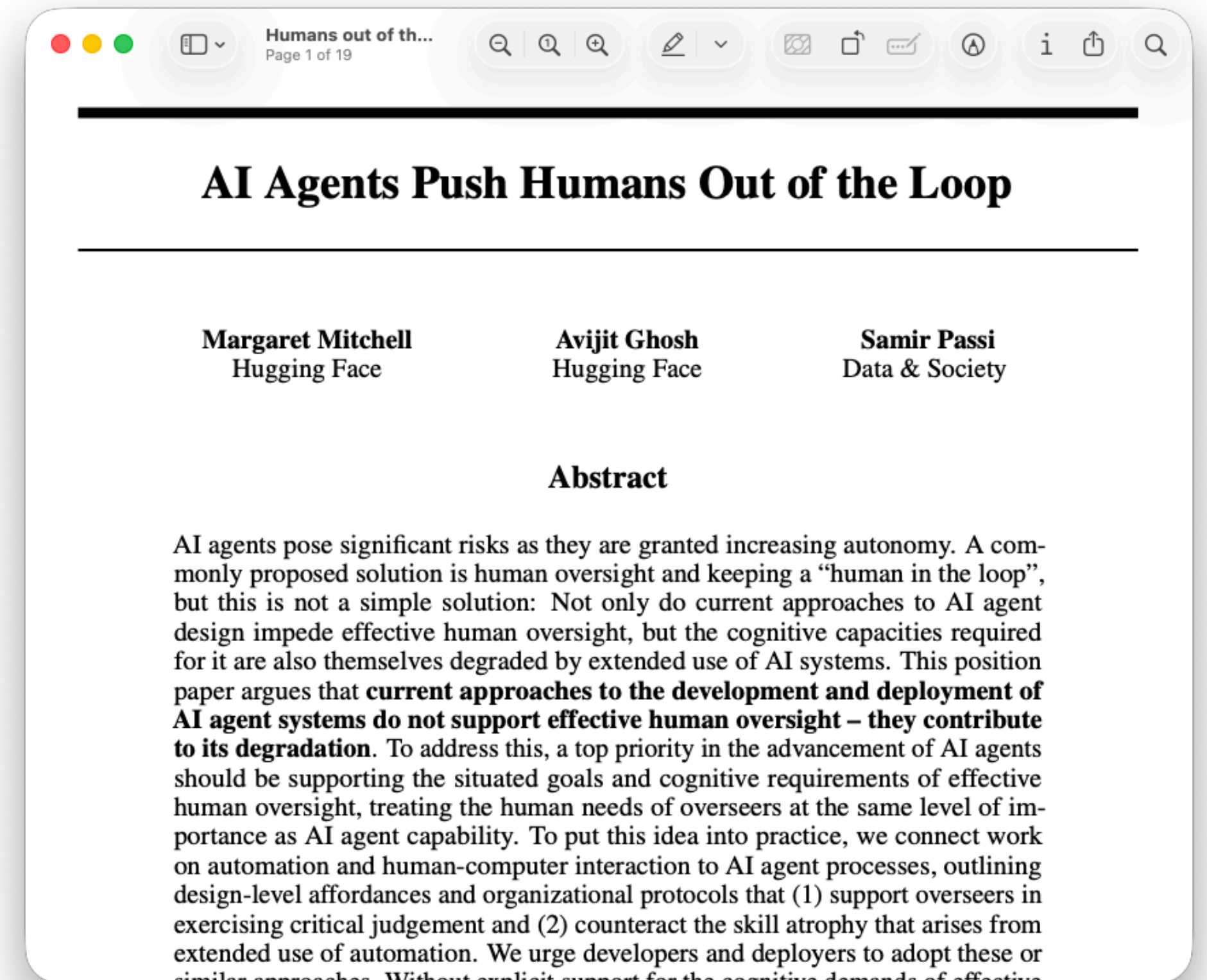
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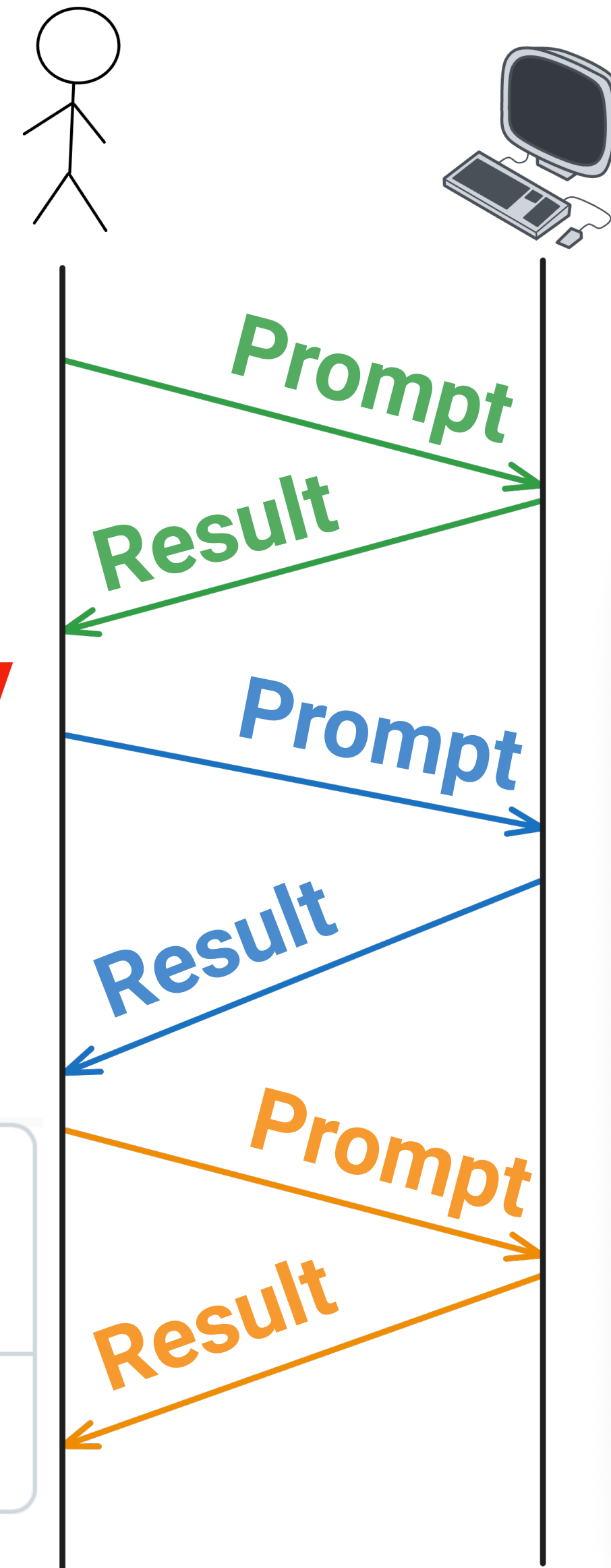
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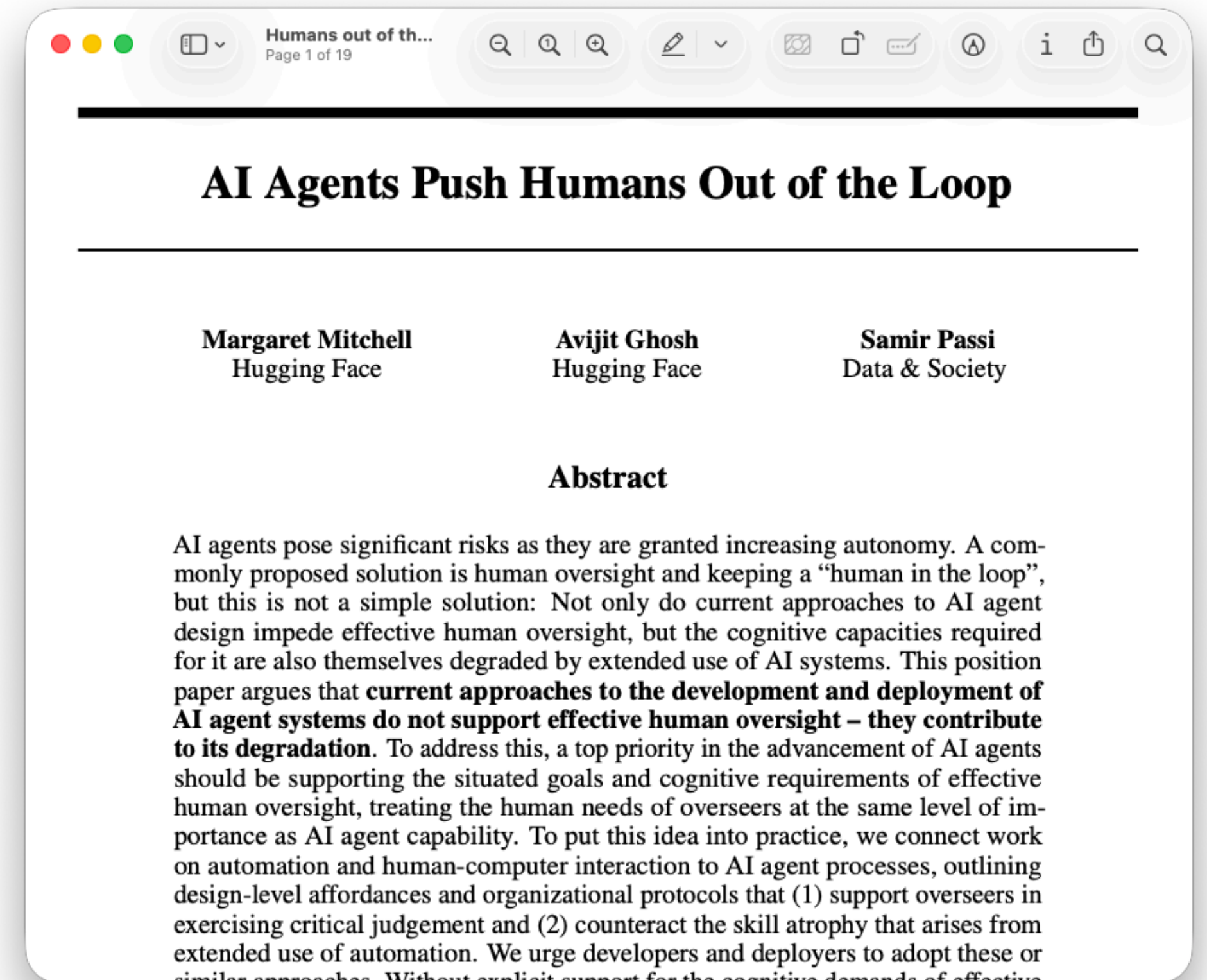
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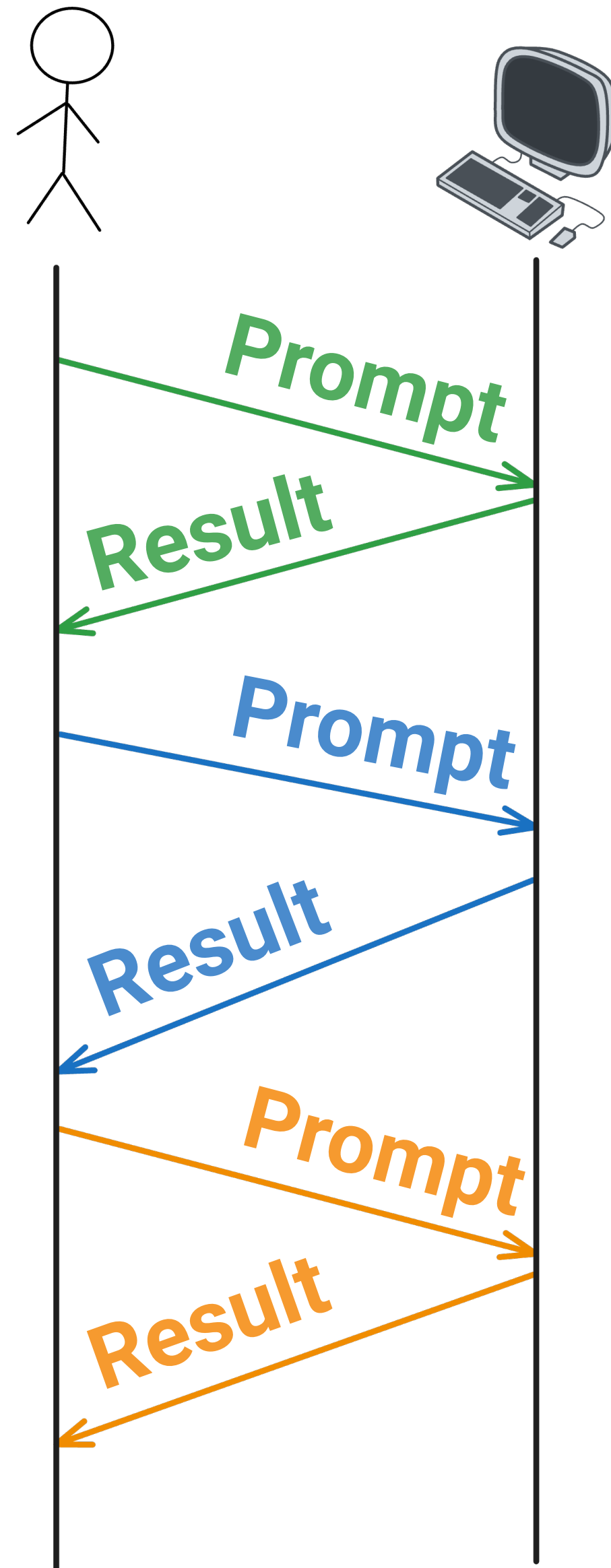


Active Participant

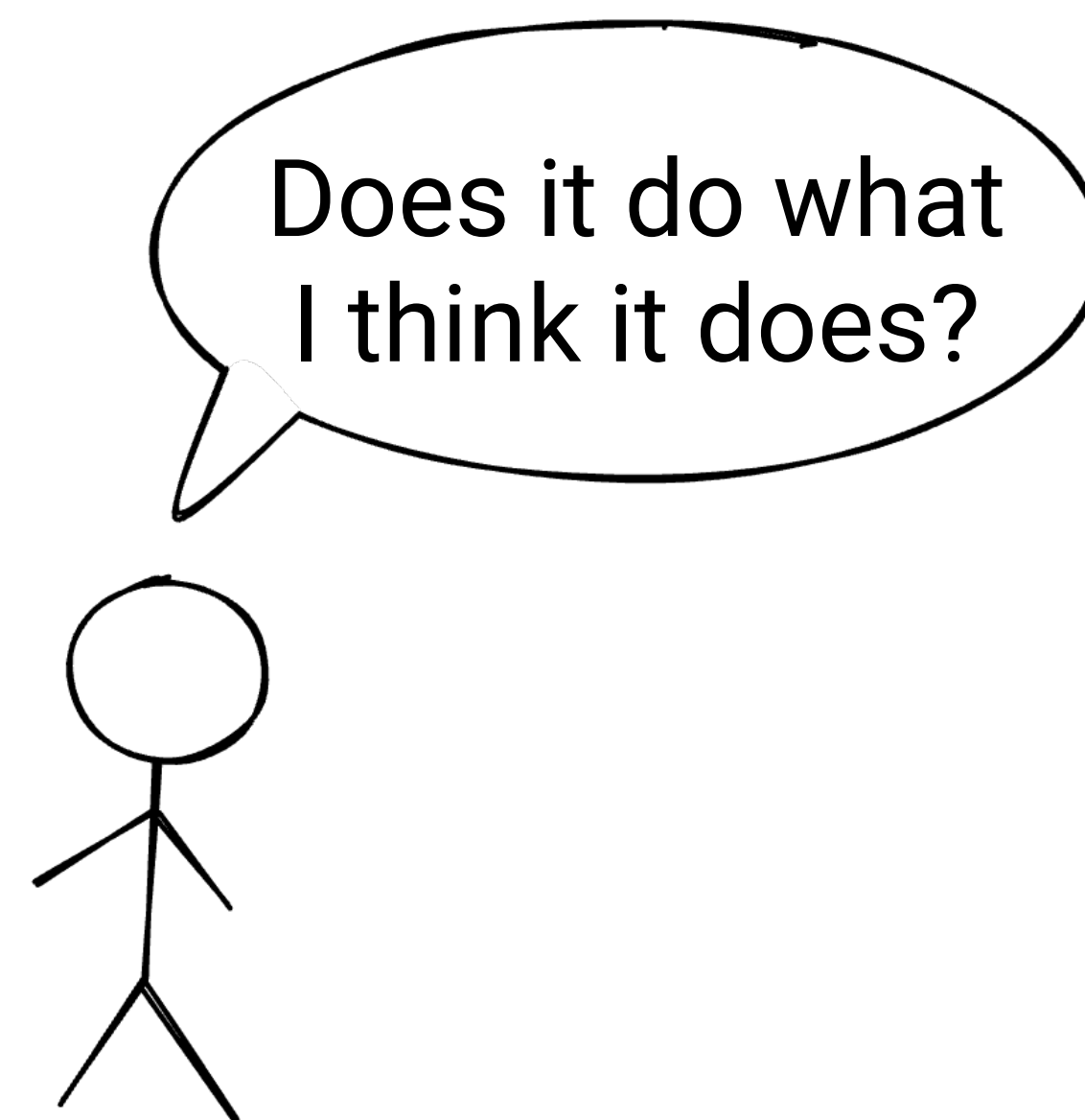
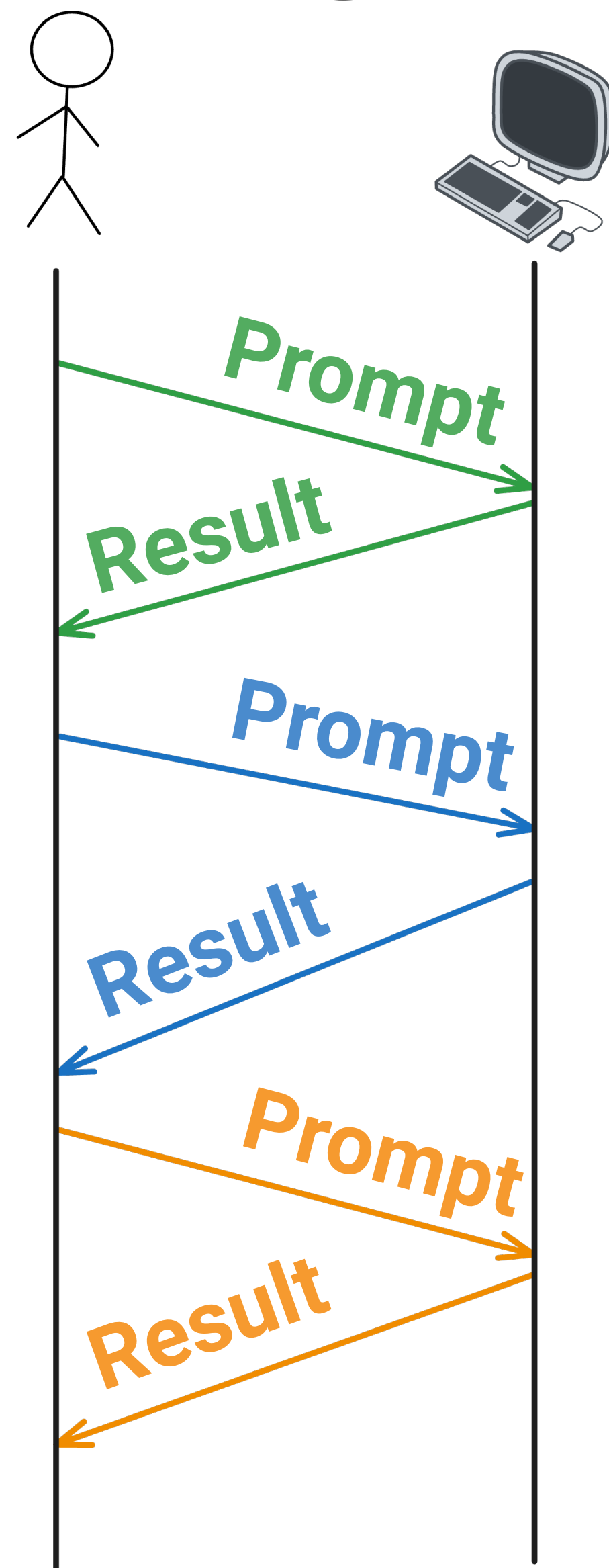
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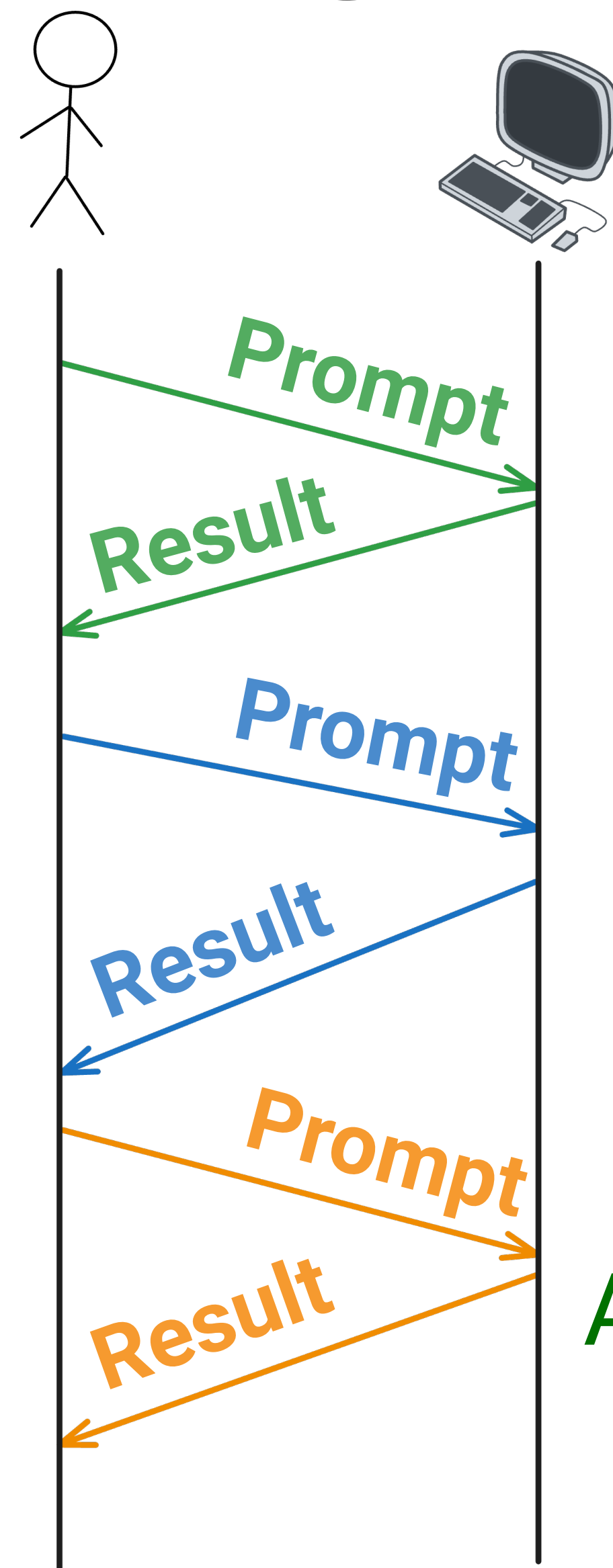
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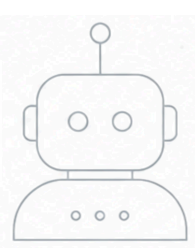


Playing To Our Strengths



Does it do what
I think it does?

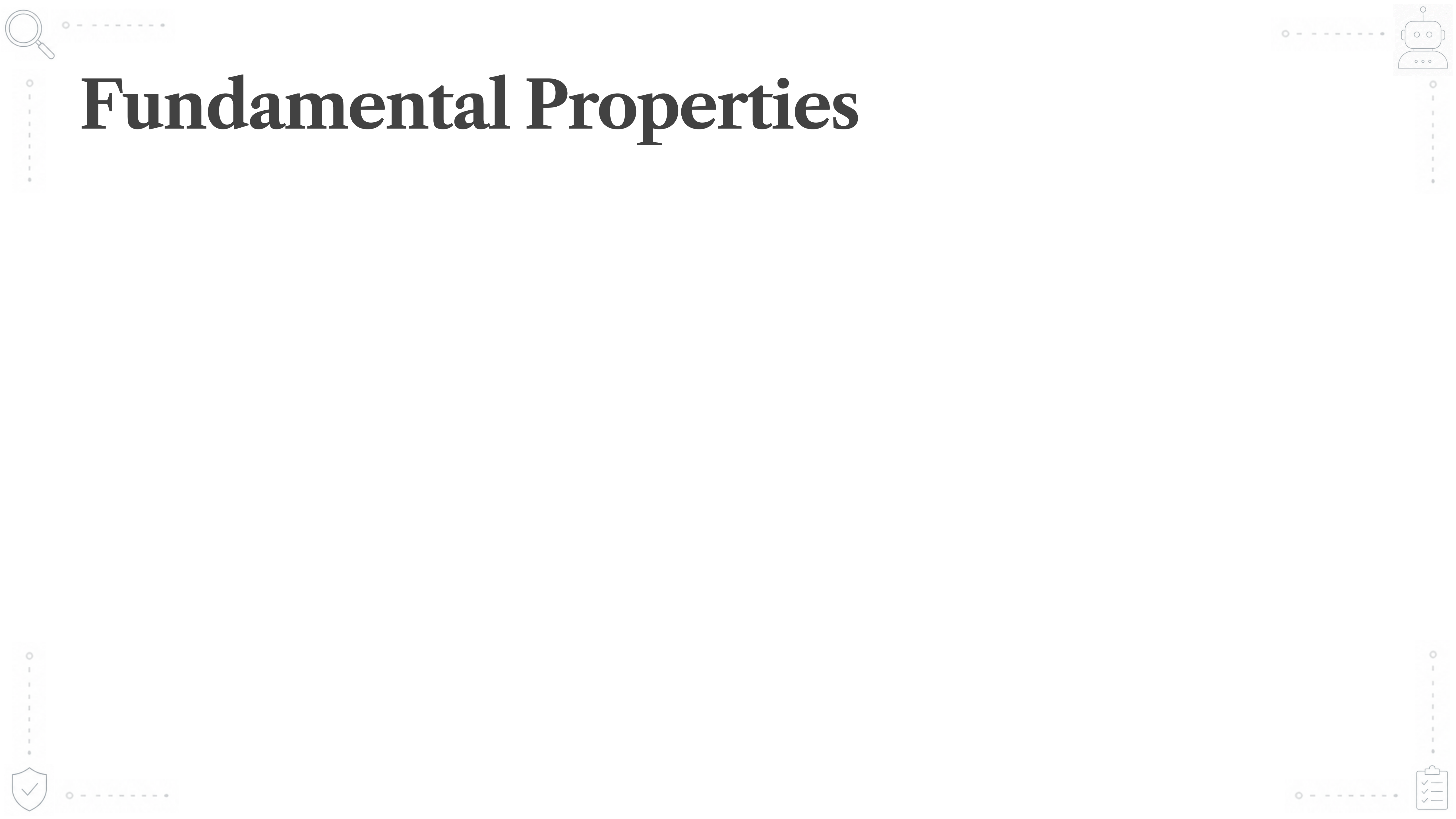
Comprehension:
Assessing the fundamentals



Embedding Comprehension

into Software Engineering





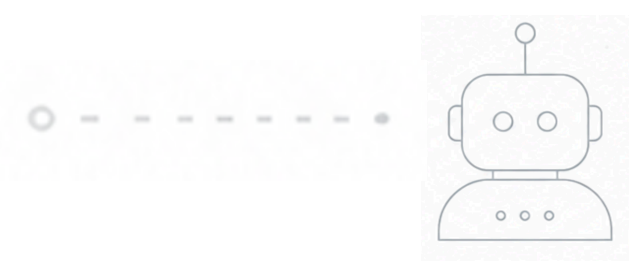
Fundamental Properties



Fundamental Properties

- A user never receives stack traces

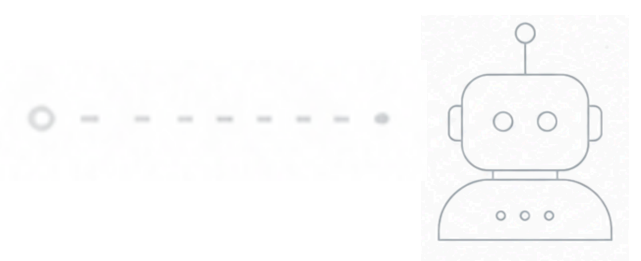




Fundamental Properties

- A user never receives stack traces
- Every element in a sorted list $\leq$ the next element

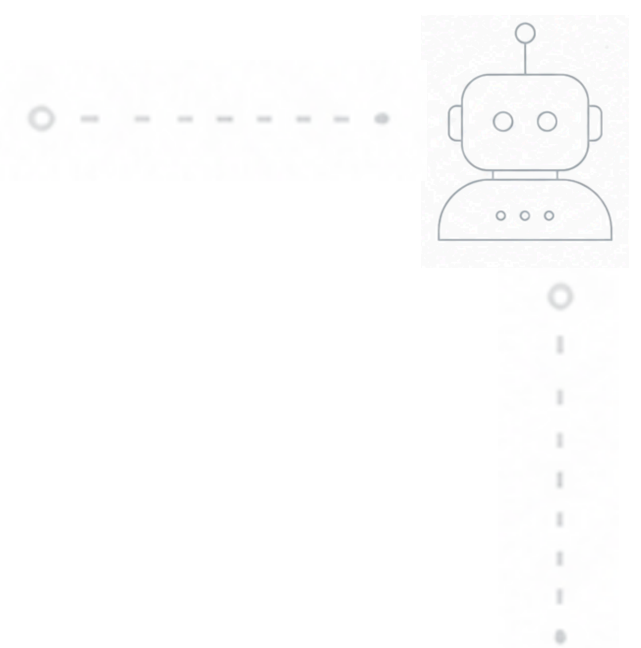




Fundamental Properties

- A user never receives stack traces
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- One person cannot be in two meetings at the same time





Fundamental Properties

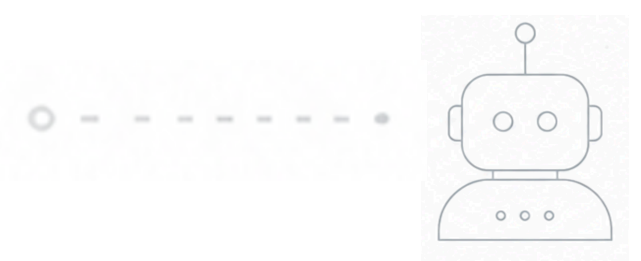
- A user never receives stack traces
- Every element in a sorted list $\leq$ the next element
- One person cannot be in two meetings at the same time
- The amount of money remains the same after account transfers





Fundamental Properties of Addition

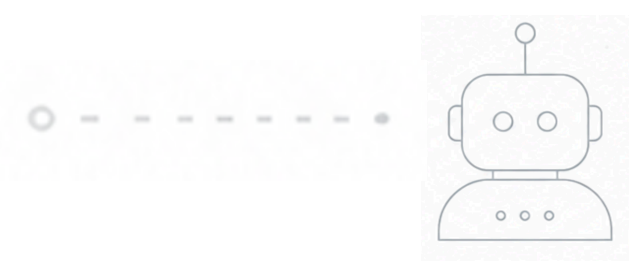




Fundamental Properties of Addition

- Zero is the additive identity
 - $a + 0 = a$

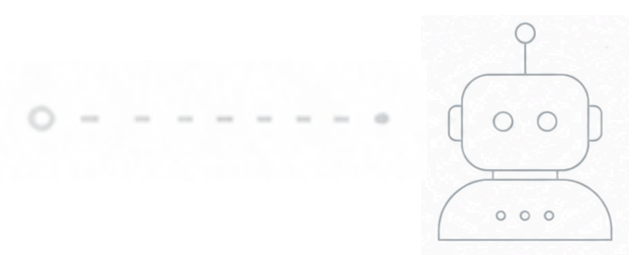




Fundamental Properties of Addition

- Zero is the additive identity
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- Adding something to its inverse gives back the identity
 - $a + (-a) = 0$

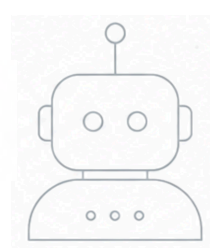




Fundamental Properties of Addition

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- Addition is commutative
 - $a + b = b + a$





Fundamental Properties of Addition

- Zero is the additive identity
 - $a + 0 = a$
- Adding something to its inverse gives back the identity
 - $a + (-a) = 0$
- Addition is commutative
 - $a + b = b + a$
- Addition is associative
 - $(a + b) + c = a + (b + c)$





Testing the Properties of Addition

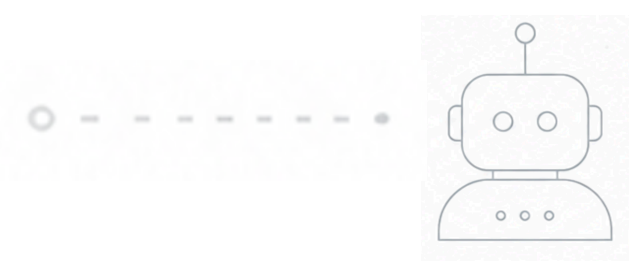




Testing the Properties of Addition

```
assertEquals(a, add(a, 0), "Additive Identity");
```

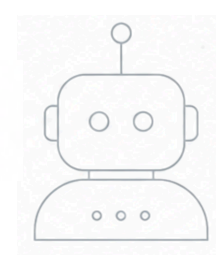




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assertEquals(add(a, b), add(b, a), "Commutativity");
```



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assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
```

Testing the Properties of Addition

```
        @ForAll double a, @ForAll double b
assertEquals(a, add(a, 0), "Additive Identity");
assertEquals(0, add(a, -a), "Additive Inverse");
assertEquals(add(a, b), add(b, a), "Commutativity");
assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");
```


Testing the Properties of Addition

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
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Testing the Properties of Addition

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}
```

@Test

```
void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(99, add(-1, 100));  
    assertEquals(230.5, add(130.5, 100));  
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}
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The problem space is explored

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    assertEquals(-101.8, add(-100, -1.8));  
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Testing the Properties of Addition

@Property

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
    assertEquals(add(a, b), add(b, a), "Commutativity");  
    assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");  
}
```

Verifies for multiple values
of **a** and **b**

@Test

```
void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(99, add(-1, 100));  
    assertEquals(230.5, add(130.5, 100));  
}
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}
```

Only verifies for 1.5 and 0

Testing the Properties of Addition

@Property

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@Test

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    assertEquals(230.5, add(130.5, 100));  
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```



Testing the Properties of Addition

@Property

void proper

assertEq

assertEq

assertEq

assertEq

}

@Test

void exampl

assertEq

assertEq

assertEq

assertEq

}

tries = 285

checks = 285

generation = RANDOMIZED

...

Original Sample

a: 15.6

b: -1.49

Original Error

org.opentest4j.AssertionFailedError:

Associativity $\implies$ expected: <15.11> but was: <15.1100000000000001>

|-----jqwik-----|

| # of calls to property

| # of not rejected calls

| parameters are randomly generated

Testing the Properties of Addition

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Testing the Properties of Addition

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void proper
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Original Sample

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jsell>

Testing the Properties of Addition

@Property

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    assertEquals
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```

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tries = 285
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generation = RANDOMIZED
...
```

Original Sample

```
a: 15.6
b: -1.49
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@Test

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void exampl
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
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Original Error

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Associativity $\Rightarrow$ expected: <15.11> but was: <15.1100000000000001>

```
|-----jqwik-----
| # of calls to property
| # of not rejected calls
| parameters are randomly generated
```

```
jshe11> 1 + (15.6 + -1.49)
$1  $\Rightarrow$  15.11
```

Testing the Properties of Addition

@Property

```
void proper
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

```
tries = 285
checks = 285
generation = RANDOMIZED
...
```

Original Sample

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a: 15.6
b: -1.49
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@Test

```
void exampl
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

Original Error

org.opentest4j.AssertionFailedError:

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-----jqwik-----
| # of calls to property
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jsHELL> 1 + (15.6 + -1.49)
$1  $\Rightarrow$  15.11
```

```
jsHELL> (1 + 15.6) + -1.49
$2  $\Rightarrow$  15.1100000000000001
```

Testing the Properties of Addition

@Property

```
void proper
    assertEquals
    assertEquals
    assertEquals
    assertEquals
}
```

```
tries = 285
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...
```

Original Sample

```
-----
a: 15.6
b: -1.49
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@Test

```
void exampl
    assertEquals
    assertEquals
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    assertEquals
}
```

Original Error

```
-----
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Associativity $\Rightarrow$ expected: <15.11> but was: <15.1100000000000001>

```
|-----jqwik-----
| # of calls to property
| # of not rejected calls
| parameters are randomly generated
```

```
jsHELL> 1 + (15.6 + -1.49)
$1  $\Rightarrow$  15.11
```

```
jsHELL> (1 + 15.6) + -1.49
$2  $\Rightarrow$  15.1100000000000001
```

```
jsHELL> 15.6d == 15.6f
$3  $\Rightarrow$  false
```


Testing the Properties of Addition

@Property

```
void propertyBasedAdditionTest(@ForAll double a, @ForAll double b) {  
    assertEquals(a, add(a, 0), "Additive Identity");  
    assertEquals(0, add(a, -a), "Additive Inverse");  
    assertEquals(add(a, b), add(b, a), "Commutativity");  
    assertEquals(add(1, add(a, b)), add(add(1, a), b), "Associativity");  
}
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@Test

```
void exampleBasedAdditionTest() {  
    assertEquals(1.5, add(1.5, 0));  
    assertEquals(-101.8, add(-100, -1.8));  
    assertEquals(99, add(-1, 100));  
    assertEquals(230.5, add(130.5, 100));  
}
```

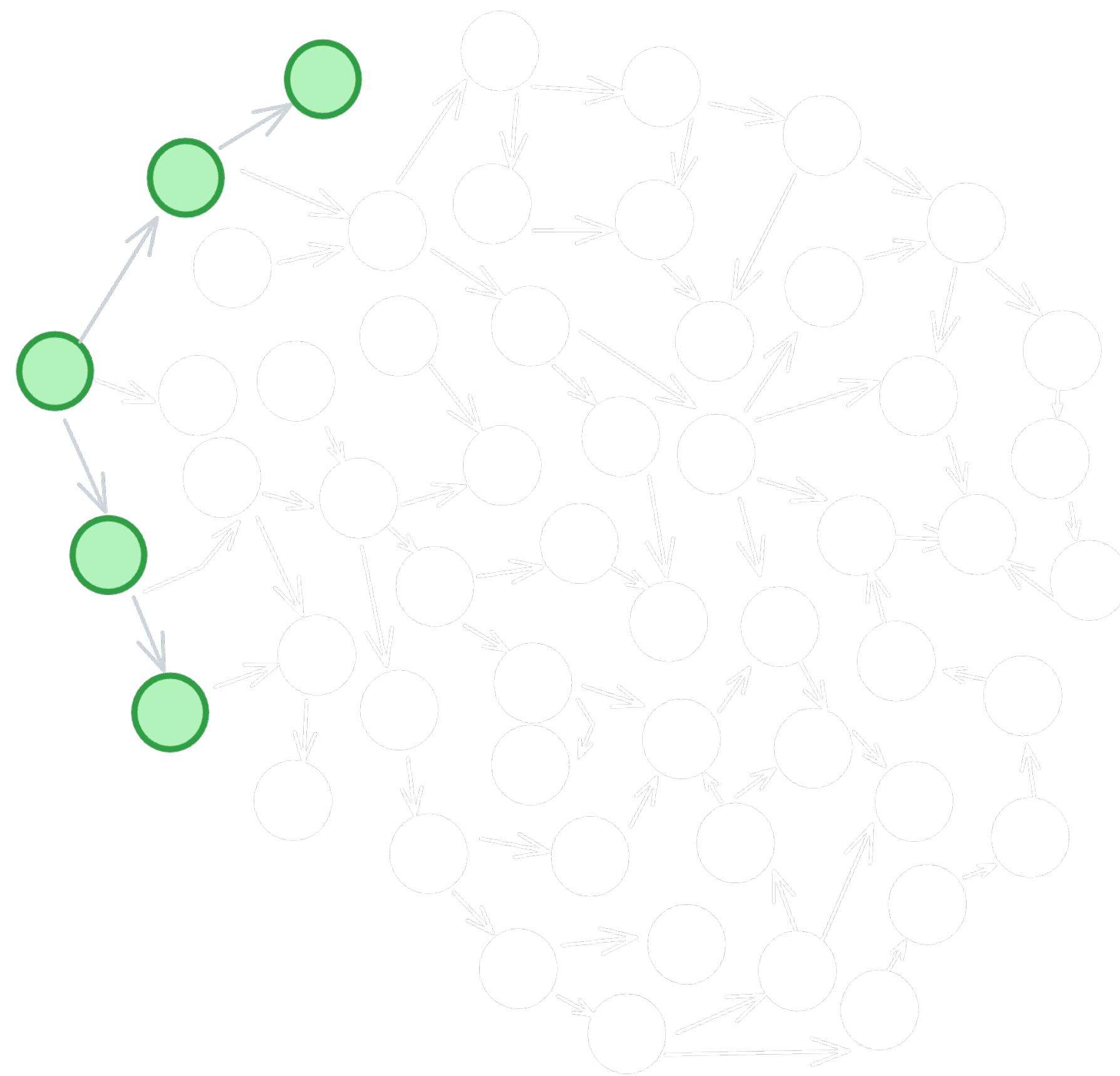




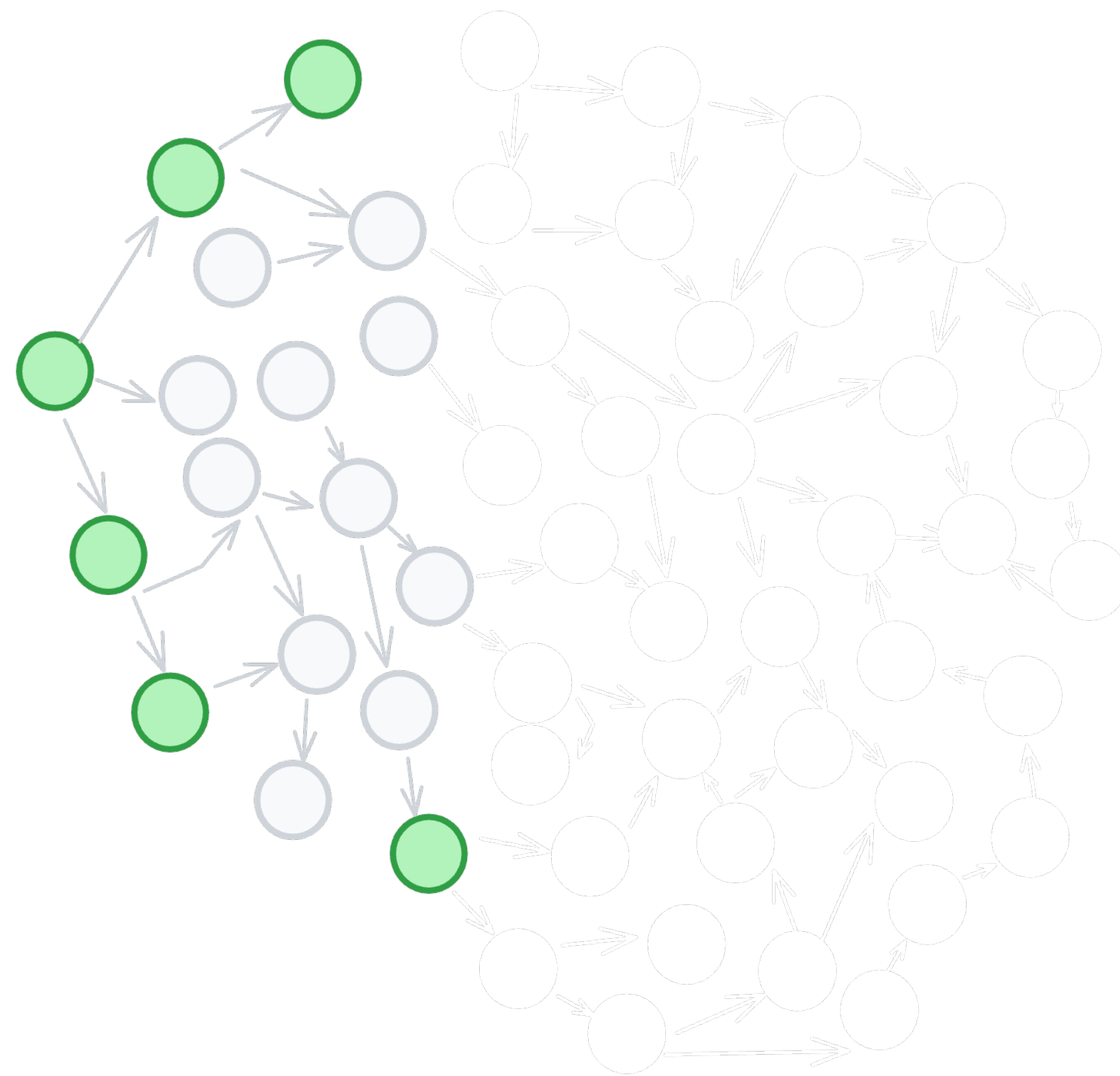
Examples → Exploration



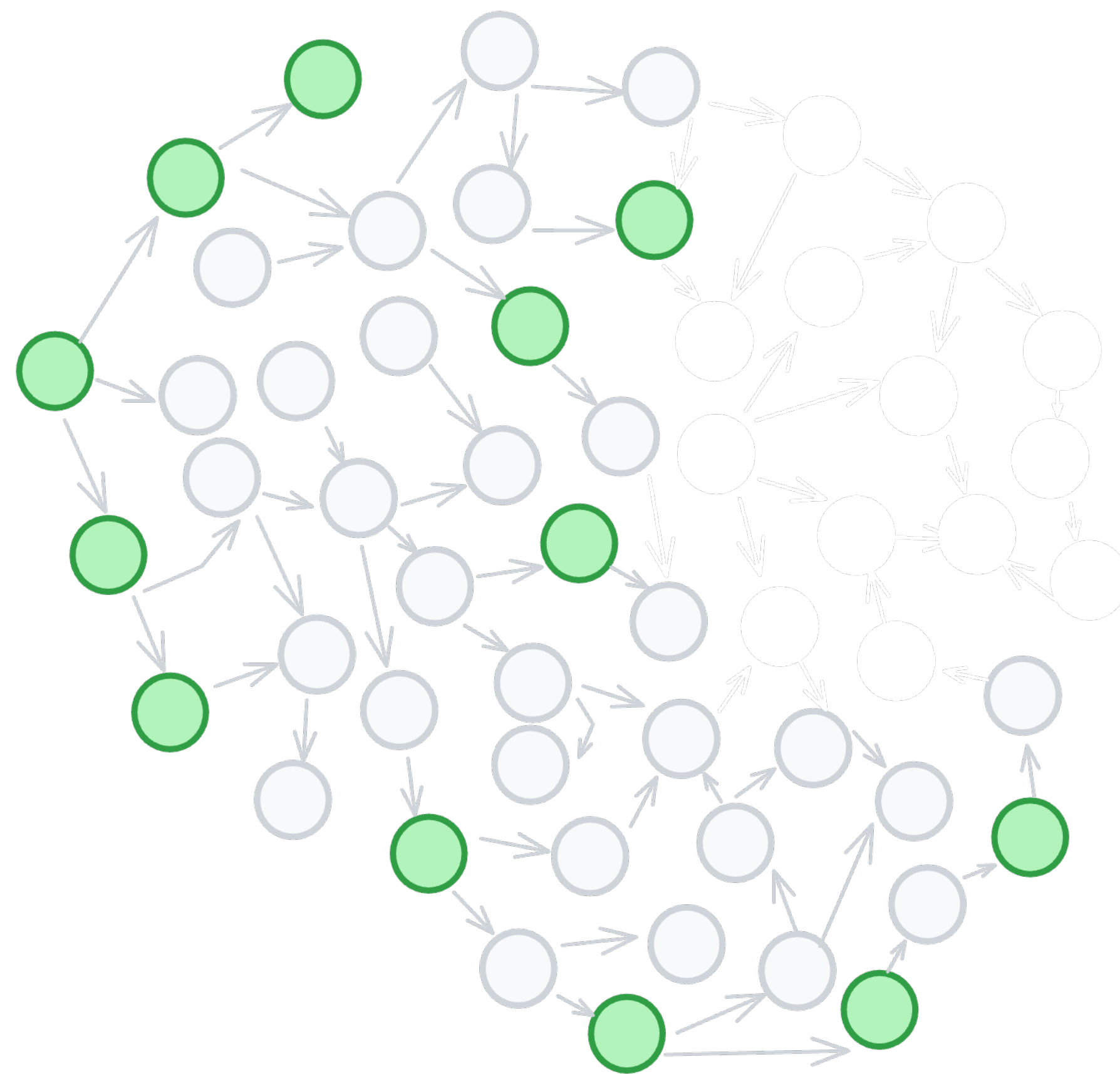
Examples → Exploration



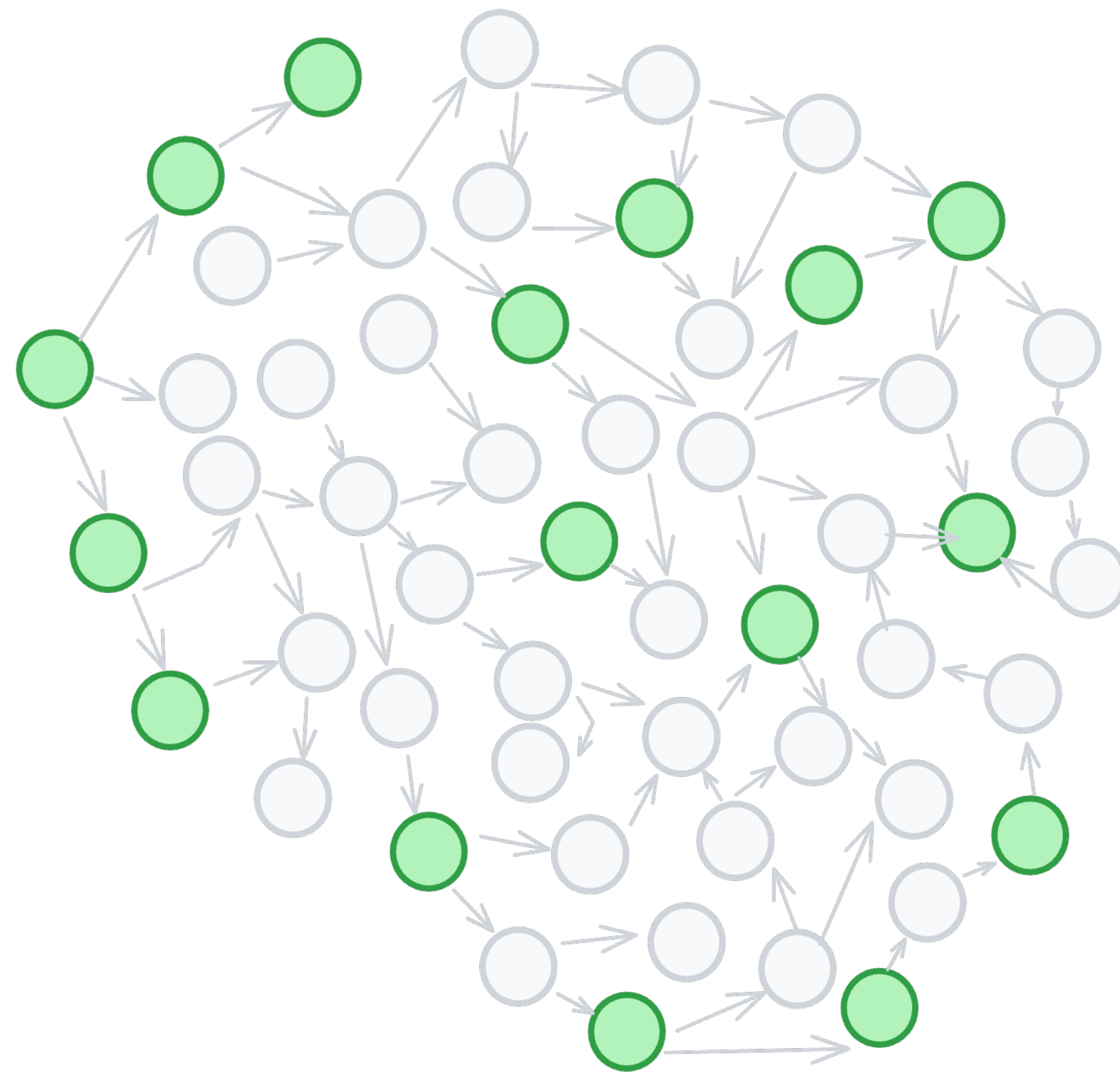
Examples → Exploration



Examples → Exploration

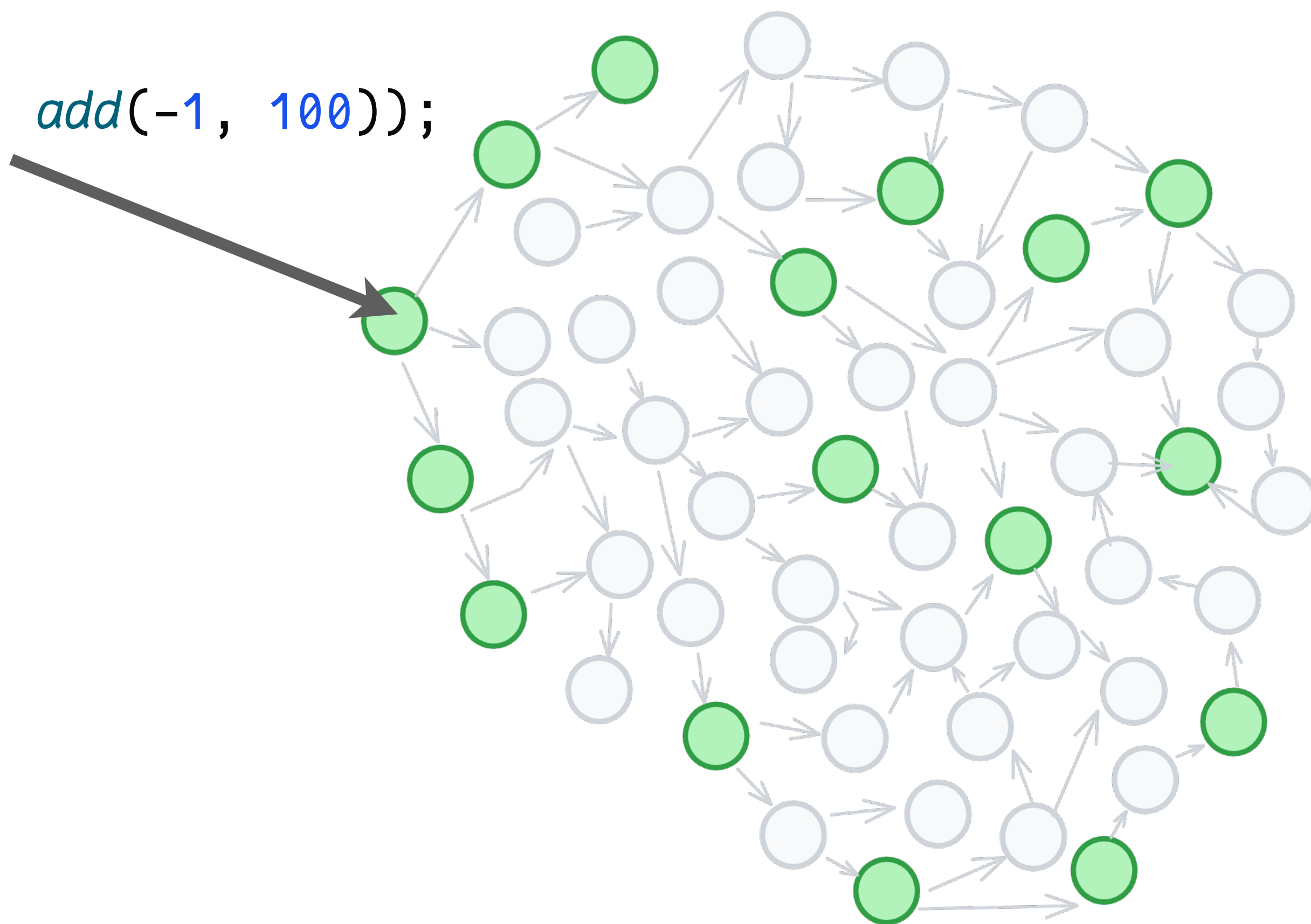


Examples → Exploration



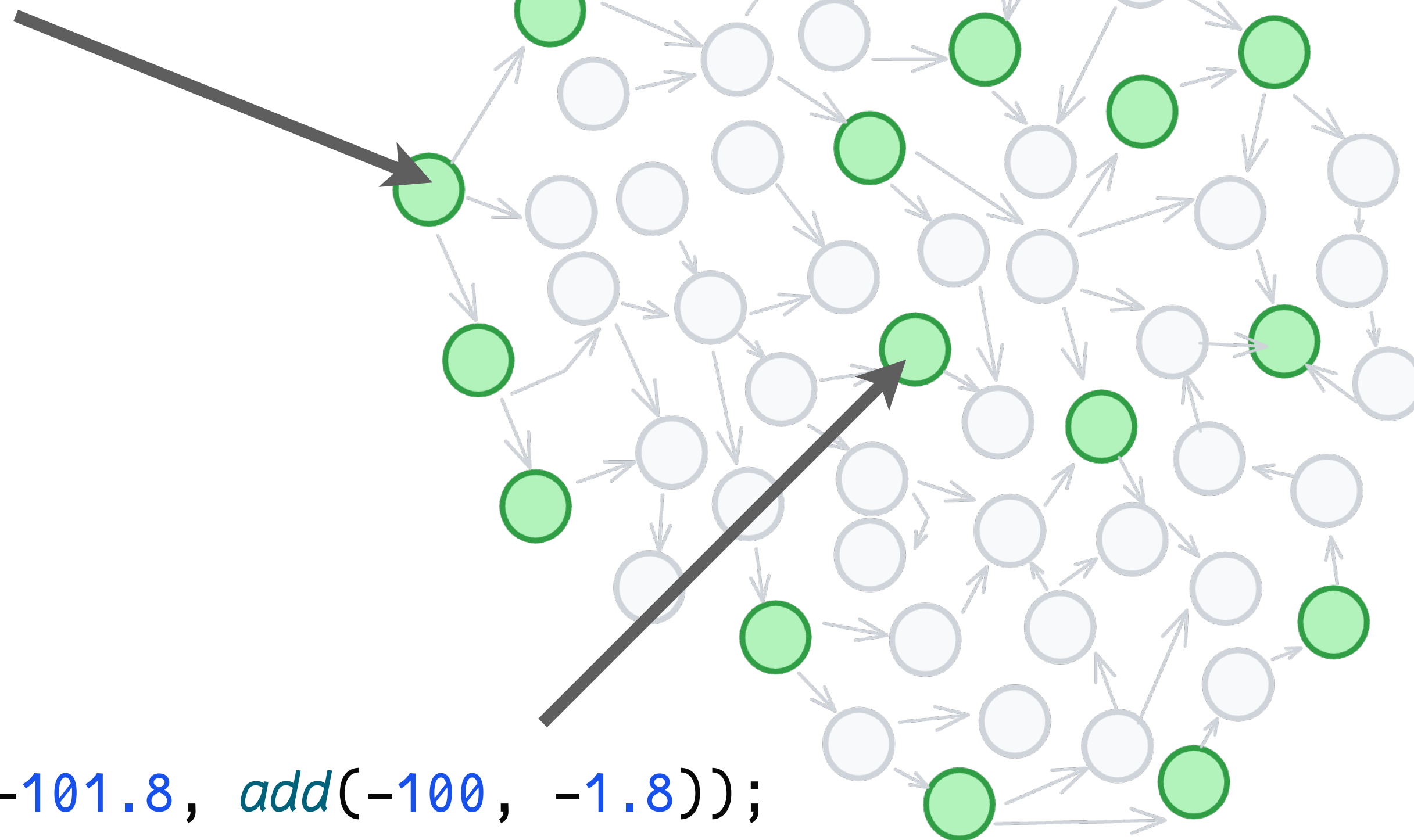
Examples → Exploration

`assertEquals(99, add(-1, 100));`



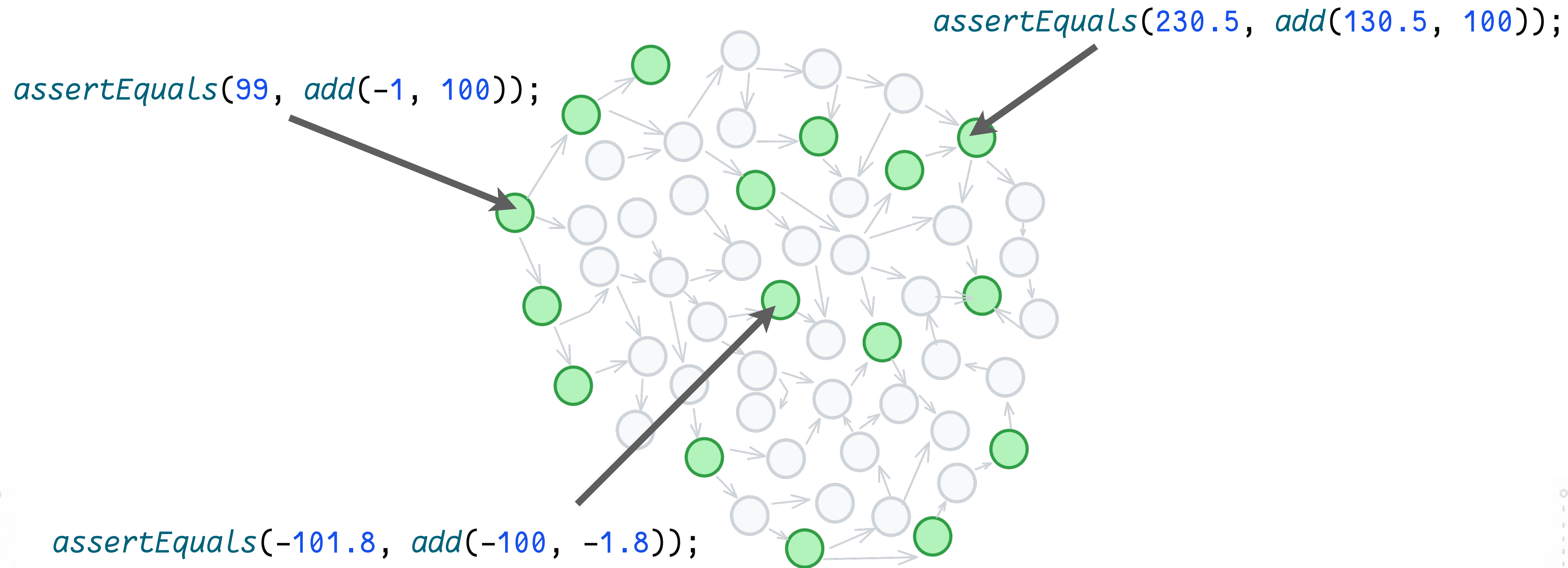
Examples → Exploration

`assertEquals(99, add(-1, 100));`

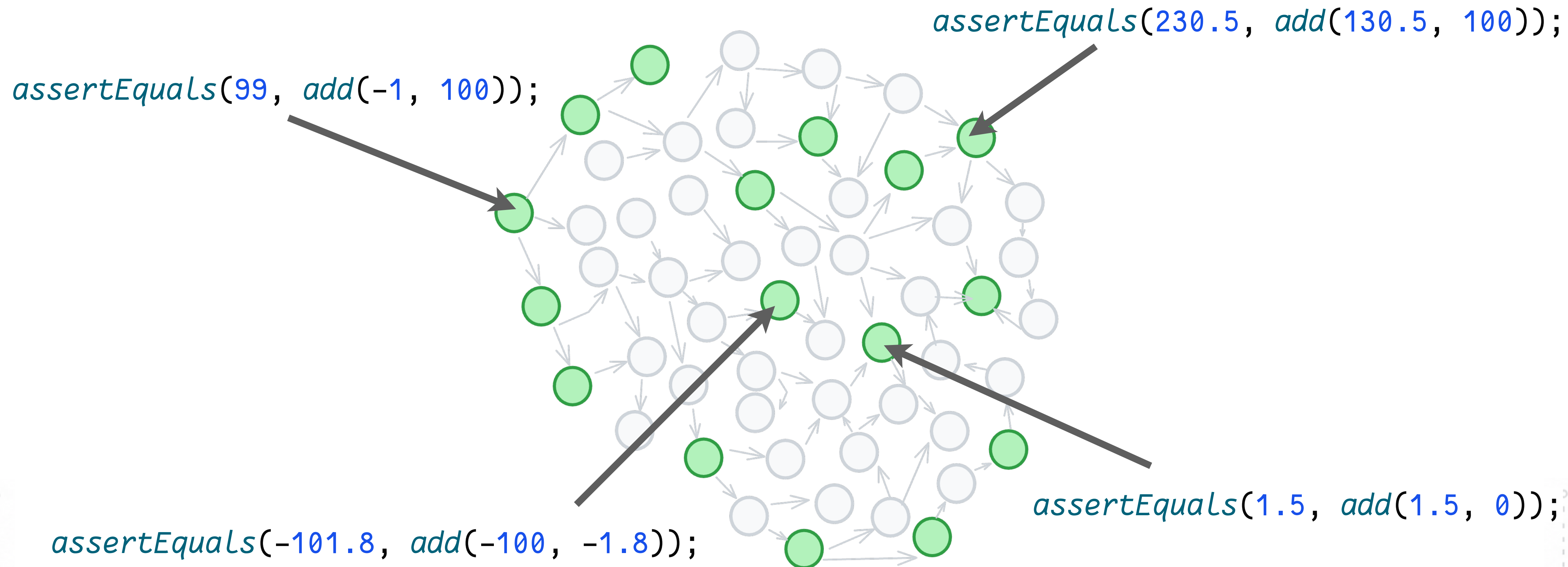


`assertEquals(-101.8, add(-100, -1.8));`

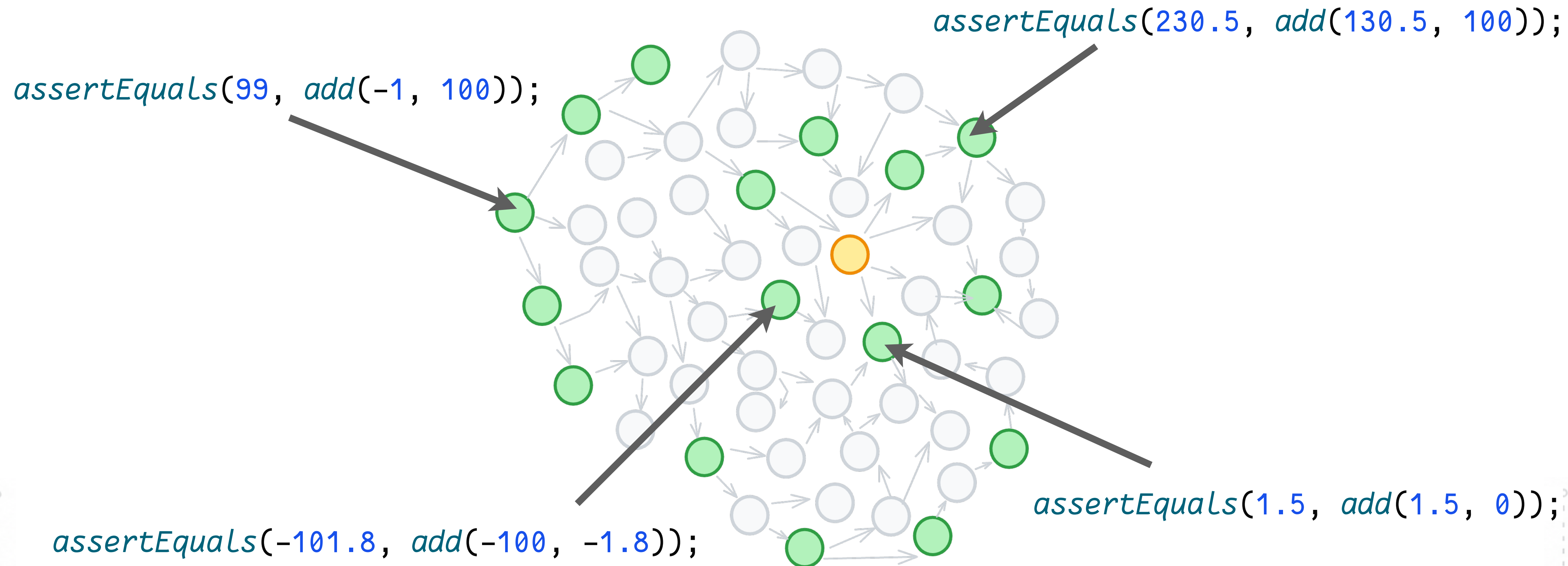
Examples → Exploration



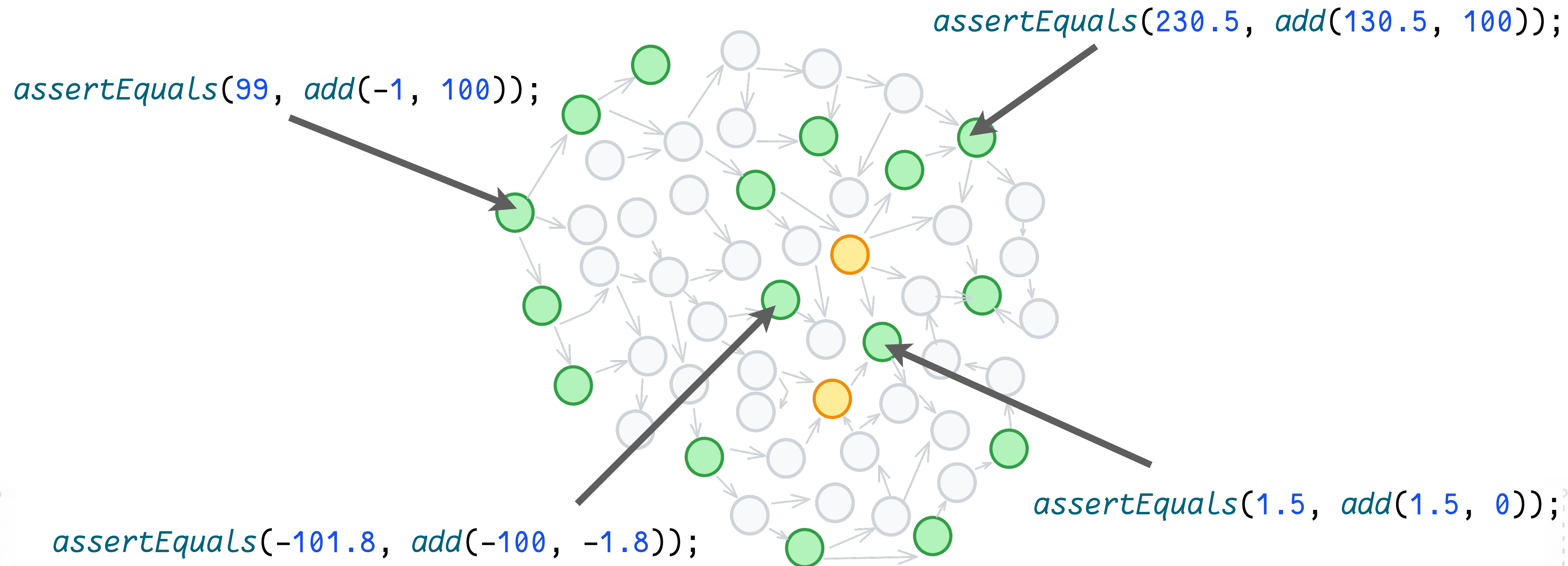
Examples → Exploration



Examples → Exploration

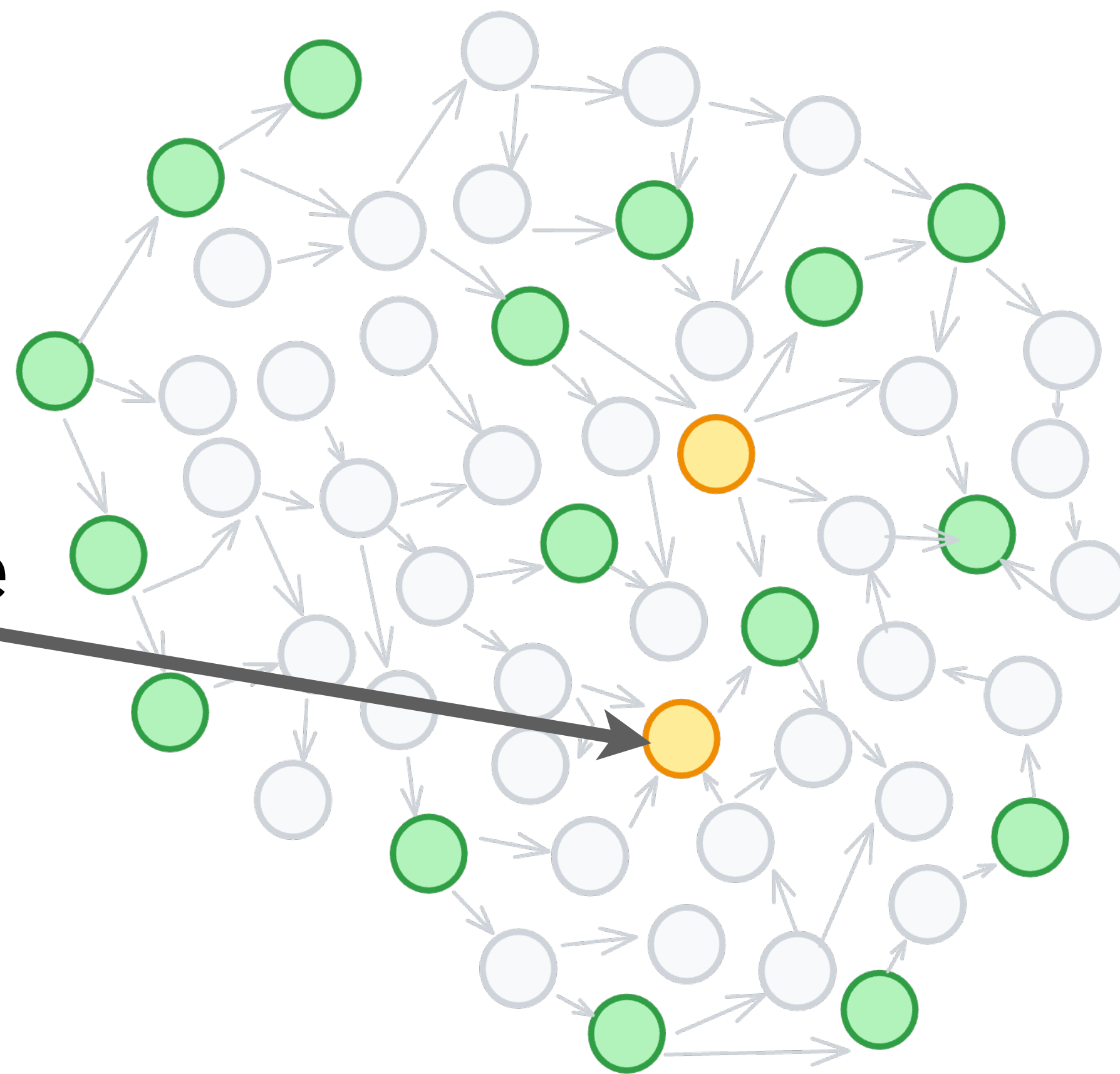


Examples → Exploration



Examples → Exploration

What if the LLM didn't write
a test case for this?





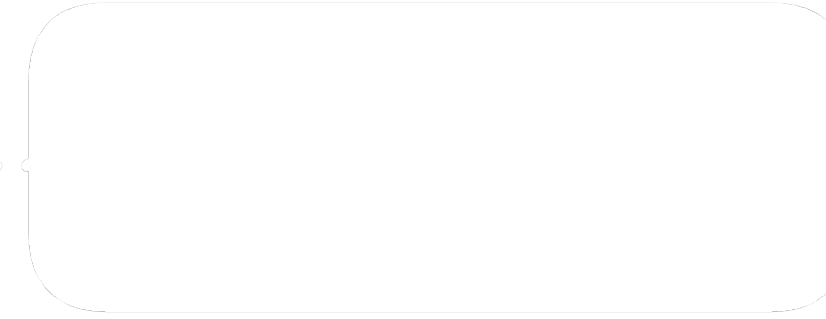
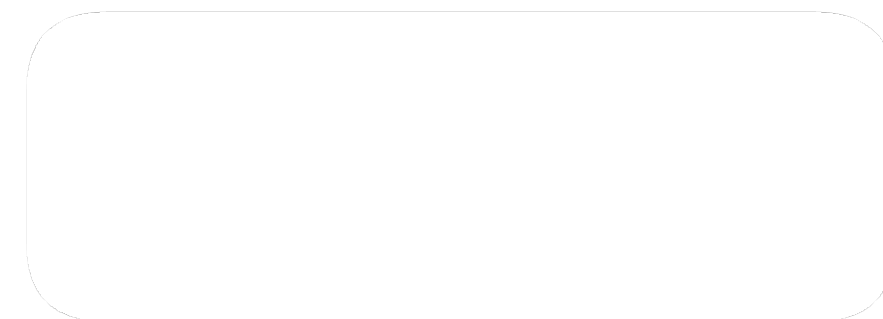
Examples → Exploration



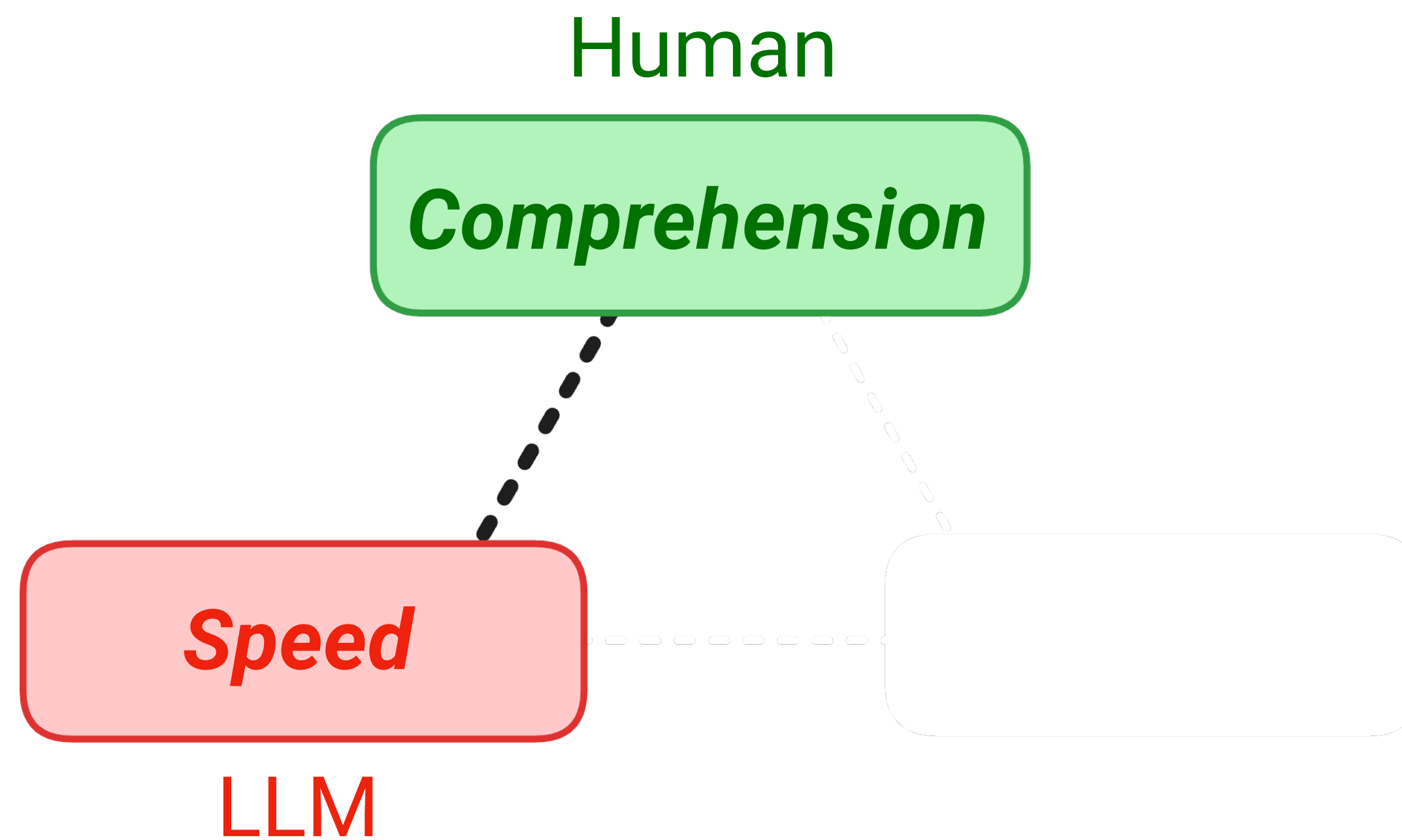
Examples → Exploration

Human

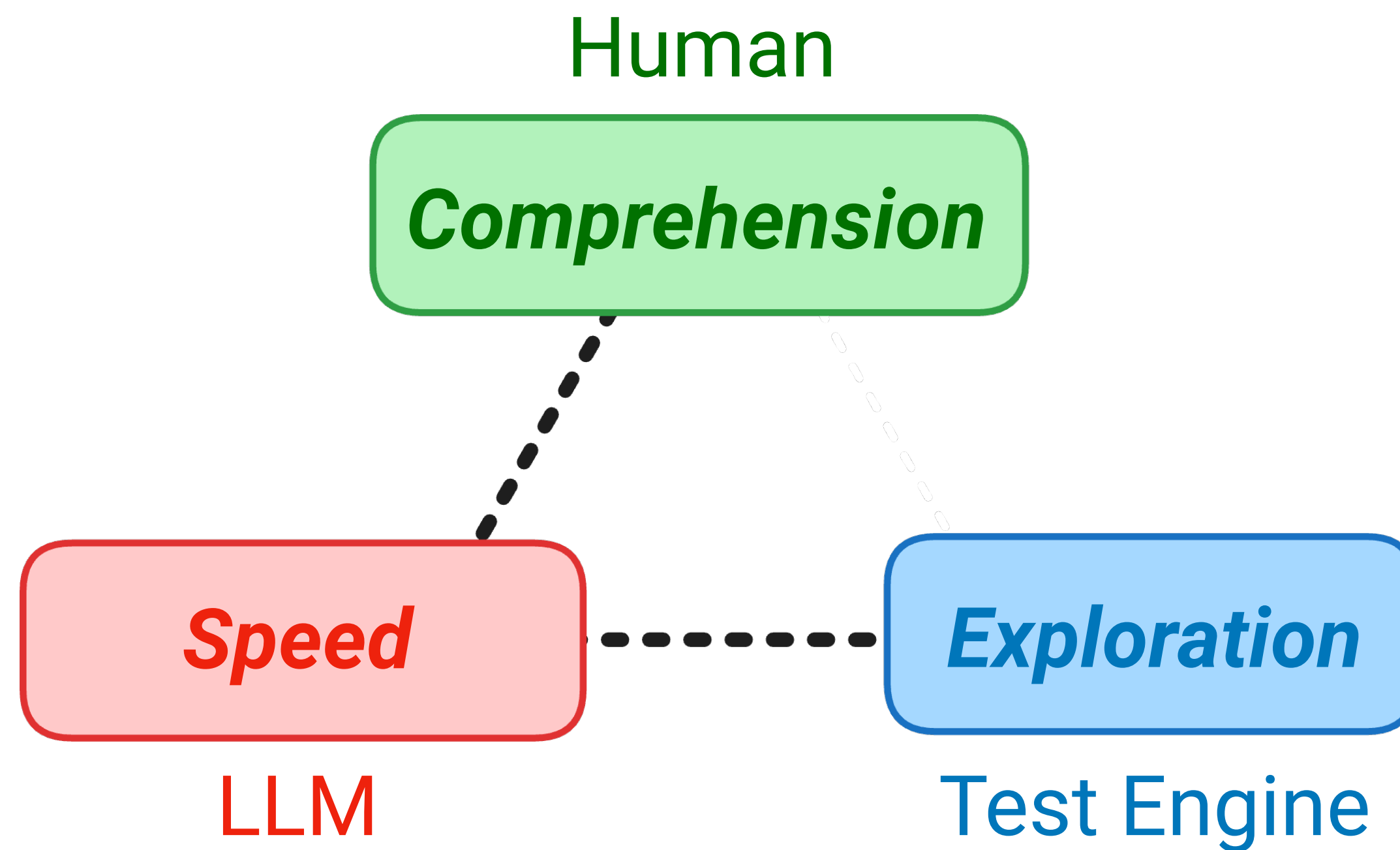
Comprehension



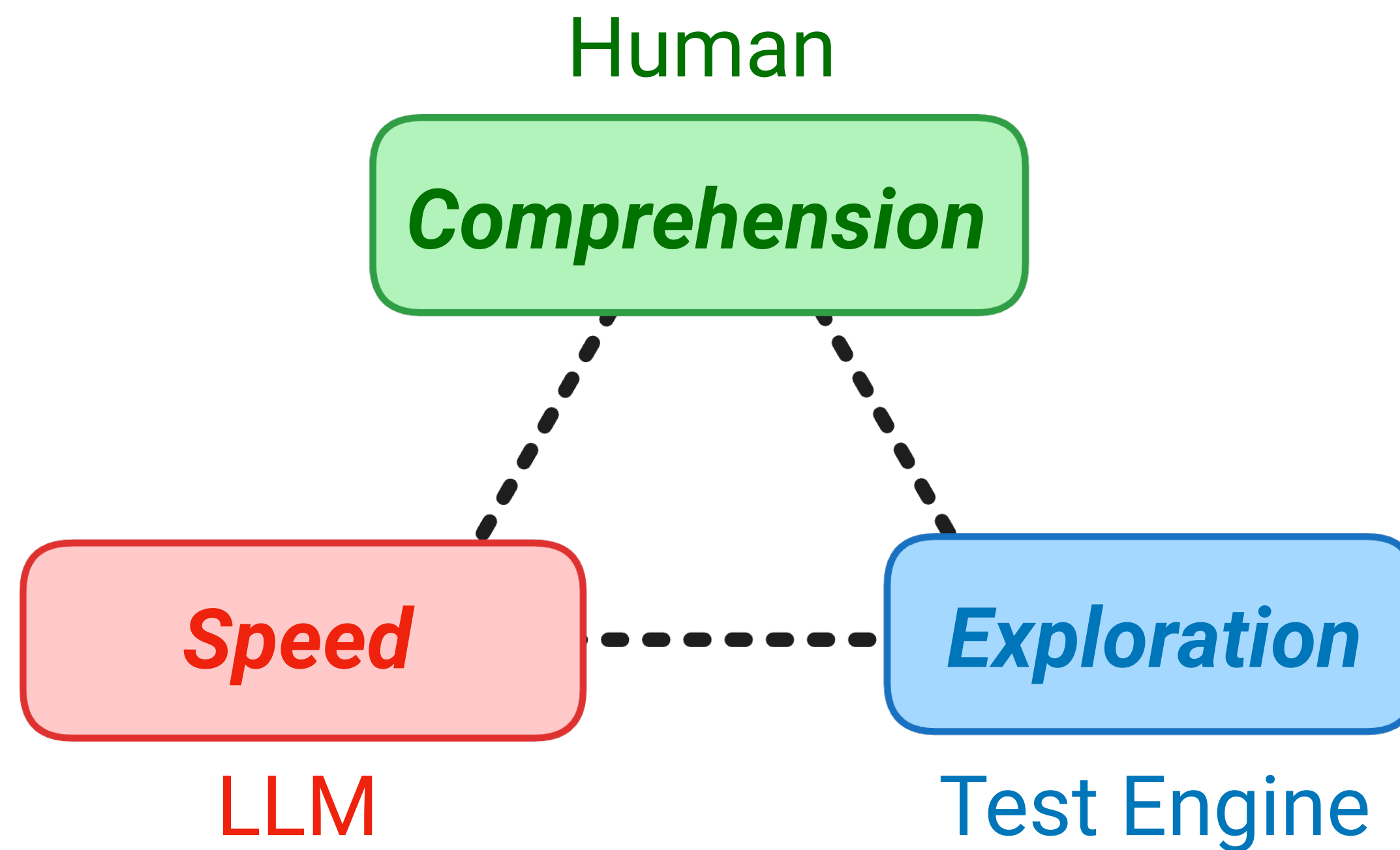
Examples → Exploration



Examples → Exploration



Examples → Exploration

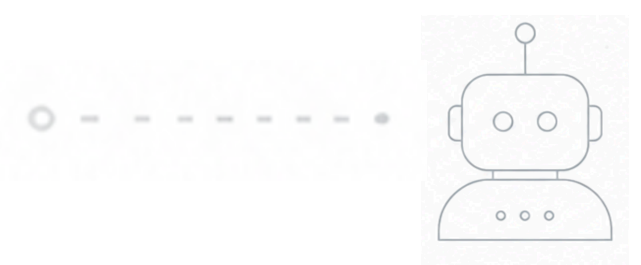




Examples → Exploration

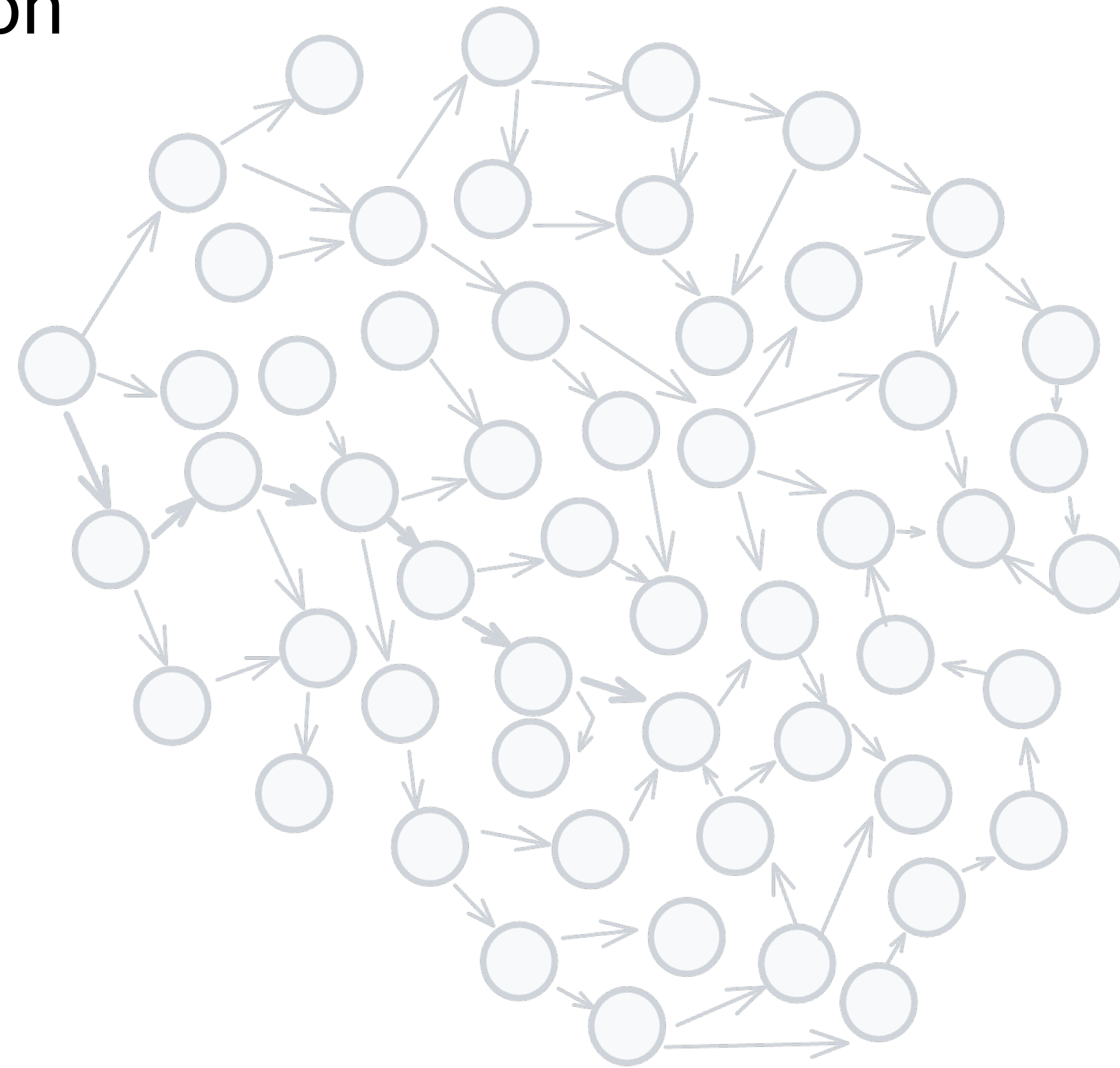
LLMs generate a solution





Examples → Exploration

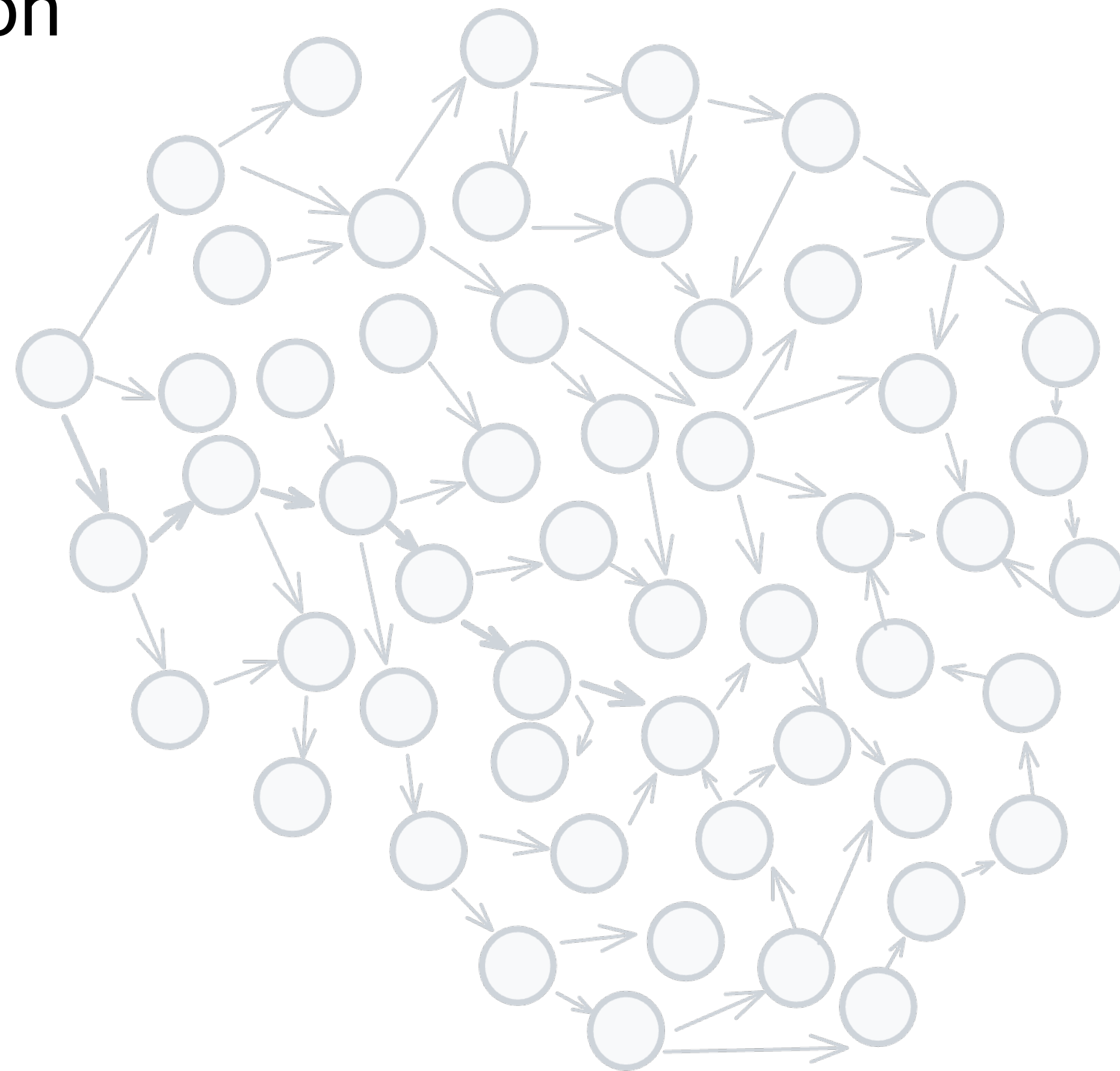
LLMs generate a solution



Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

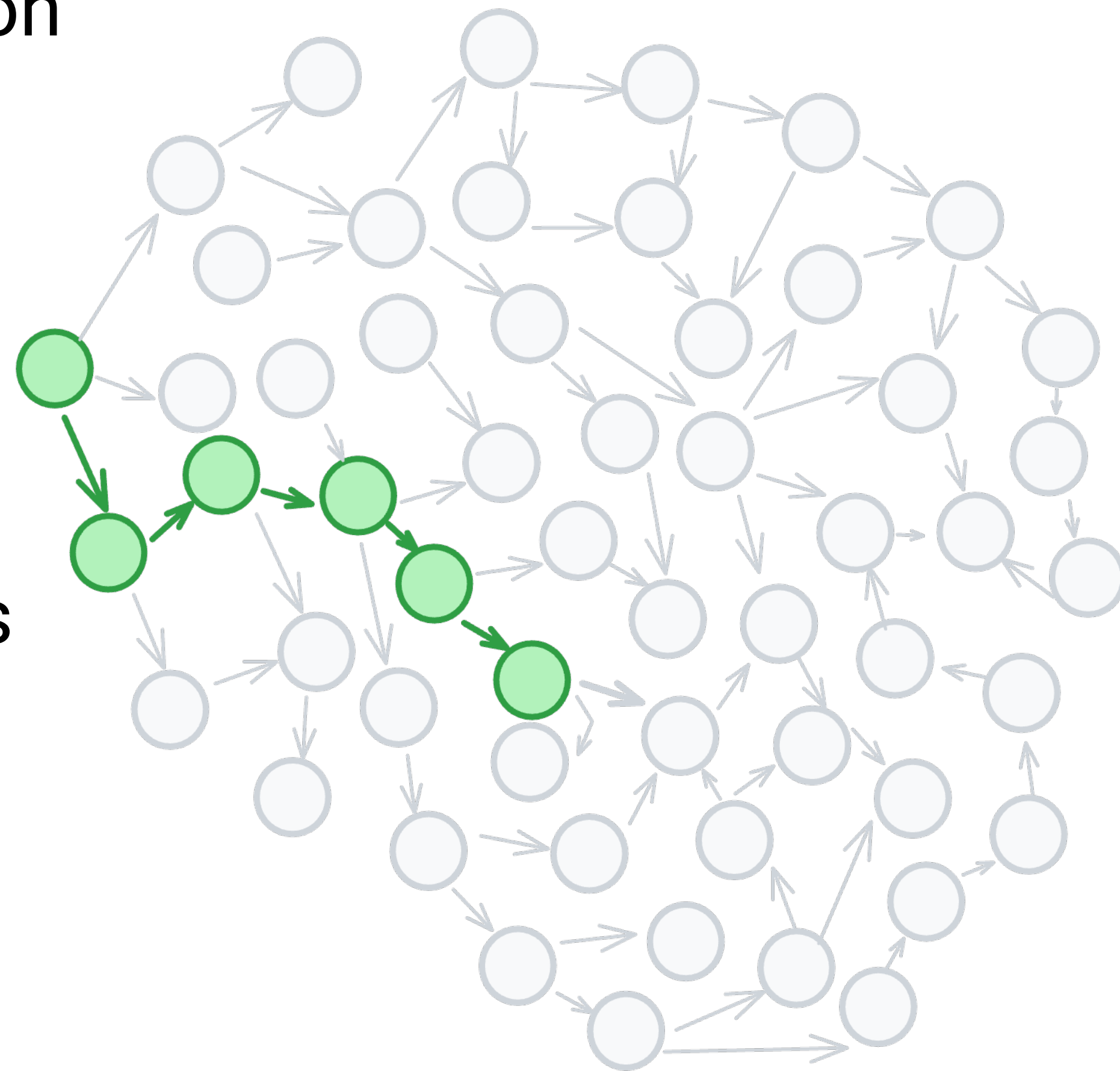


Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

The test engine explores
the problem space

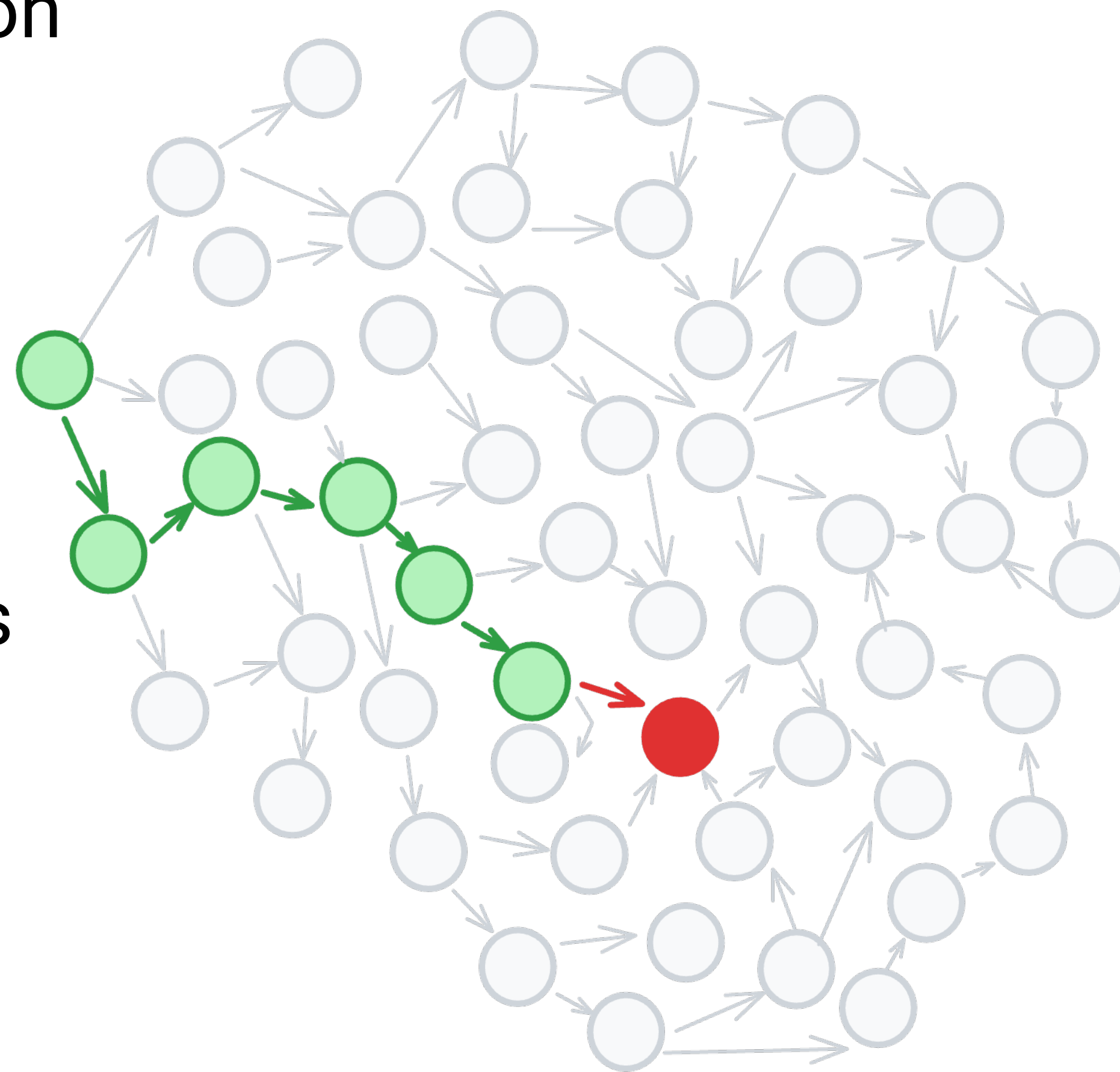


Examples → Exploration

LLMs generate a solution

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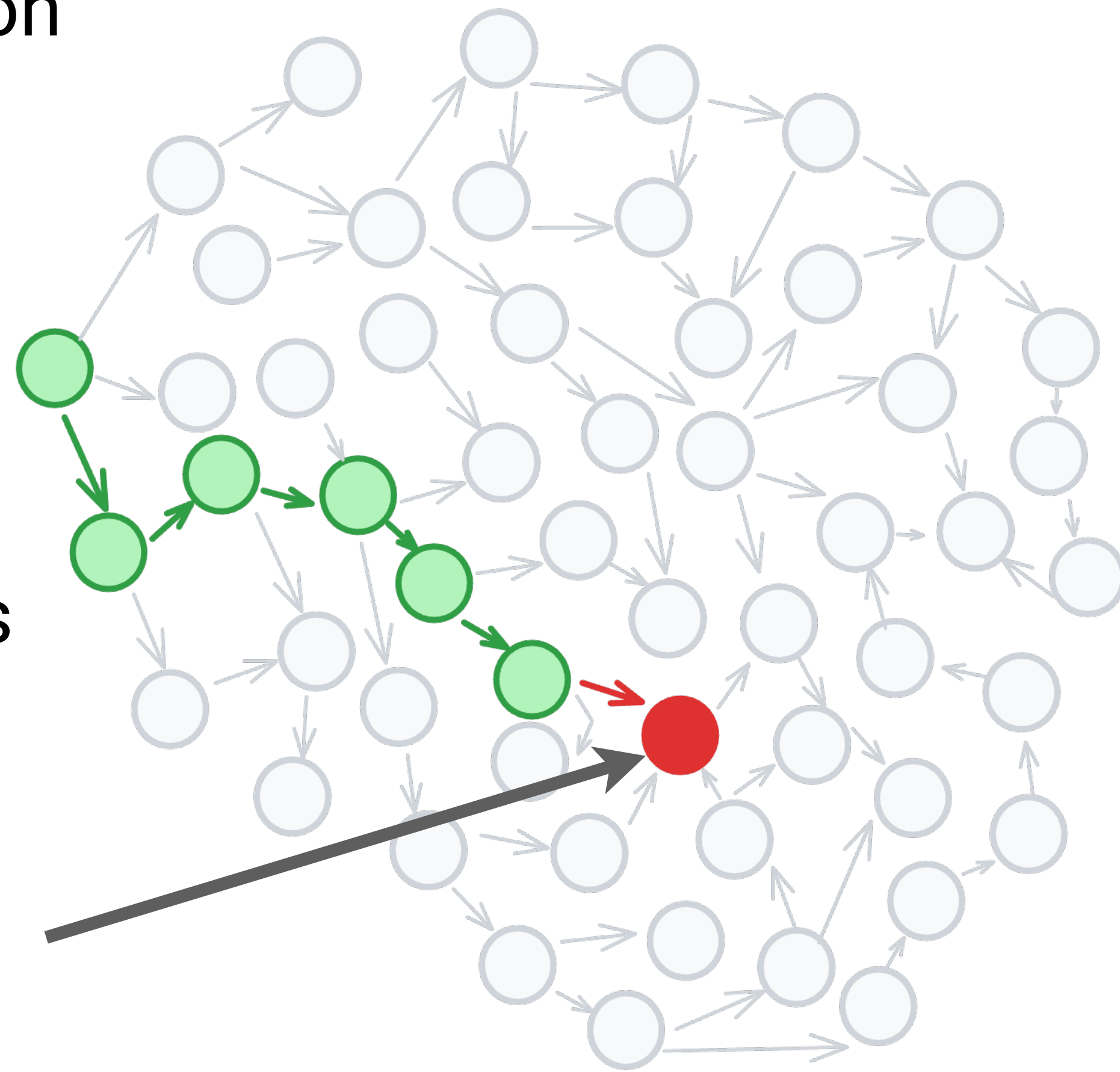
Examples → Exploration

LLMs generate a solution

Humans define what
must always hold true

The test engine explores
the problem space

The test engine finds
a counter example



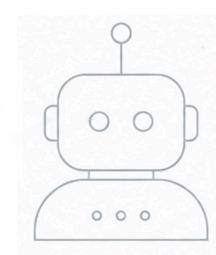


Failure



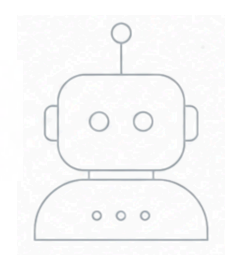
Failure → Shrinking





Failure → Shrinking → Understanding





Failure → Shrinking → Understanding

Original Sample

```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,  
           249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,  
           8045, -373, 147465, -3, -58, 514, -177, 97, 1386116950, 2953, 3804]
```

Expecting actual:

1386116950

to be less than or equal to:

3804



Failure → Shrinking → Understanding

Original Sample

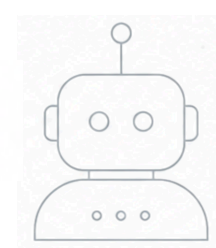
```
integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,  
           249644599, -5995, 2396, -1997, 24, 23, -121416, -868149, -38934,  
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Expecting actual:

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to be less than or equal to:

3804



Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

Original Sample

integers: [-778383, 138, -958125, -9, -2125, 1430, -2, 16, 1, -50110, -91381,
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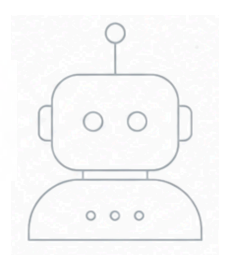
Expecting actual:

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to be less than or equal to:

3804





Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```



Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

-1

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for (int i = 0; i < n - 1; i++) {  
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            arr.set(j, arr.get(j + 1));  
            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted

Failure → Shrinking → Understanding

Shrunk Sample (11 steps)

integers: [0, -1]

Expecting actual:

0

to be less than or equal to:

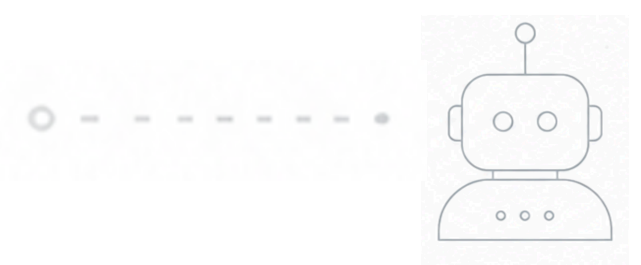
-1

```
for (int i = 0; i < n - 1; i++) {  
    for (int j = 0; j < n - i - 2; j++) {  
        if (arr.get(j) > arr.get(j + 1)) {  
            int temp = arr.get(j);  
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            arr.set(j + 1, temp);  
        }  
    }  
}
```

Last item never gets sorted
Shrunk samples are easier to **understand**



How Did I Miss That?



Meeting Scheduling

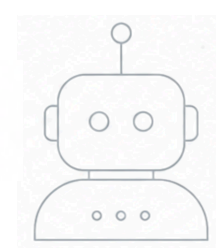




Meeting Scheduling

Existing Meeting



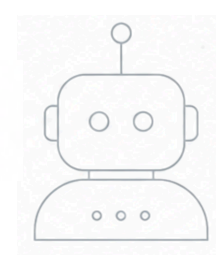


Meeting Scheduling

Existing Meeting

New Meeting





Meeting Scheduling

Existing Meeting

New Meeting

New Meeting



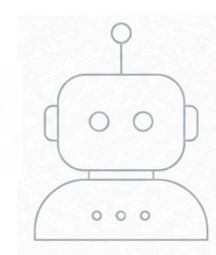


Meeting Scheduling

Existing Meeting

New Meeting





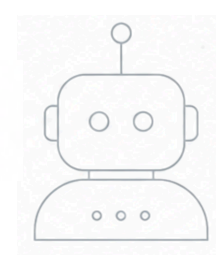
Meeting Scheduling

Existing Meeting

New Meeting

New Meeting





Meeting Scheduling

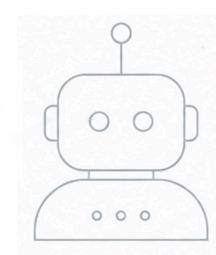
Existing Meeting

New Meeting

New Meeting

New Meeting





Meeting Scheduling

Calendar 1

Existing Meeting

New Meeting

New Meeting

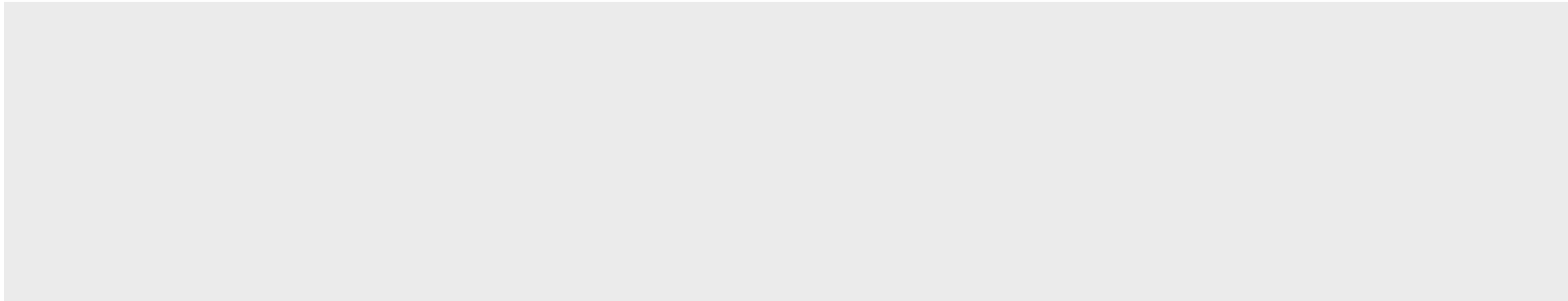
New Meeting



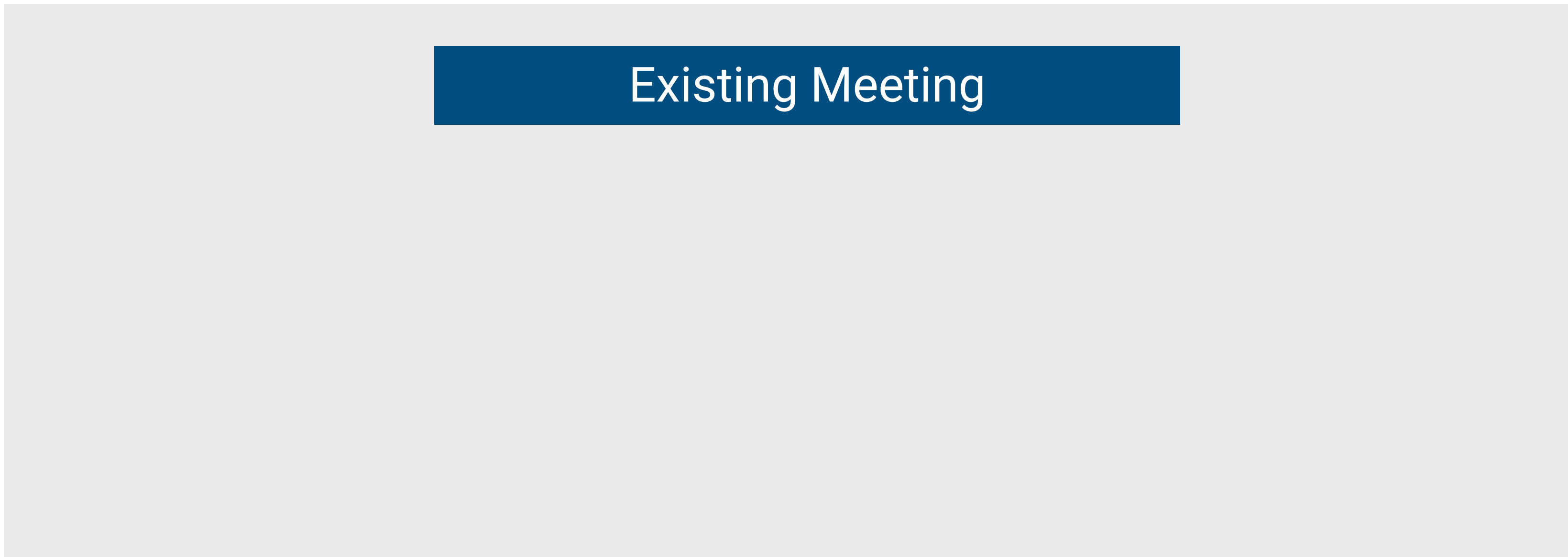


Meeting Scheduling

Calendar 2

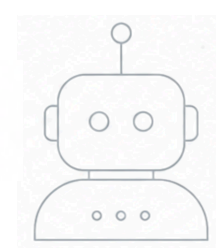


Calendar 1



Existing Meeting





Meeting Scheduling

Calendar 2

New Meeting

Calendar 1

Existing Meeting



Meeting Scheduling

Calendar 2

New Meeting

Calendar 1

~~Existing Meeting~~

**Auto-reject lower
priority conflicts**

Meeting Scheduling, Vibe Coded



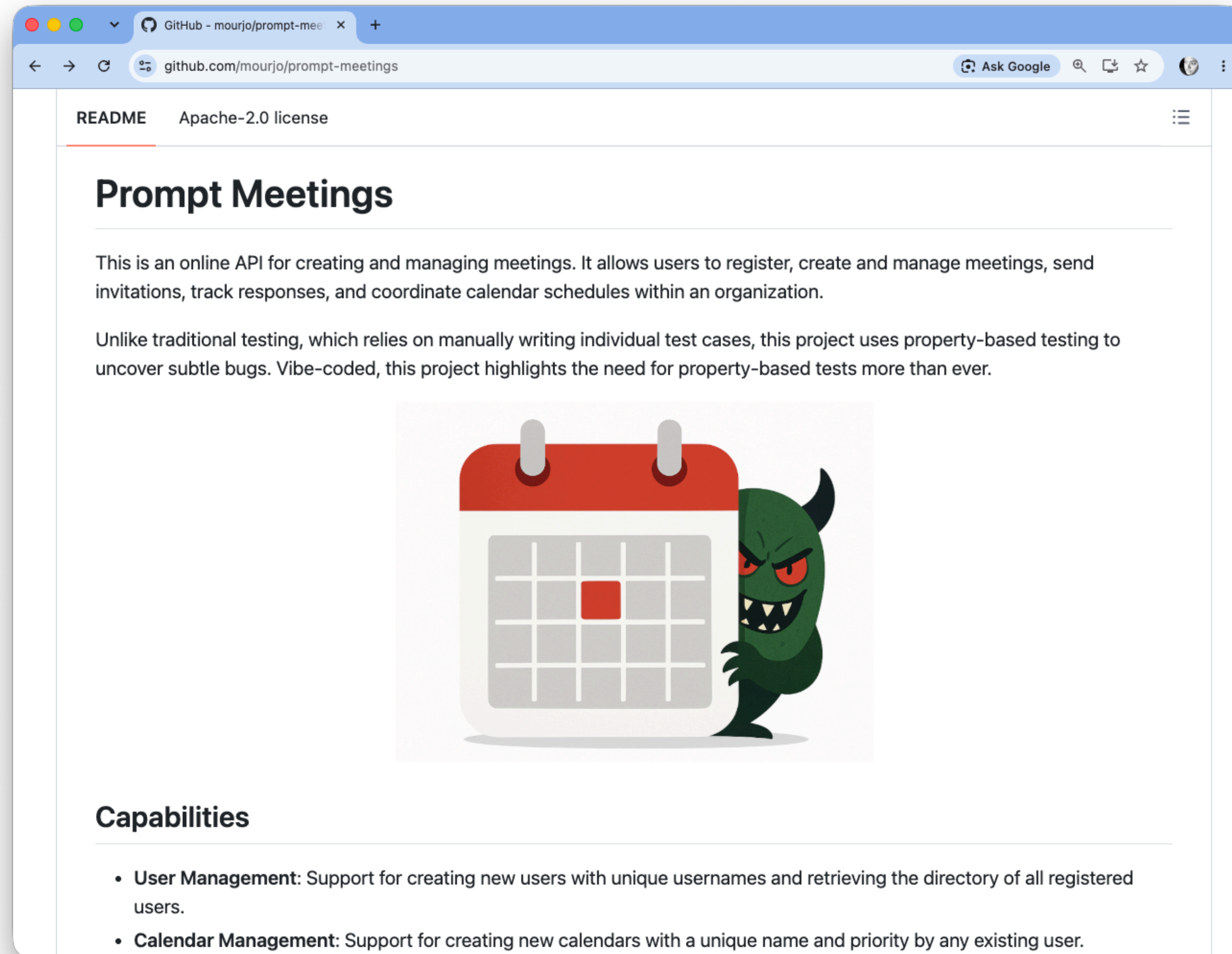
Antigravity CLI 1.1.27
~/.repos/prompt-meetings@gmail.com (Google AI Pro)
Claude Opus 4.6 (Thinking)
~/repos/prompt-meetings

> This application called "prompt-meetings" is designed to be used for creating/managing meetings. Create the following endpoints:

- create a user with a unique username
- view all users in the system
- create a meeting with a start and end time
- invite other users by user name to a meeting
- accept an invite
- reject and invite
- view pending invitations for a user
- view all meetings
- view all pending invitations received by a user

Claude Opus 4.6 (Thinking)

Meeting Scheduling, Vibe Coded



Source: github.com/mourjo/prompt-meetings



Meeting Scheduling, Vibe Coded





Meeting Scheduling, Vibe Coded

“

This application called "prompt-meetings" is designed to be used for creating/managing meetings.

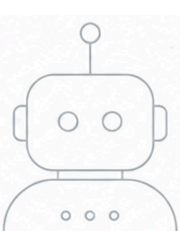
create the following endpoints:

- create a user with a unique username*
- view all users in the system*
- create a meeting with a start and end time*

...

”





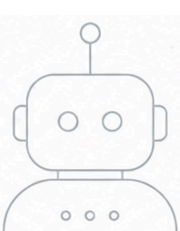
Meeting Scheduling, Vibe Coded

“

*I want to introduce now a **concept of calendars**. A calendar contains meetings. Every meeting must be part of a calendar. Calendars have priorities denoted by a floating point number. Migrate the existing meetings to a default calendar. Update the database schema. Update the create-meeting endpoints...*

”





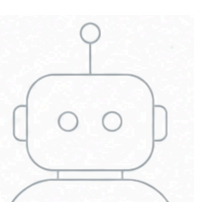
Meeting Scheduling, Vibe Coded

“

I want to introduce now a concept of calendars. A calendar contains meetings. Every meeting must be part of a calendar. Calendars have priorities denoted by a floating point number. Migrate the existing meetings to a default calendar. Update the database schema. Update the create-meeting endpoints...

”





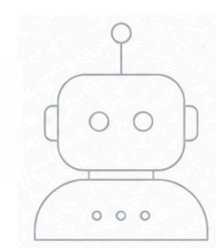
Meeting Scheduling, Vibe Coded

“

Now I want to use the priority of calendars – when a meeting is being created, check if any of my meetings are conflicting with this. If the conflicting meeting is originating from a calendar of a lower priority, reject that meeting. If the conflict is coming from a calendar of the same priority or higher, do not allow its creation...

”





Meeting Scheduling, Vibe Coded

```
$ mvn clean test
```

```
[INFO] -----
```

```
[INFO]  T E S T S
```

```
[INFO] -----
```

```
...
```

```
[INFO] Results:
```

```
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] -----
```

```
[INFO] BUILD SUCCESS
```

```
[INFO] -----
```

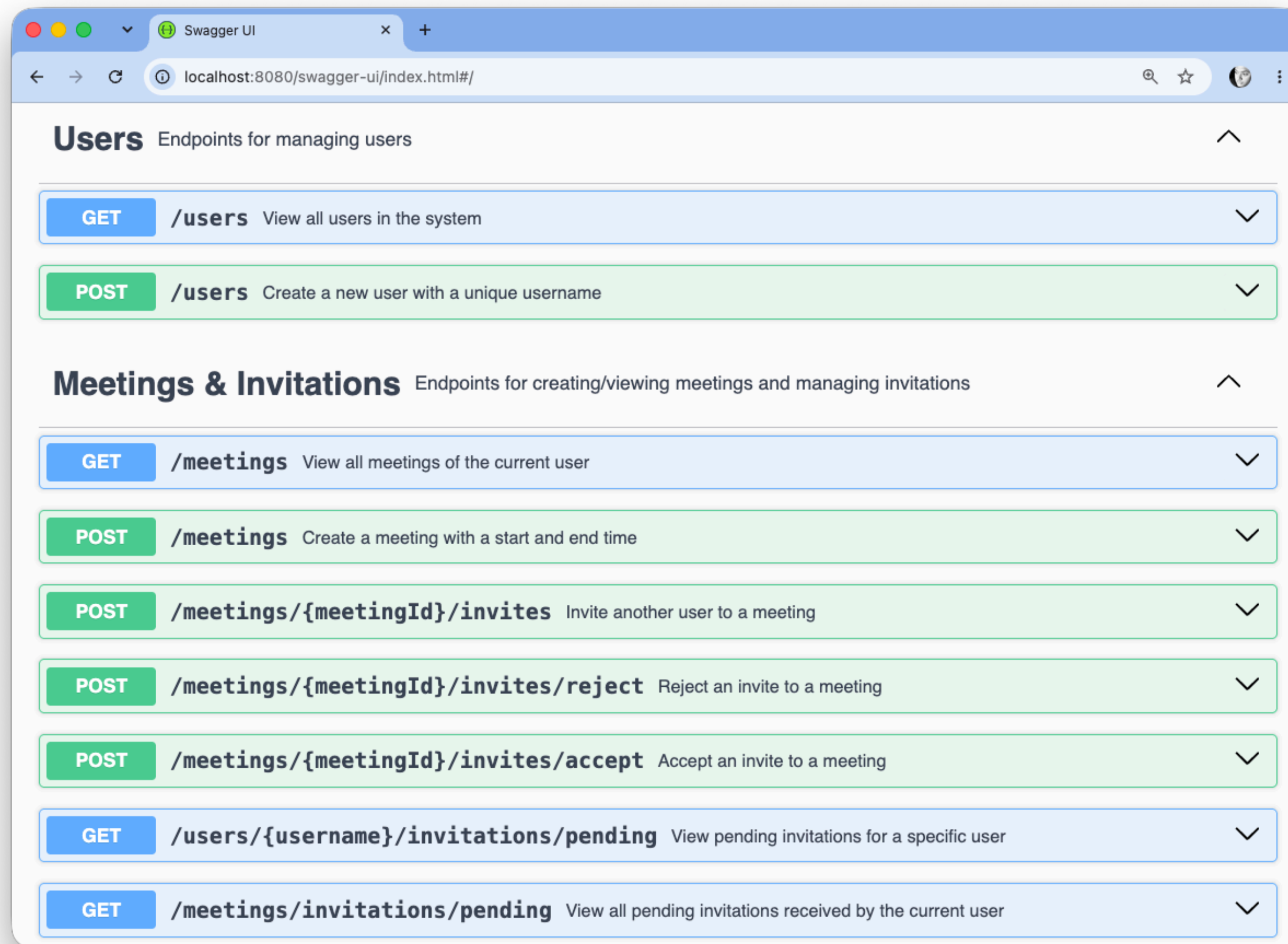
```
[INFO] Total time:  6.349 s
```

```
[INFO] Finished at: 2026-08-16T17:54:28+02:00
```

```
[INFO] -----
```



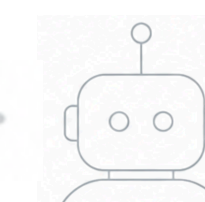
Meeting Scheduling, Vibe Coded



Source: github.com/mourjo/prompt-meetings

Meeting Scheduling, Vibe Coded





Meeting Scheduling, Vibe Coded

“

I want to add property based tests using jqwik. Use version 1.9.3. Use the spring boot extension.

- There should be 4 users in the system (this should be a constant, do this in the setup)*
- There should be 3 calendars of priorities default, medium, high, this should be a constant, do this in the setup...*

”





Meeting Scheduling, Vibe Coded

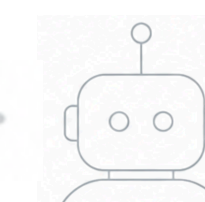
“

...the property based test should perform sequences of actions on behalf of users

- create-meeting*
- invite someone else to a meeting*
- accept an invitation*
- reject an invitation*

”





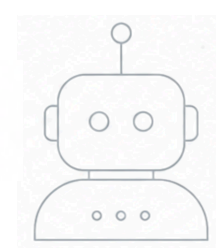
Meeting Scheduling, Vibe Coded

“

...the invariant should always hold true: no user can ever part of two meetings at the same time – no sequence of actions should ever allow this.

”

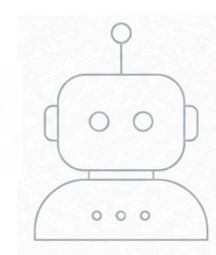




Meeting Scheduling, Vibe Coded

```
# ~/repos/prompt-meetings → mvn clean test
[INFO] Scanning for projects...
...
[INFO]
[INFO] -----
[INFO]  T E S T S
[INFO] -----
[INFO]
[INFO] Results:
[INFO]
[ERROR] Failures:
[ERROR]   MeetingSchedulerPropertyTests.noOverlappingMeetingsForAnyUser:90
>lambda$noOverlappingMeetingsForAnyUser$0:80 Invariant 'no-overlap' failed after the following actions: [
    alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
    alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
    alice accepts invitation to meeting Meeting-0
]
final state: me.mourjo.prompt.meetings.MeetingSchedulerPropertyTests$CalendarMeetingsState@7fdfe501
User charlie is in overlapping meetings: Meeting-0 [10:00 - 10:01] and Meeting-1 [10:00 - 10:01]
[INFO]
[ERROR] Tests run: 3, Failures: 1, Errors: 0, Skipped: 0
[INFO]
[INFO] -----
[INFO] BUILD FAILURE
```

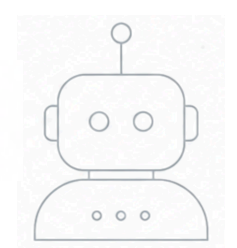




Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [
 alice creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
 alice creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
 alice accepts invitation to meeting Meeting-0
]





Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [

```
charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49
alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04
charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28
dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37
alice rejects invitation to Meeting-2
dan accepts invitation to meeting Meeting-2
dan invites alice to Meeting-2
alice invites charlie to Meeting-1
alice rejects invitation to Meeting-0
charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
dan rejects invitation to Meeting-3
charlie accepts invitation to meeting Meeting-6
dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14
alice invites dan to Meeting-4
charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01
dan invites bob to Meeting-9
charlie invites alice to Meeting-4
bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10
dan accepts invitation to meeting Meeting-4
```

]



Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [

charlie creates Meeting-0 in calendar high (UTC) from 10:41 to 10:49

alice creates Meeting-1 in calendar medium (UTC) from 13:49 to 13:50

bob creates Meeting-2 in calendar default (UTC) from 10:44 to 11:04

charlie creates Meeting-3 in calendar high (UTC) from 11:29 to 12:28

dan creates Meeting-4 in calendar medium (UTC) from 10:05 to 10:25

bob creates Meeting-5 in calendar default (UTC) from 10:19 to 10:37

alice rejects invitation to Meeting-2

dan accepts invitation to meeting Meeting-2

dan invites alice to Meeting-2

alice invites charlie to Meeting-1

alice rejects invitation to Meeting-0

charlie creates Meeting-6 in calendar medium (UTC) from 15:40 to 16:09

dan rejects invitation to Meeting-3

charlie accepts invitation to meeting Meeting-6

dan creates Meeting-7 in calendar high (UTC) from 10:10 to 10:14

alice invites dan to Meeting-4

charlie creates Meeting-8 in calendar high (UTC) from 13:15 to 14:01

dan invites bob to Meeting-9

charlie invites alice to Meeting-4

bob creates Meeting-9 in calendar default (UTC) from 10:07 to 10:10

dan accepts invitation to meeting Meeting-4

]

Meeting Scheduling, Vibe Coded

Invariant 'no-overlap' failed after the following actions: [

charlie **creates** Meeting-0 in calendar high (UTC) from 10:41 to 10:49
alice **creates** Meeting-1 in calendar medium (UTC) from 13:49 to 13:50
bob **creates** Meeting-2 in calendar default (UTC) from 10:44 to 11:04
charlie **creates** Meeting-3 in calendar high (UTC) from 11:29 to 12:28
dan **creates** Meeting-4 in calendar medium (UTC) from 10:05 to 10:25
bob **creates** Meeting-5 in calendar default (UTC) from 10:19 to 10:37
alice **rejects** invitation to Meeting-2
dan **accepts** invitation to meeting Meeting-2
dan **invites** alice to Meeting-2
alice **invites** charlie to Meeting-1
alice **rejects** invitation to Meeting-0
charlie **creates** Meeting-6 in calendar medium (UTC) from 15:40 to 16:09
dan **rejects** invitation to Meeting-3
charlie **accepts** invitation to meeting Meeting-6
dan **creates** Meeting-7 in calendar high (UTC) from 10:10 to 10:14
alice **invites** dan to Meeting-4
charlie **creates** Meeting-8 in calendar high (UTC) from 13:15 to 14:01
dan **invites** bob to Meeting-9
charlie **invites** alice to Meeting-4
bob **creates** Meeting-9 in calendar default (UTC) from 10:07 to 10:10
dan **accepts** invitation to meeting Meeting-4

]



Meeting Scheduling, Vibe Coded



Invariant 'no-overlap' failed after the following actions: [

charlie creates **Meeting-0** in calendar high (UTC) from 10:41 to 10:49

alice creates **Meeting-1** in calendar medium (UTC) from 13:49 to 13:50

bob creates **Meeting-2** in calendar default (UTC) from 10:44 to 11:04

charlie creates **Meeting-3** in calendar high (UTC) from 11:29 to 12:28

dan creates **Meeting-4** in calendar medium (UTC) from 10:05 to 10:25

bob creates **Meeting-5** in calendar default (UTC) from 10:19 to 10:37

alice rejects invitation to **Meeting-2**

dan accepts invitation to meeting **Meeting-2**

dan invites alice to **Meeting-2**

alice invites charlie to **Meeting-1**

alice rejects invitation to **Meeting-0**

charlie creates **Meeting-6** in calendar medium (UTC) from 15:40 to 16:09

dan rejects invitation to **Meeting-3**

charlie accepts invitation to meeting **Meeting-6**

dan creates **Meeting-7** in calendar high (UTC) from 10:10 to 10:14

alice invites dan to **Meeting-4**

charlie creates **Meeting-8** in calendar high (UTC) from 13:15 to 14:01

dan invites bob to **Meeting-9**

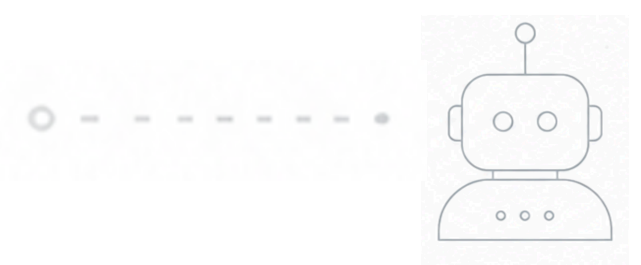
charlie invites alice to **Meeting-4**

bob creates **Meeting-9** in calendar default (UTC) from 10:07 to 10:10

dan accepts invitation to meeting **Meeting-4**

]

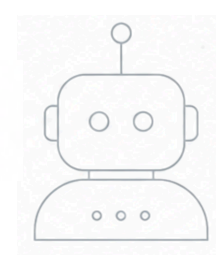




Meeting Scheduling, Vibe Coded

@Property





Meeting Scheduling, Vibe Coded

@Property

```
@ForAll("actions") ActionChain<CalendarState> chain
```



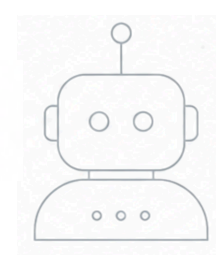
Meeting Scheduling, Vibe Coded

@Property

@ForAll("actions") ActionChain<CalendarState> chain

@Provide

```
Arbitrary<ActionChain<CalendarState>> actions() {  
    return ActionChain.startWith(CalendarMeetingsState::new)  
        .withAction(new CreateMeetingAction())  
        .withAction(new InviteUserAction())  
        .withAction(new AcceptInviteAction())  
        .withAction(new RejectMeetingAction());  
}
```

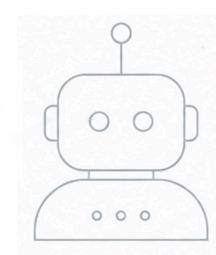



Meeting Scheduling, Vibe Coded

@Property

```
@ForAll("actions") ActionChain<CalendarState> chain
```



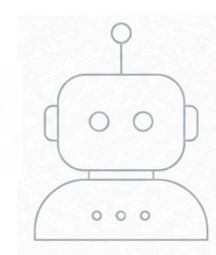


Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
```





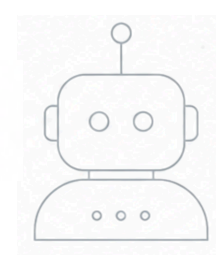
Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {
```

```
    }).run();
```





Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {
```

```
        for (String user : USERS) {
```

```
        }
```

```
    }).run();
```





Meeting Scheduling, Vibe Coded

@Property

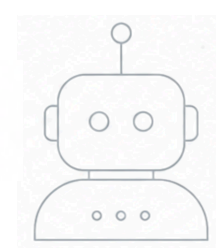
```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {  
  
        for (String user : USERS) {  
            var meetings = meetingService.getMyMeetings(user).stream()  
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))  
                .toList();  
  
        }  
    }).run();  
}
```



Meeting Scheduling, Vibe Coded

@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {  
    chain.withInvariant("no-overlap", sut → {  
  
        for (String user : USERS) {  
            var meetings = meetingService.getMyMeetings(user).stream()  
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))  
                .toList();  
  
            for (int i = 0; i < meetings.size(); i++) {  
                for (int j = i + 1; j < meetings.size(); j++) {  
  
                }  
            }  
        }  
    }).run();  
}
```



Meeting Scheduling, Vibe Coded

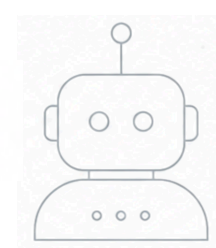
@Property

```
void noOverlappingMeetingsForAnyUser(@ForAll("actions") ActionChain<CalendarState> chain) {
    chain.withInvariant("no-overlap", sut → {

        for (String user : USERS) {
            var meetings = meetingService.getMyMeetings(user).stream()
                .filter(m → "ACCEPTED".equalsIgnoreCase(m.userStatus()))
                .toList();

            for (int i = 0; i < meetings.size(); i++) {
                for (int j = i + 1; j < meetings.size(); j++) {
                    if (overlaps(meetings.get(i), meetings.get(j))) {
                        throw new AssertionError(...);
                    }
                }
            }
        }
    }).run();
}
```

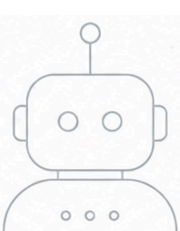




Meeting Scheduling, Vibe Coded

```
[ERROR]    Invariant 'no-overlap' failed after the following actions: [  
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01  
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01  
    charlie accepts invitation to meeting Meeting-0  
]
```





Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0





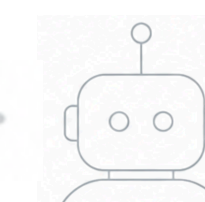
Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0

Charlie creates an overlapping meeting in a higher priority calendar





Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0
]

Charlie creates Meeting-0

Charlie creates an overlapping meeting in a higher priority calendar

Charlie accepts Meeting-0





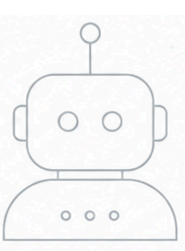
Meeting Scheduling, Vibe Coded

“

When accepting a meeting ensure that the invitation is in pending state – ie, do not allow acceptance of a meeting which is already accepted or rejected.

”

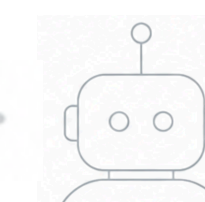




Meeting Scheduling, Vibe Coded

```
[ERROR]    Invariant 'no-overlap' failed after the following actions: [  
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01  
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
    alice invites charlie to Meeting-1  
    charlie accepts invitation to meeting Meeting-1  
]
```





Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

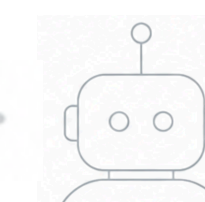


Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Alice creates an overlapping meeting



Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0

Alice creates an overlapping meeting

Alice invites Charlie to their meeting



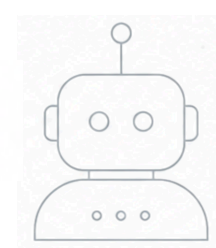


Meeting Scheduling, Vibe Coded

[ERROR] Invariant 'no-overlap' failed after the following actions: [
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

Charlie creates a Meeting-0
Alice creates an overlapping meeting
Alice invites Charlie to their meeting
Charlie accepts the invitation



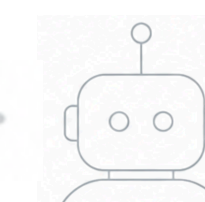


Meeting Scheduling, Vibe Coded

```
[ERROR]    Invariant 'no-overlap' failed after the following actions: [  
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01  
    alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01  
    alice invites charlie to Meeting-1  
    charlie accepts invitation to meeting Meeting-1  
]
```

```
[ERROR]    Invariant 'no-overlap' failed after the following actions: [  
    charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01  
    charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01  
    charlie accepts invitation to meeting Meeting-0  
]
```





Meeting Scheduling, Vibe Coded

[

charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1

] Four actions

Three actions

[

charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0

]



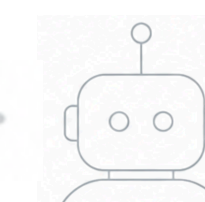
Meeting Scheduling, Vibe Coded

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
  alice invites charlie to Meeting-1
  charlie accepts invitation to meeting Meeting-1
]
```

] Two users

One user

```
[
  charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
  charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
  charlie accepts invitation to meeting Meeting-0
]
```

Meeting Scheduling, Vibe Coded

[
charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
alice creates Meeting-1 in calendar default (UTC) from 10:00 to 10:01
alice invites charlie to Meeting-1
charlie accepts invitation to meeting Meeting-1
]

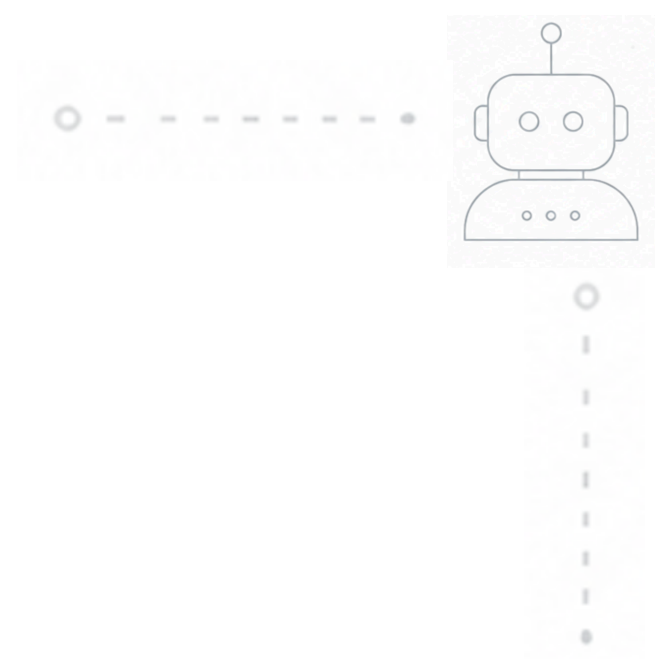
One calendar

Two calendars [

charlie creates Meeting-0 in calendar default (UTC) from 10:00 to 10:01
charlie creates Meeting-1 in calendar medium (UTC) from 10:00 to 10:01
charlie accepts invitation to meeting Meeting-0

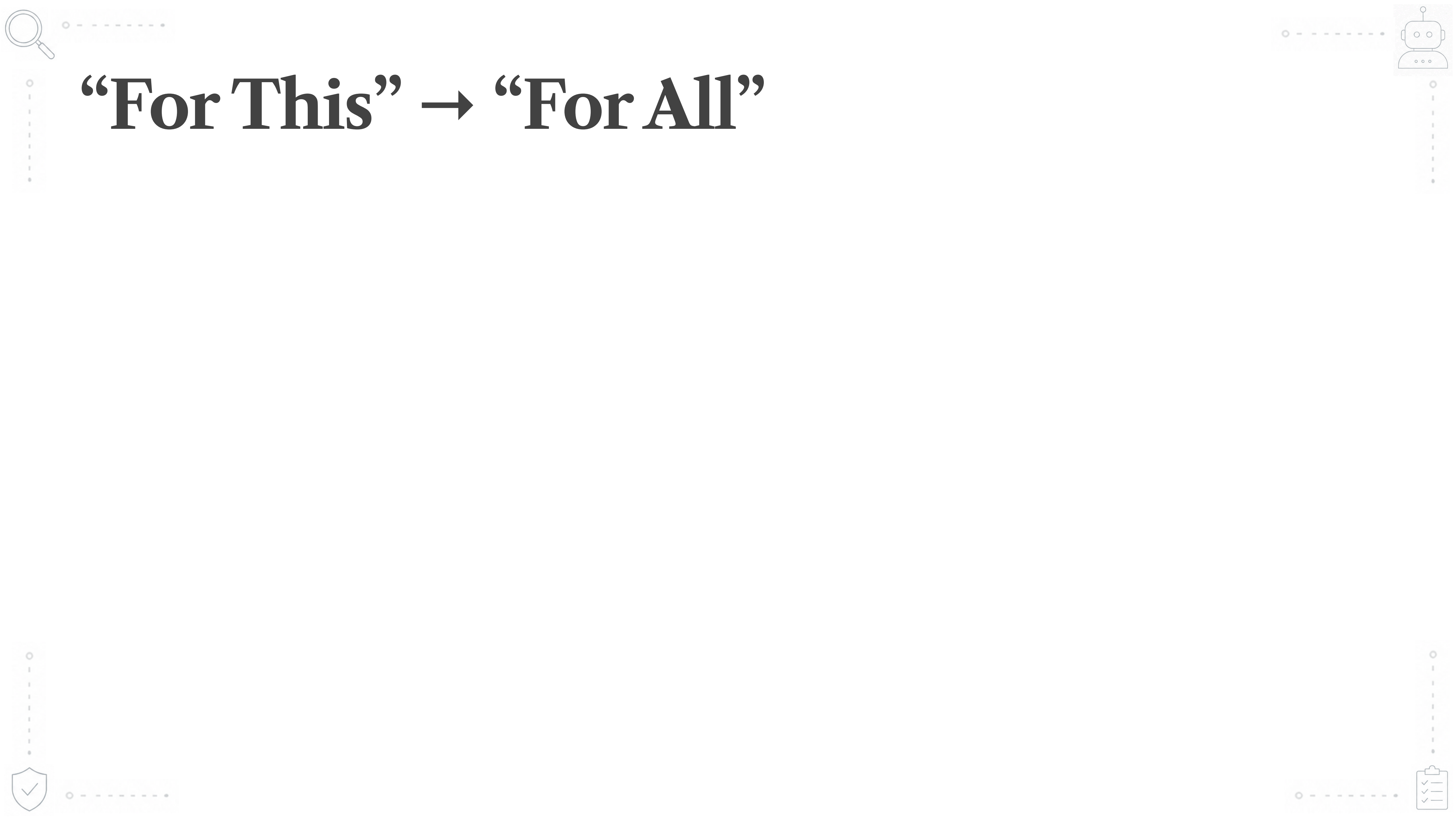
]



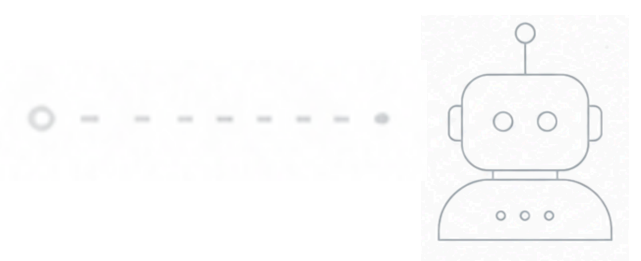


More Than Just Testing

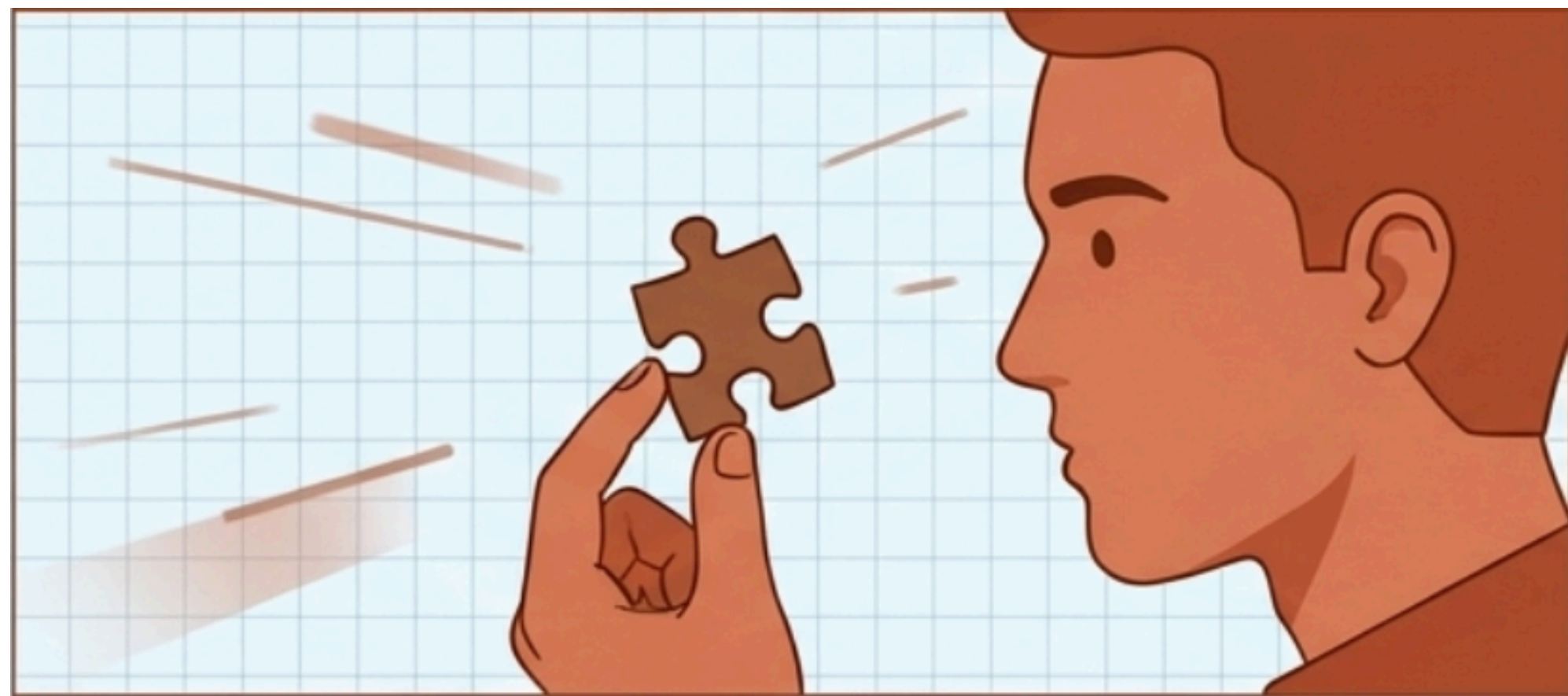




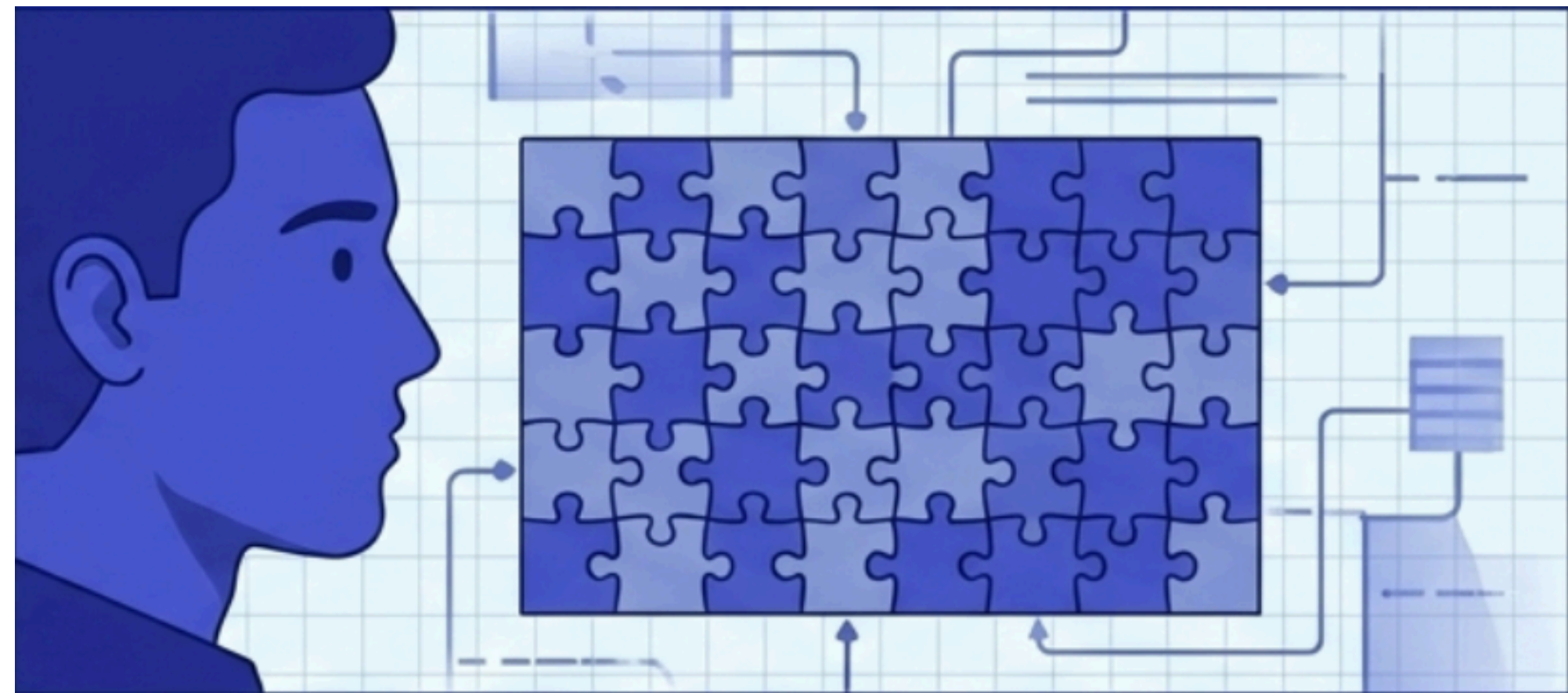
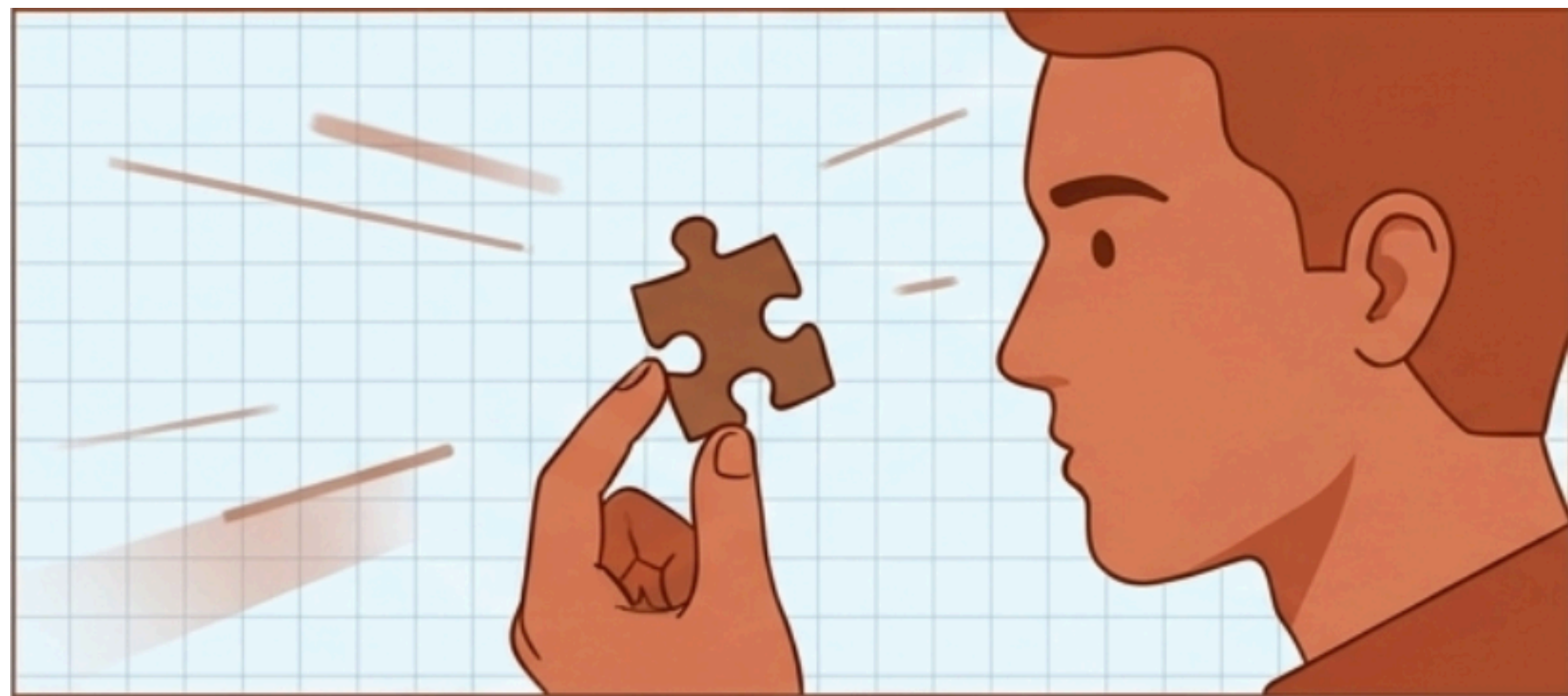
“For This” → “For All”



“For This” → “For All”

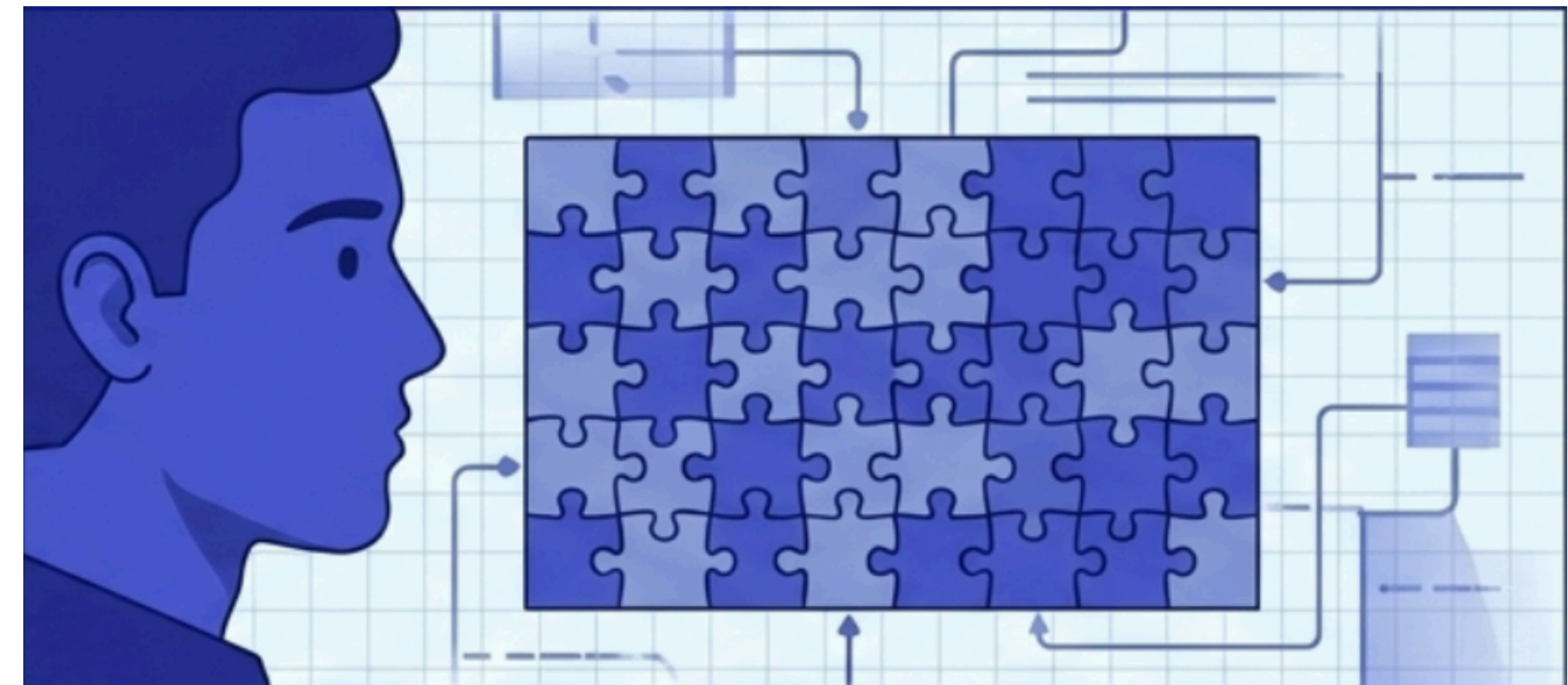
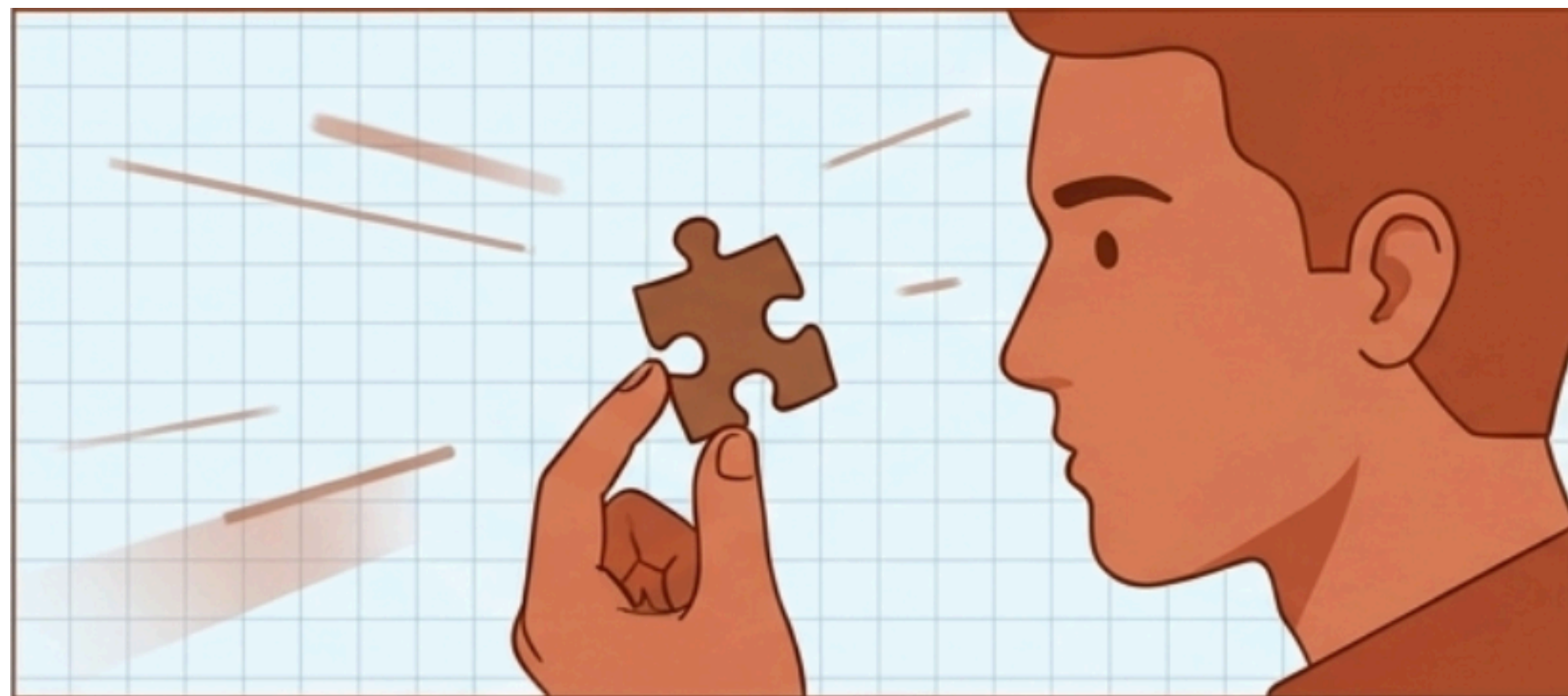


“For This” → “For All”



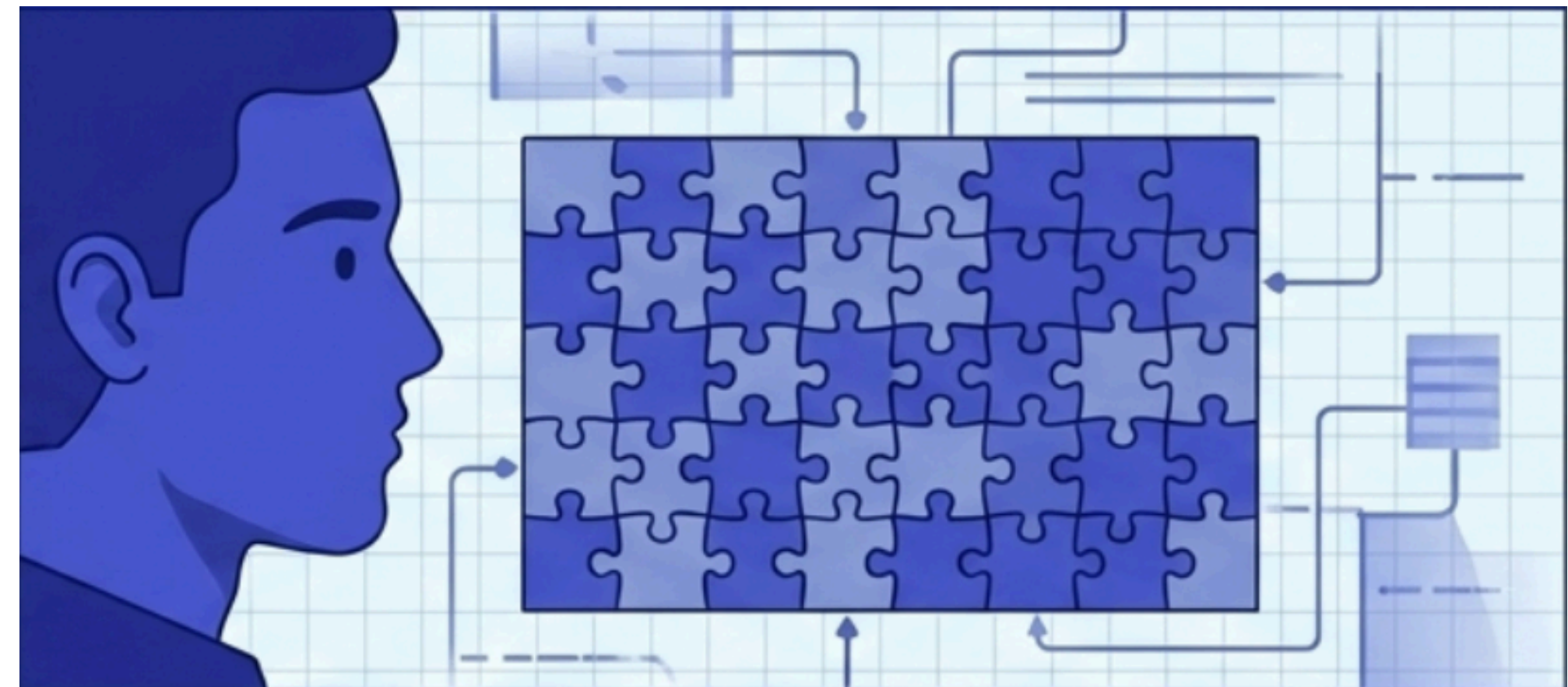
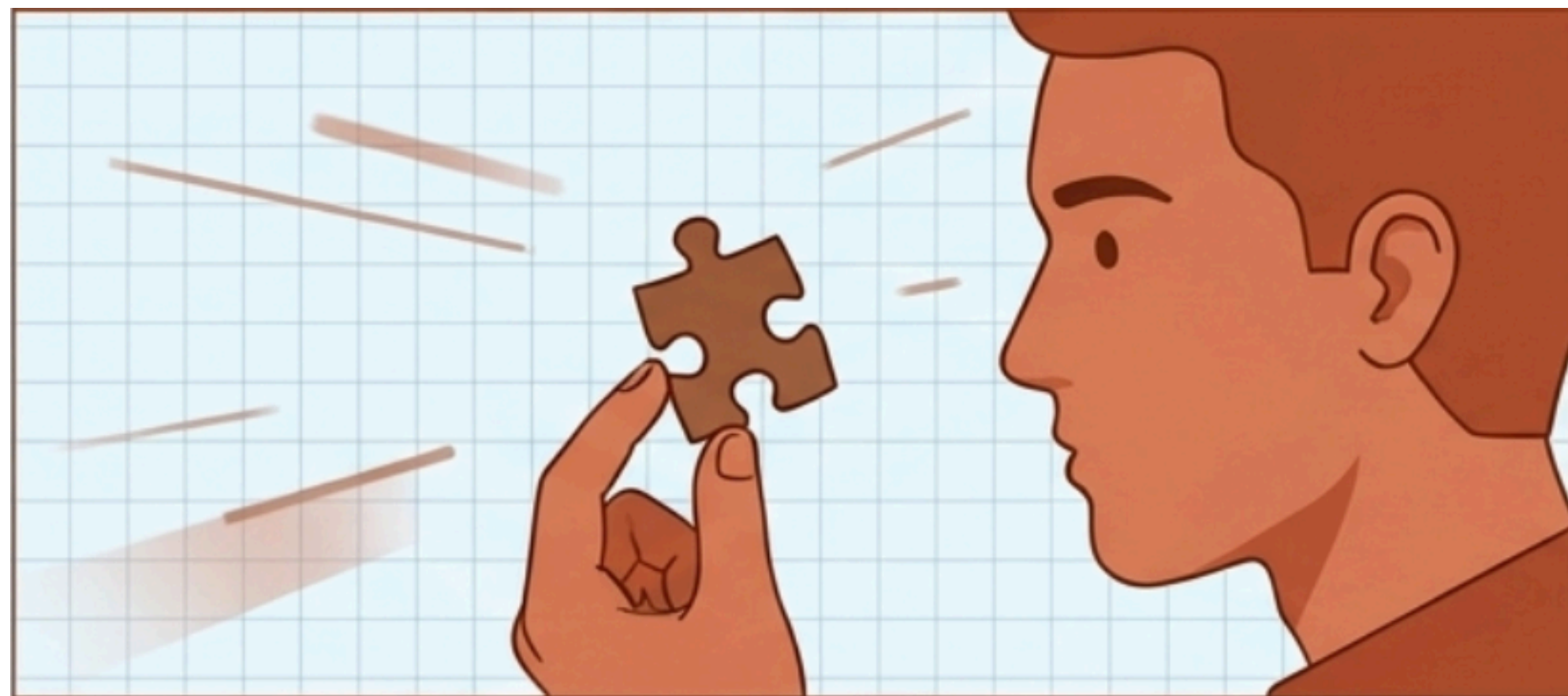
“For This” → “For All”

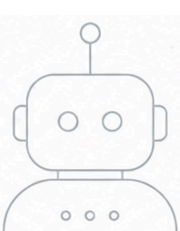
- Defining what must always hold true



“For This” → “For All”

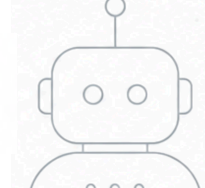
- Defining what must always hold true
- Understanding counter examples





“For This” $\rightarrow$ “For All”

- Defining what must always hold true
 - Understanding counter examples
- 
- 



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

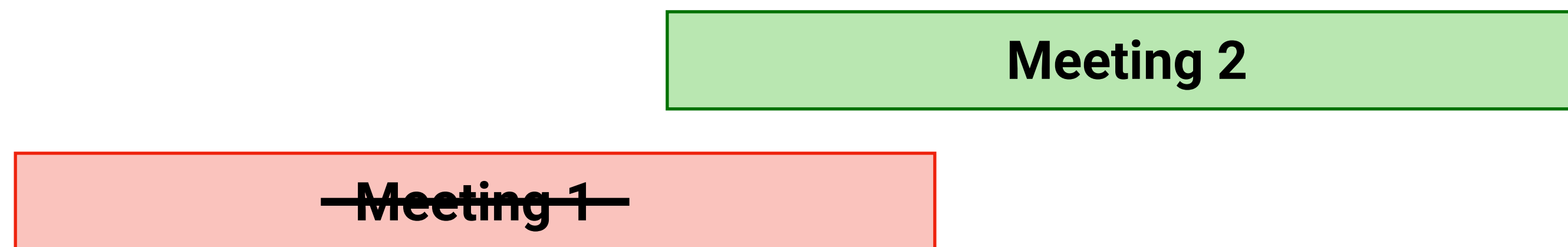
“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”

Meeting 1

“For This” → “For All”

- Defining what must always hold true
- Understanding counter examples

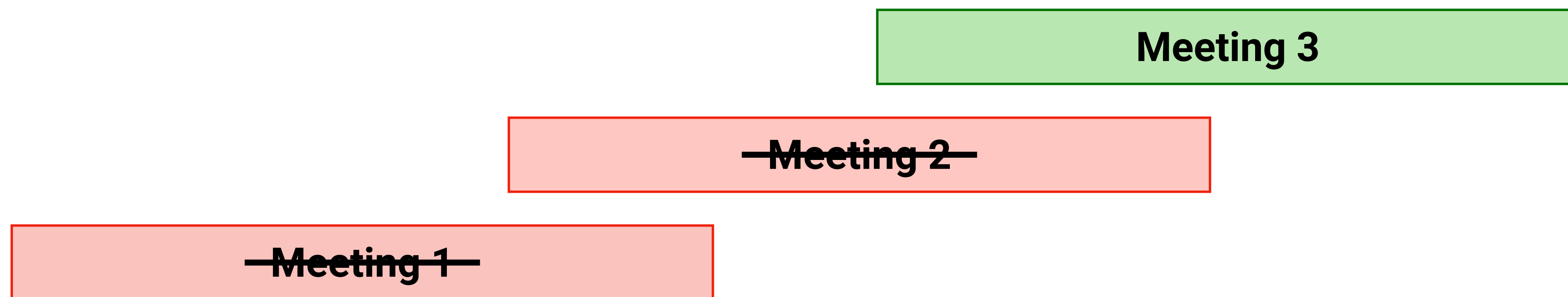
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“For This” → “For All”

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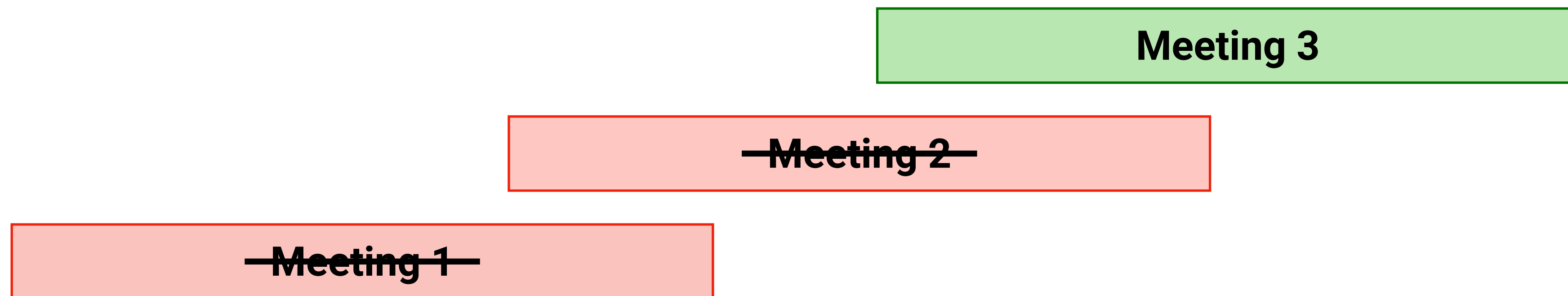
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“Add a new property based test which ensures this invariant: **for every auto rejected meeting, there should be at least one higher priority meeting that overlaps with this meeting and is accepted** by the current user.”



No overlapping accepted meeting



Counter Examples are Transformative



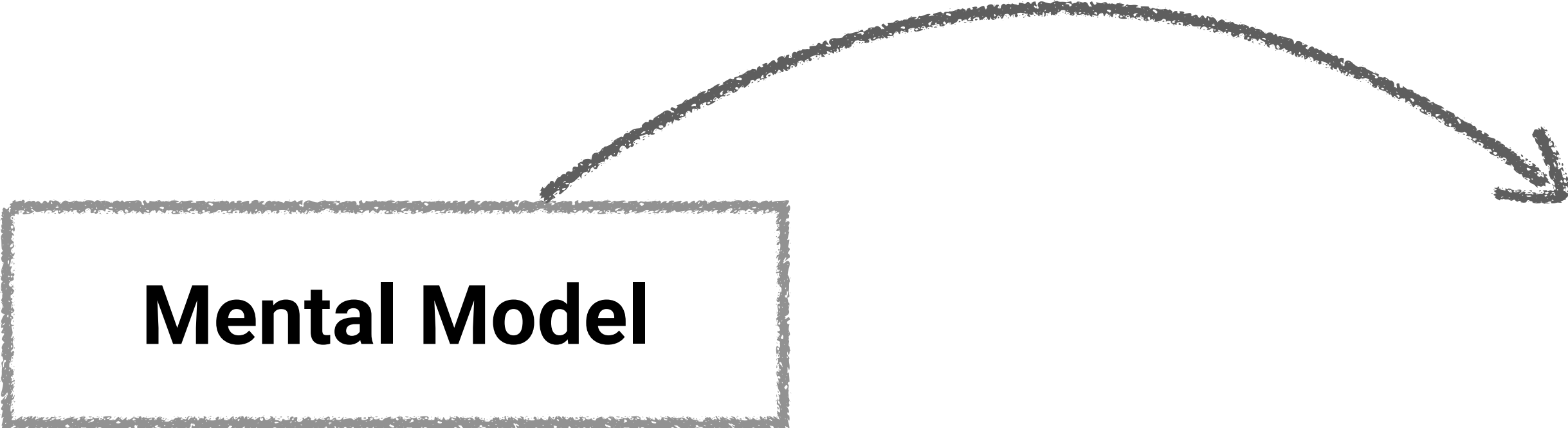
Counter Examples are Transformative

Mental Model

Counter Examples are Transformative

What must ***always*** hold true

Mental Model



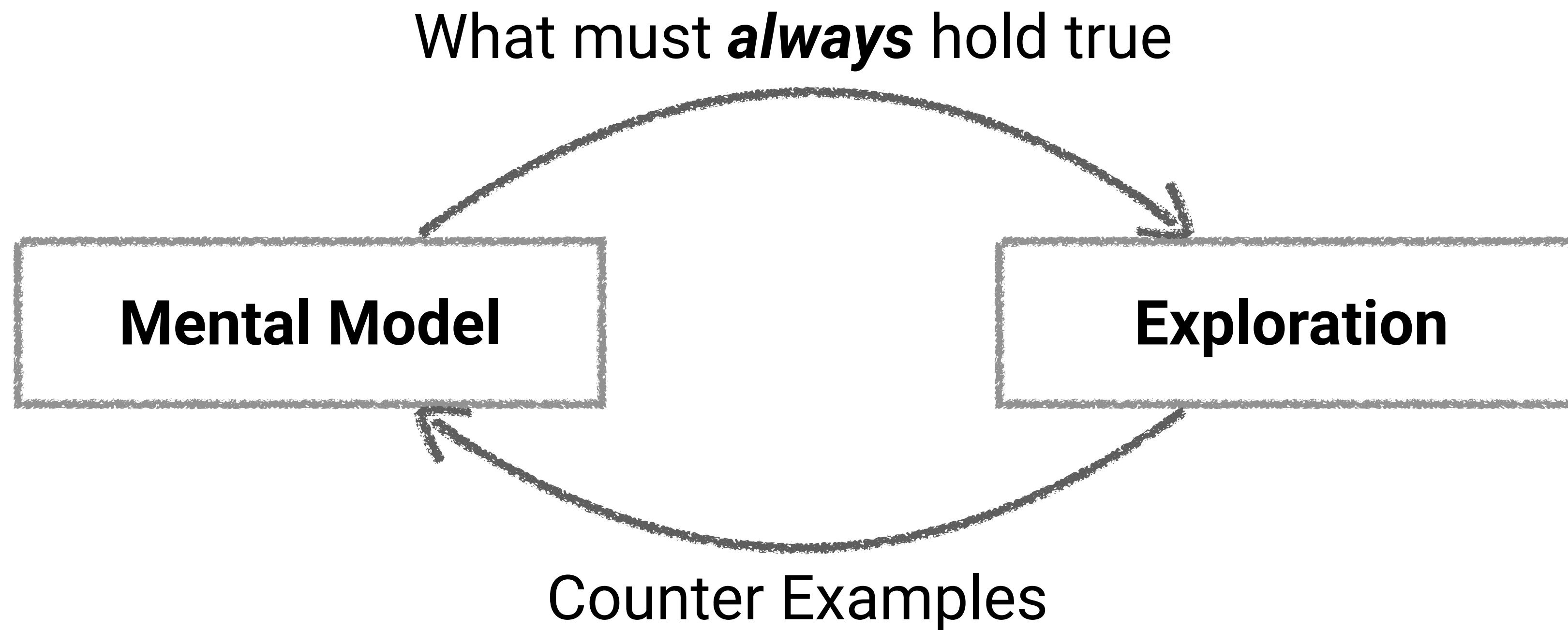
Counter Examples are Transformative

What must ***always*** hold true

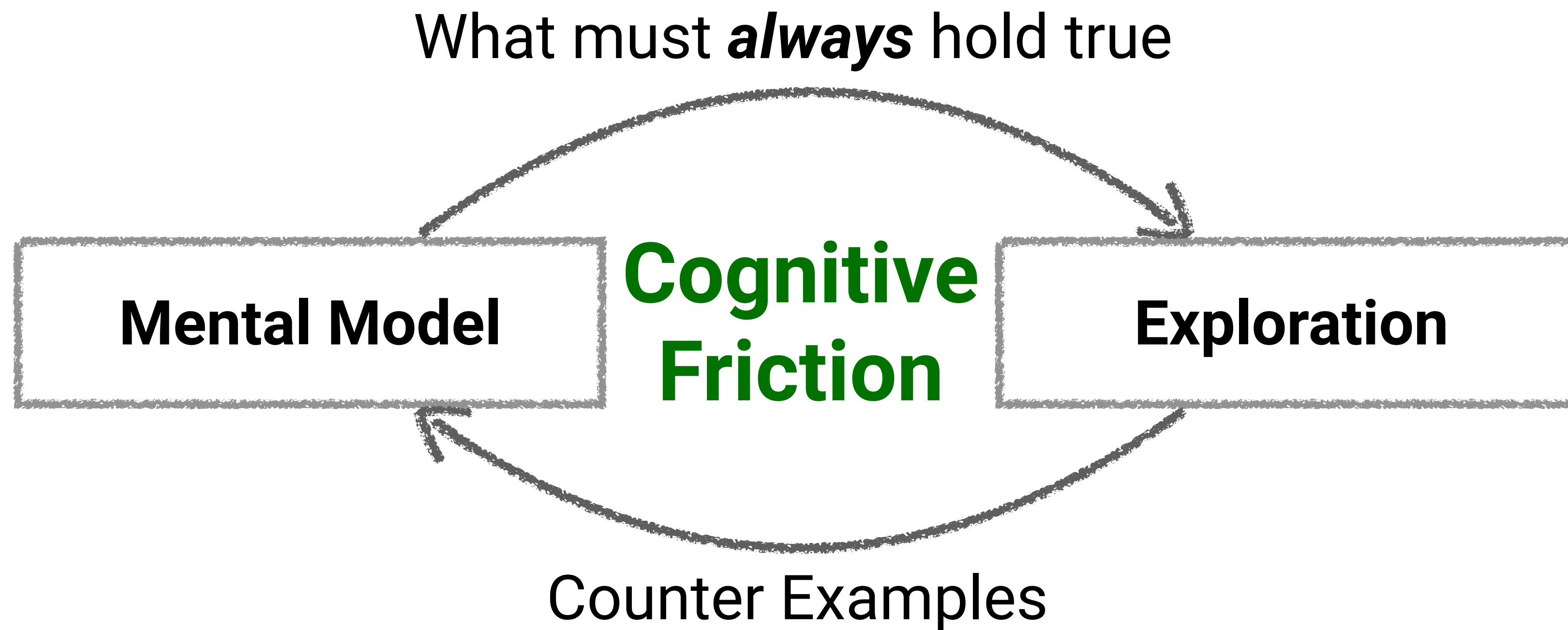
Mental Model

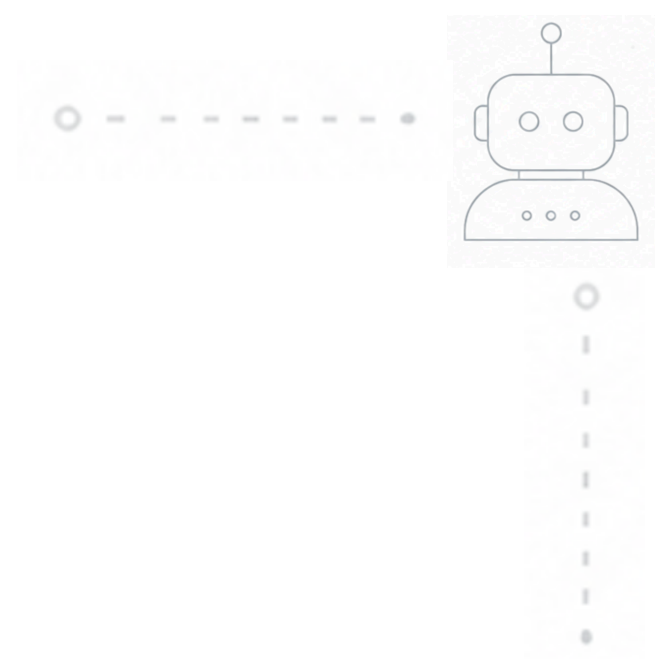
Exploration

Counter Examples are Transformative



Counter Examples are Transformative





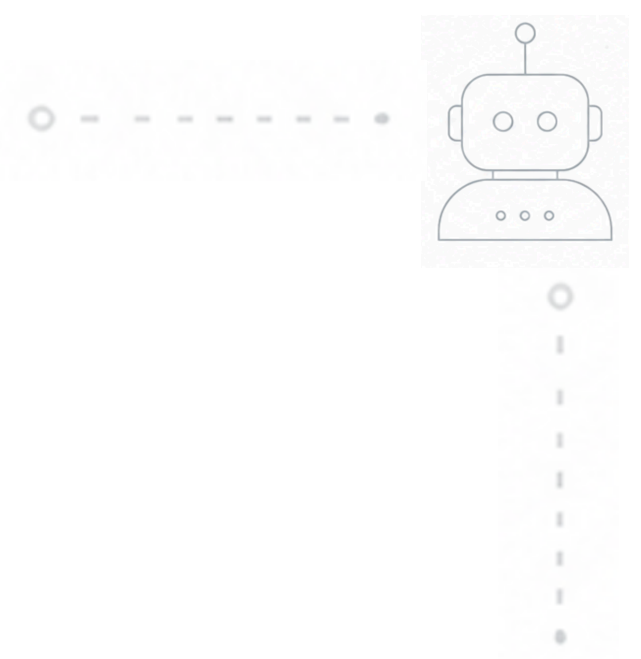
Everything's Changed





Passive Overseeing is Not Enough





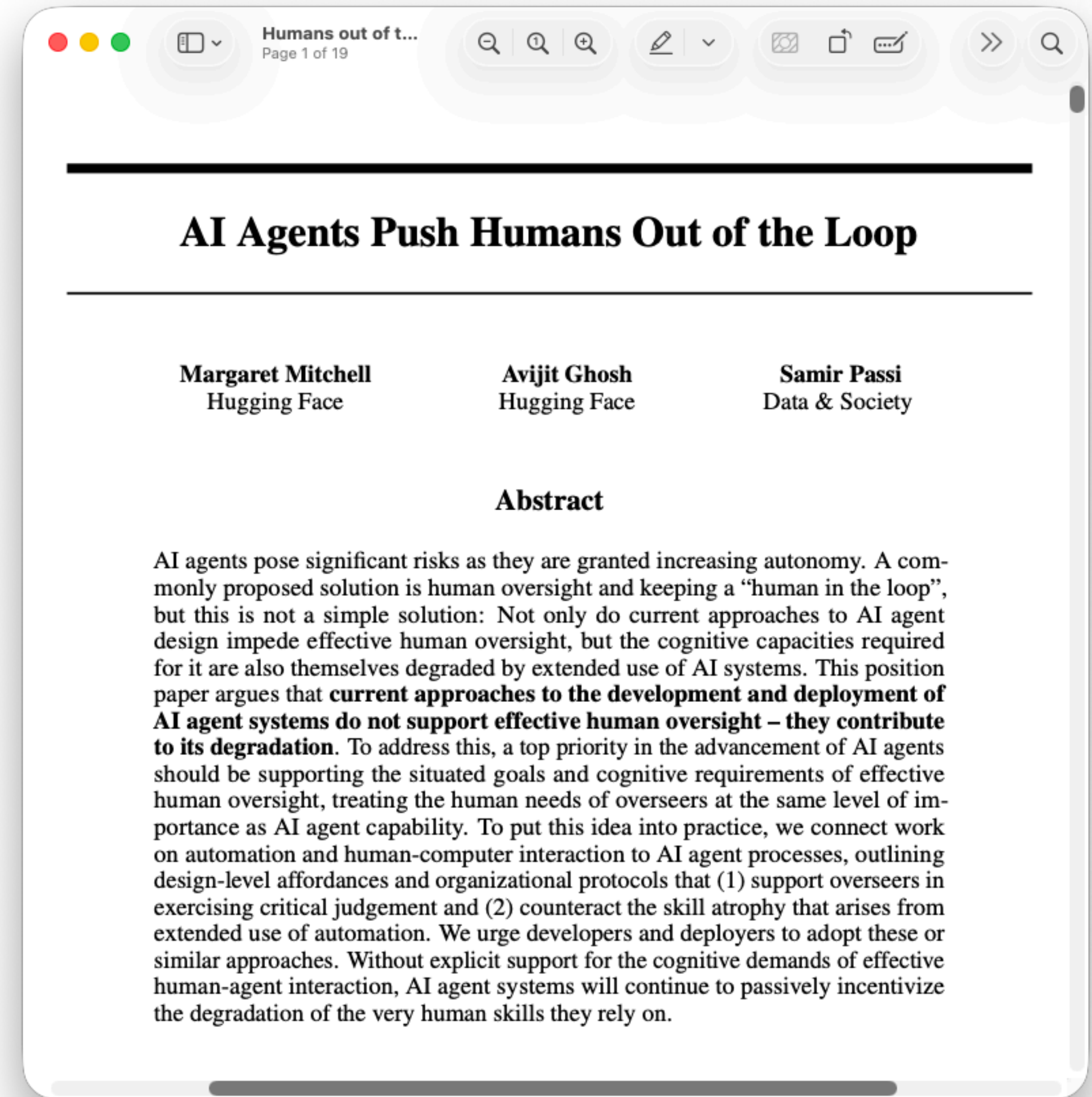
Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work



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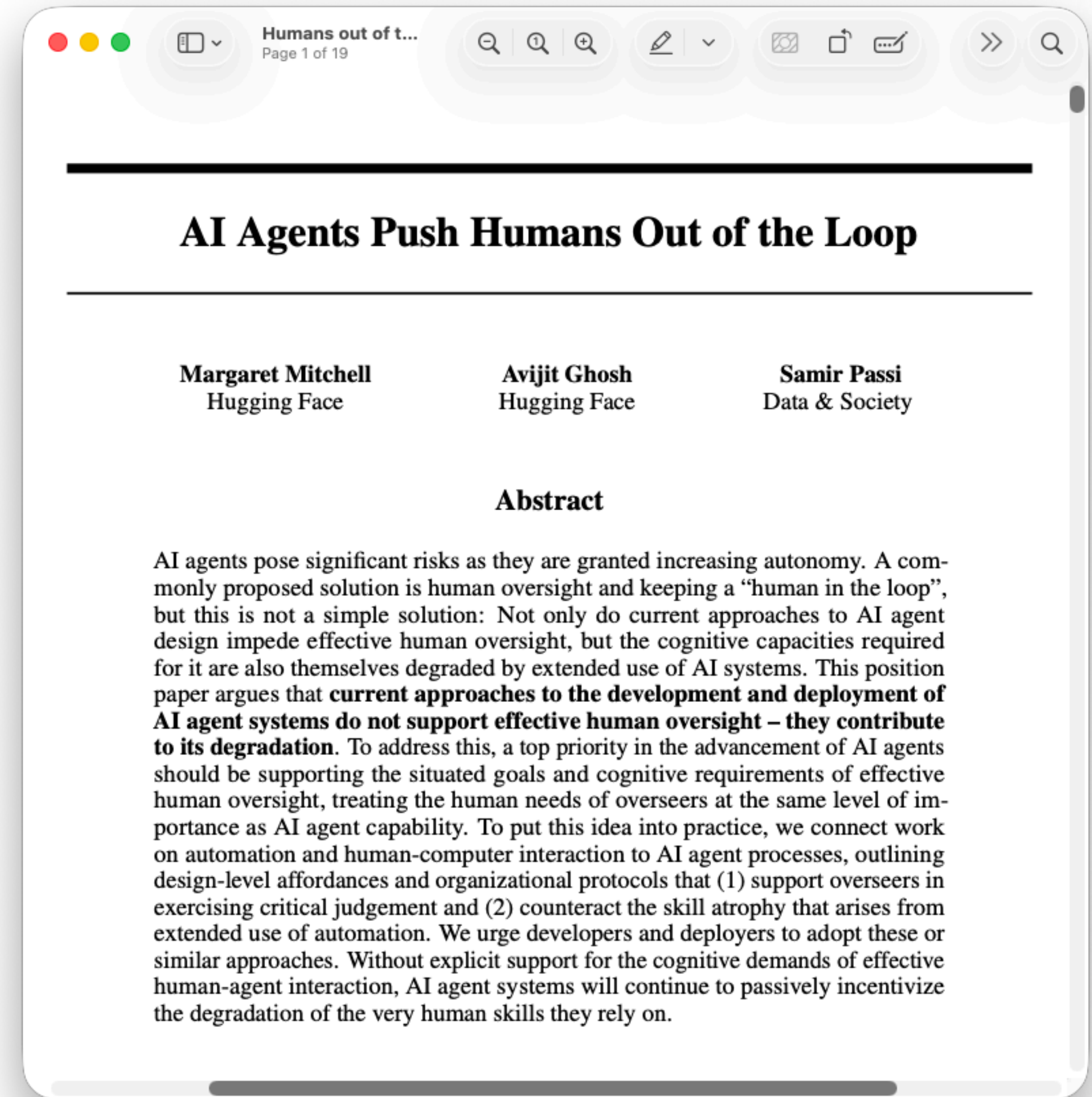


Source: <https://arxiv.org/abs/2608.23642>

Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work
- Strategic cognitive friction
 - Embed human comprehension into software engineering

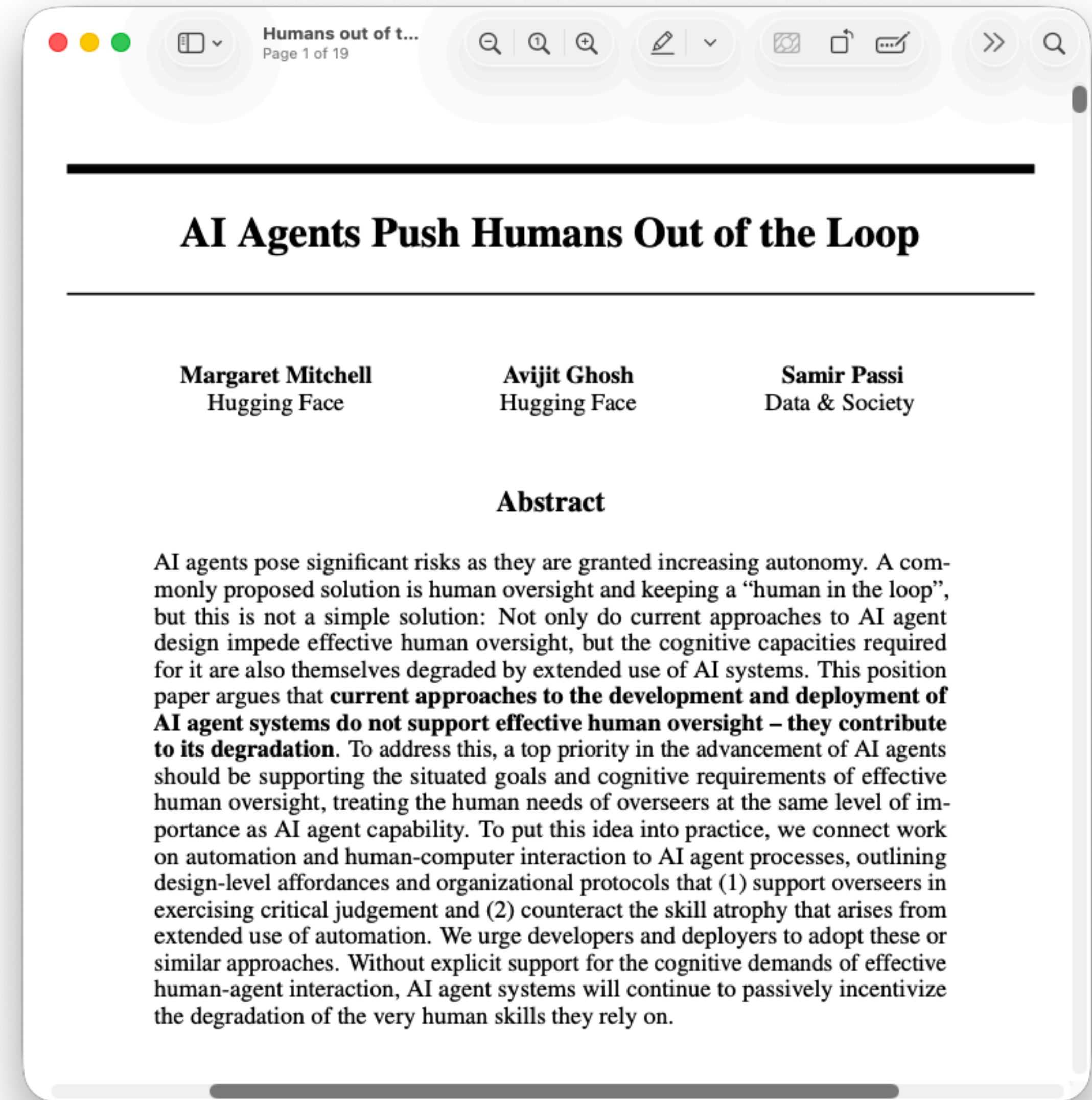
Source: <https://arxiv.org/abs/2608.23642>



Passive Overseeing is Not Enough

- LLMs code faster than we can read
 - Quality is not guaranteed
 - Just being “in the loop” doesn’t work
- Strategic cognitive friction
 - Embed human comprehension into software engineering
- Exploratory testing brings cognitive friction back

Source: <https://arxiv.org/abs/2608.23642>

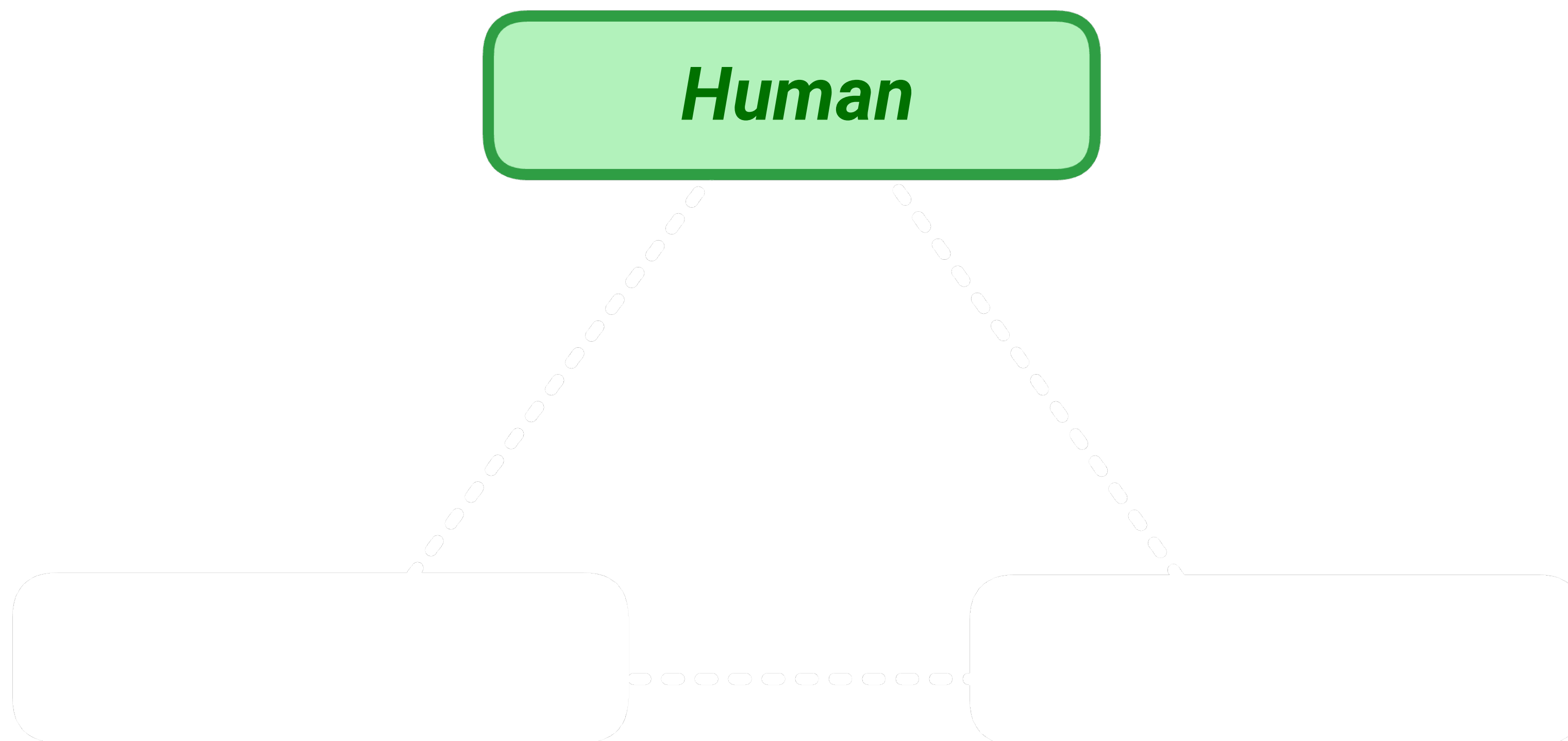




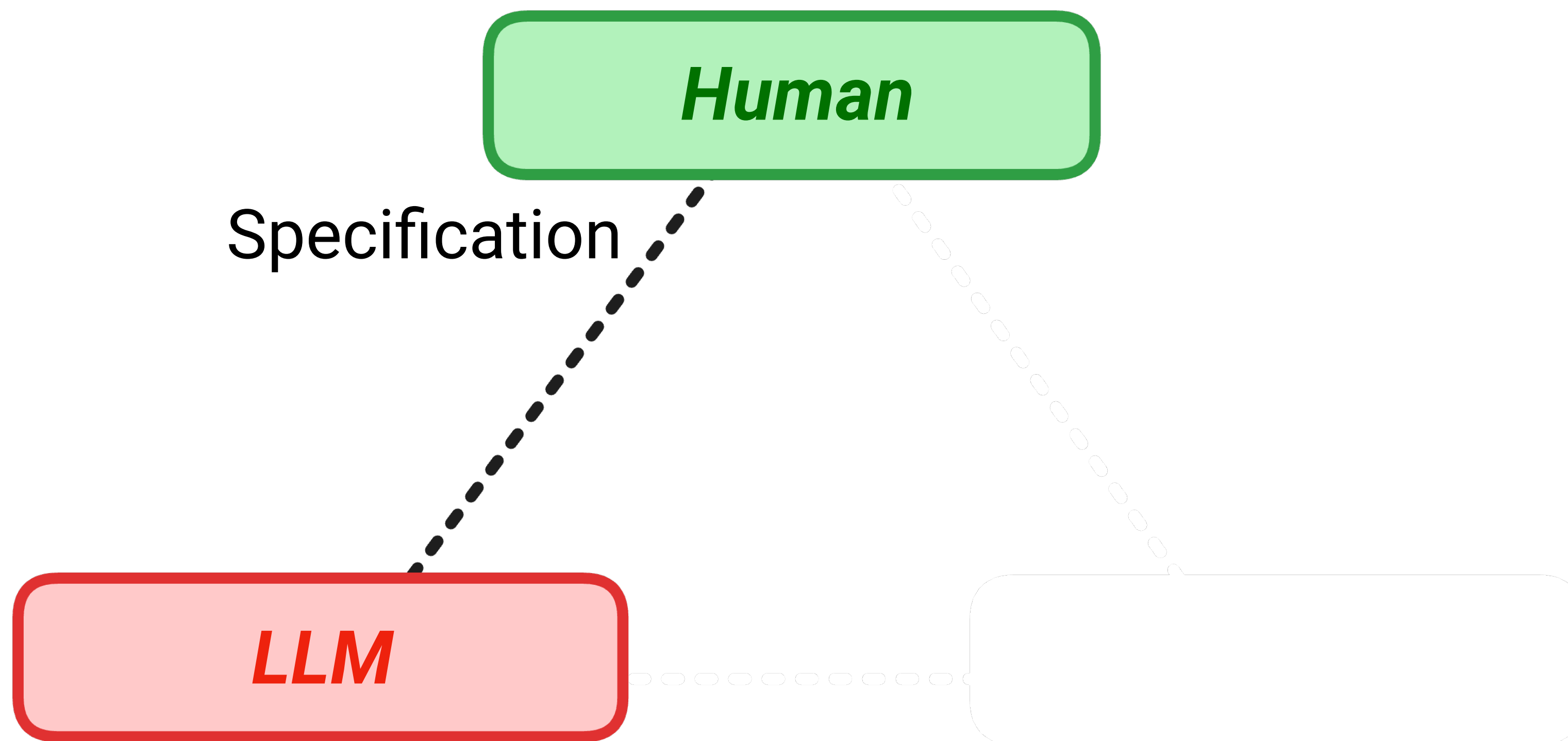
Division of Cognitive Labour



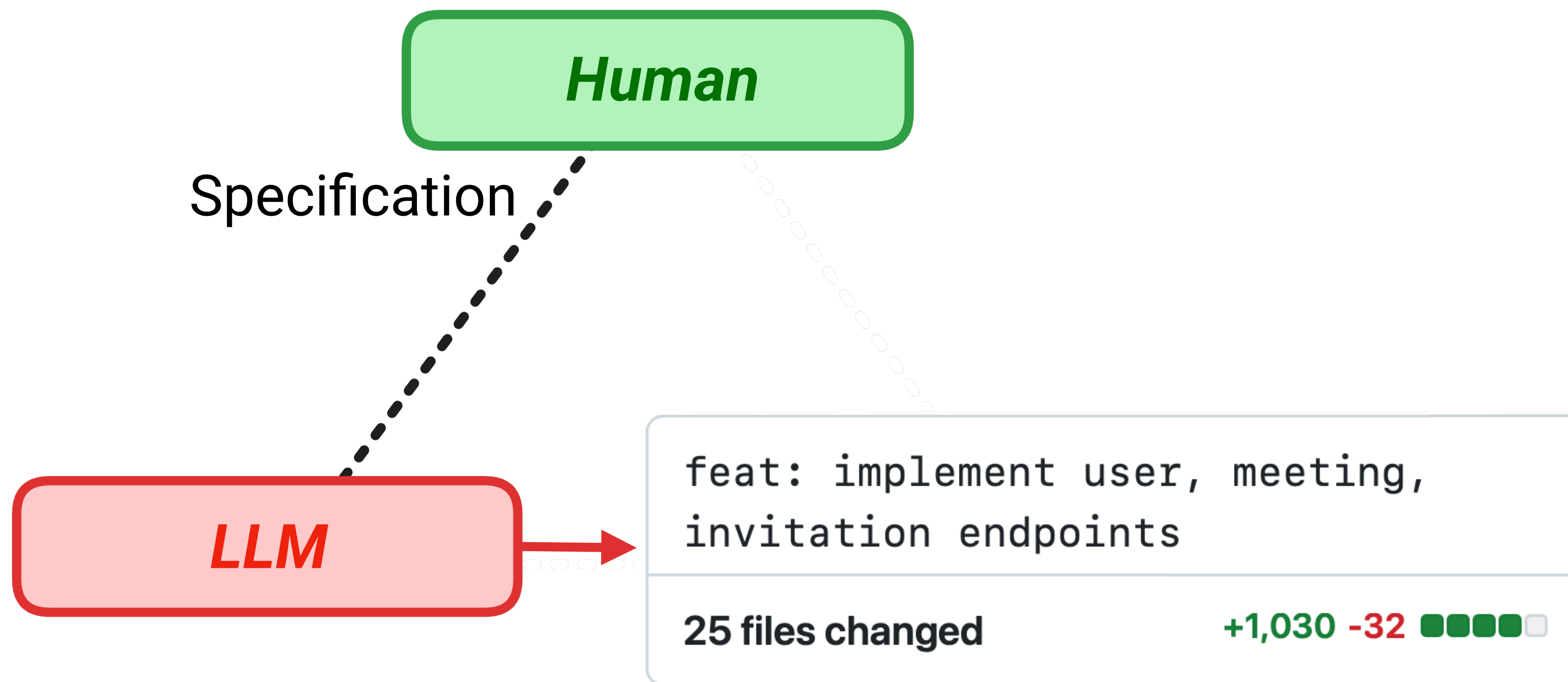
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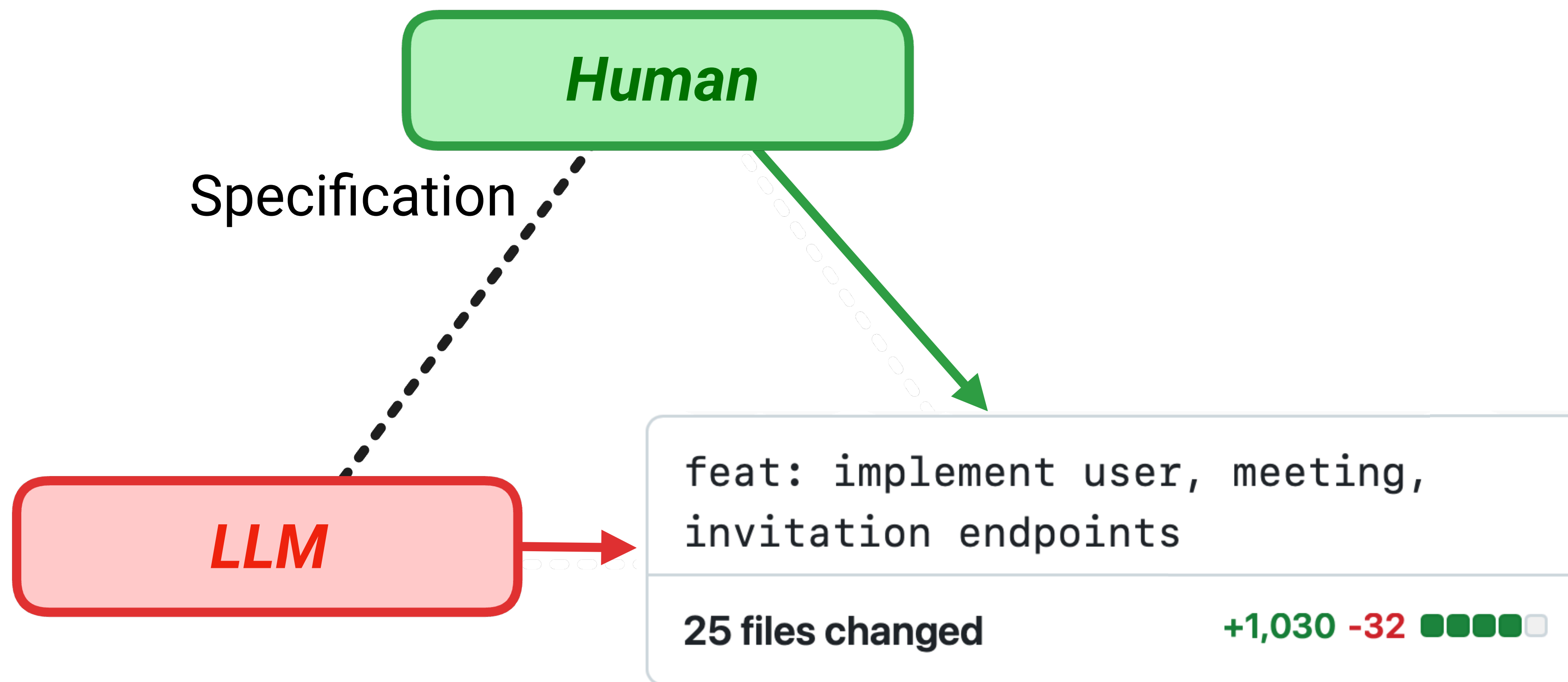
Division of Cognitive Labour



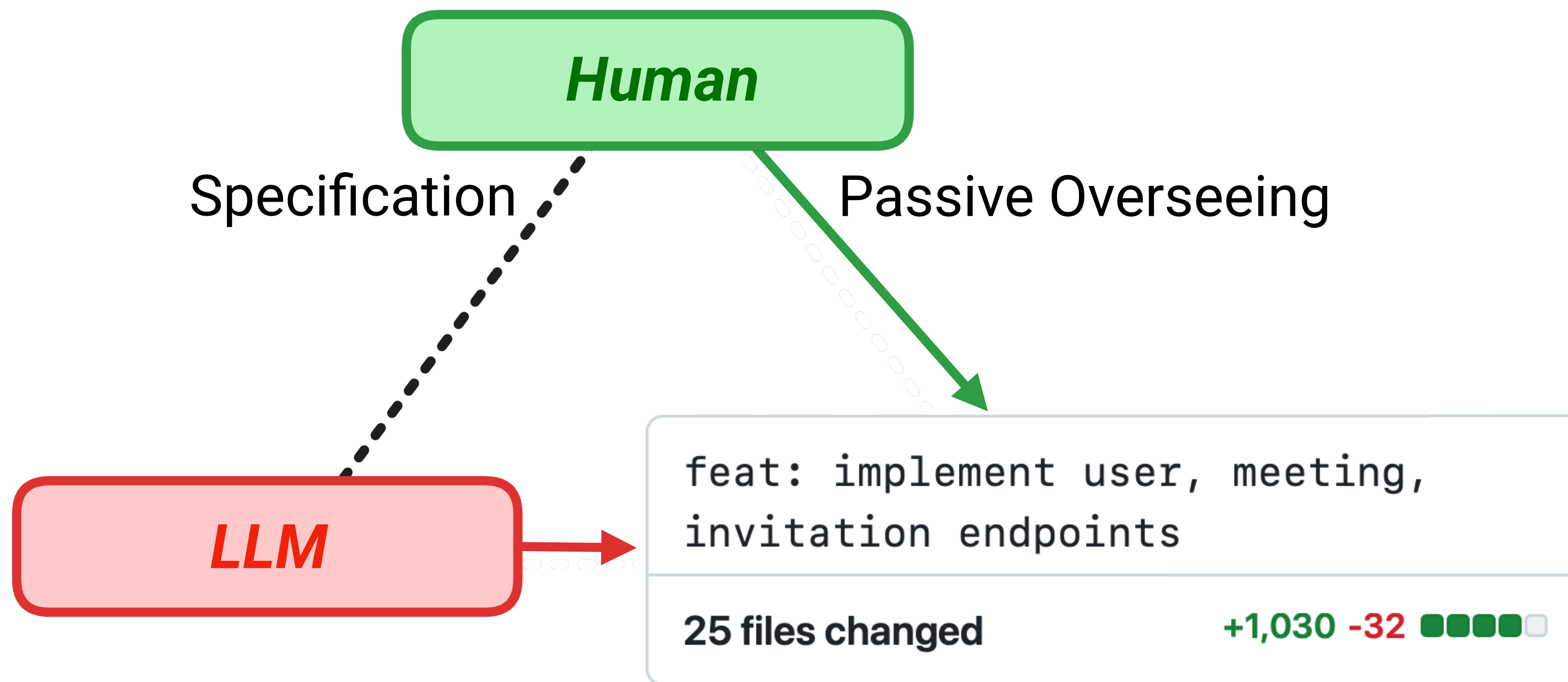
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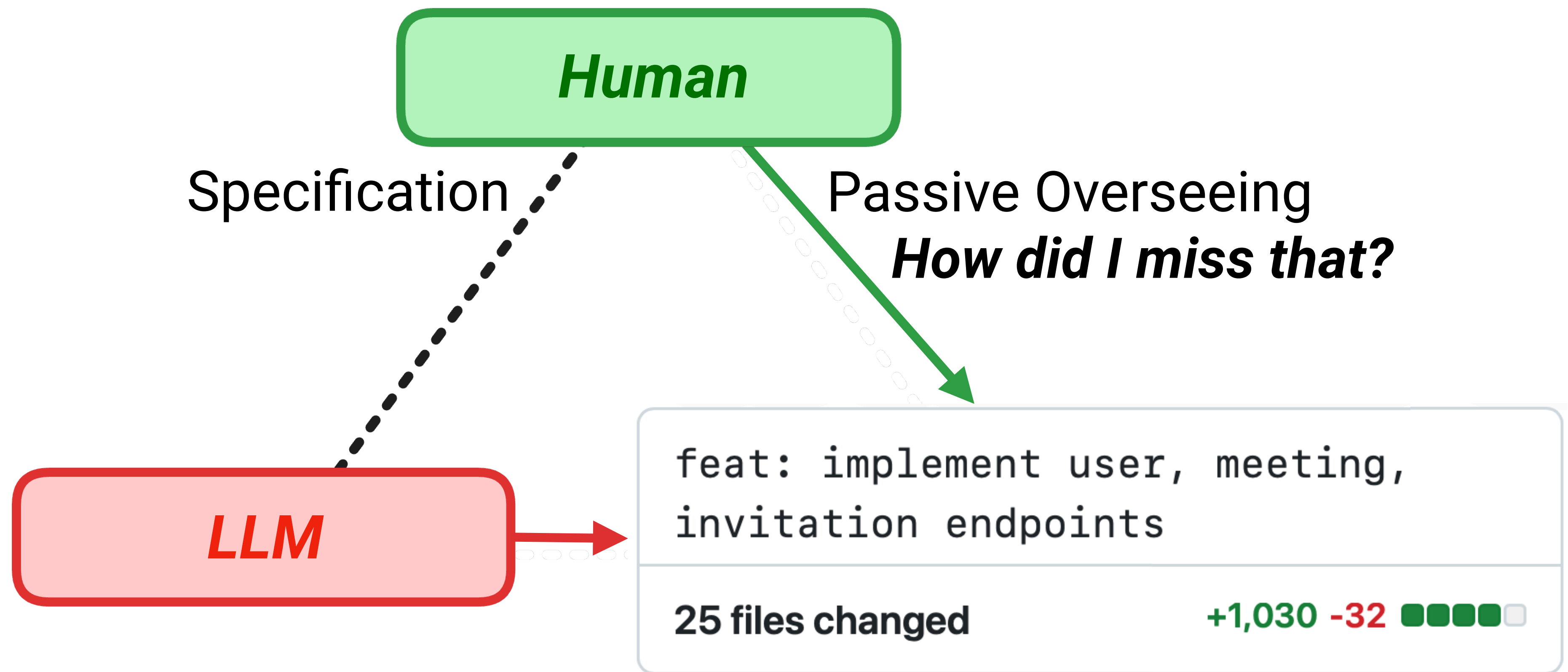
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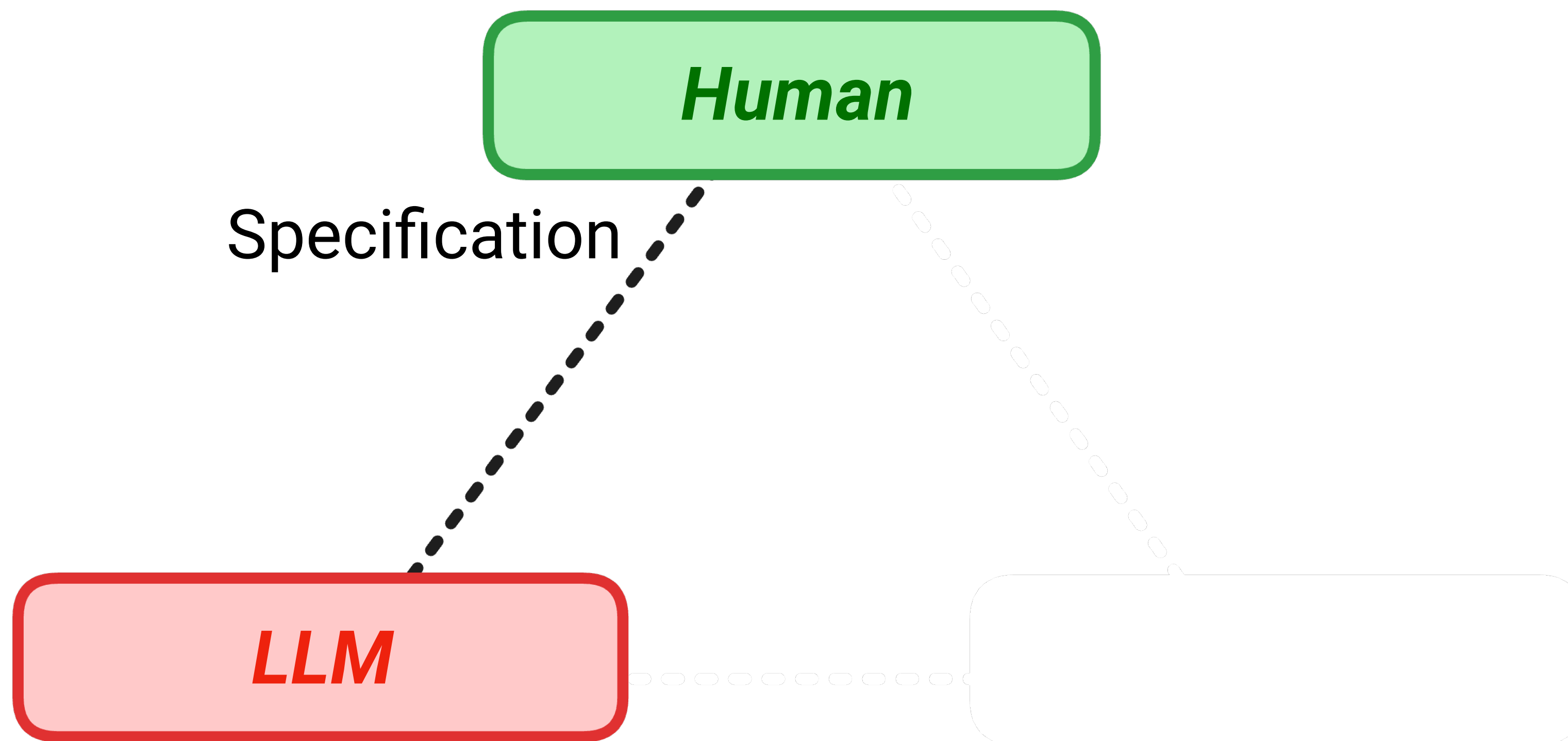
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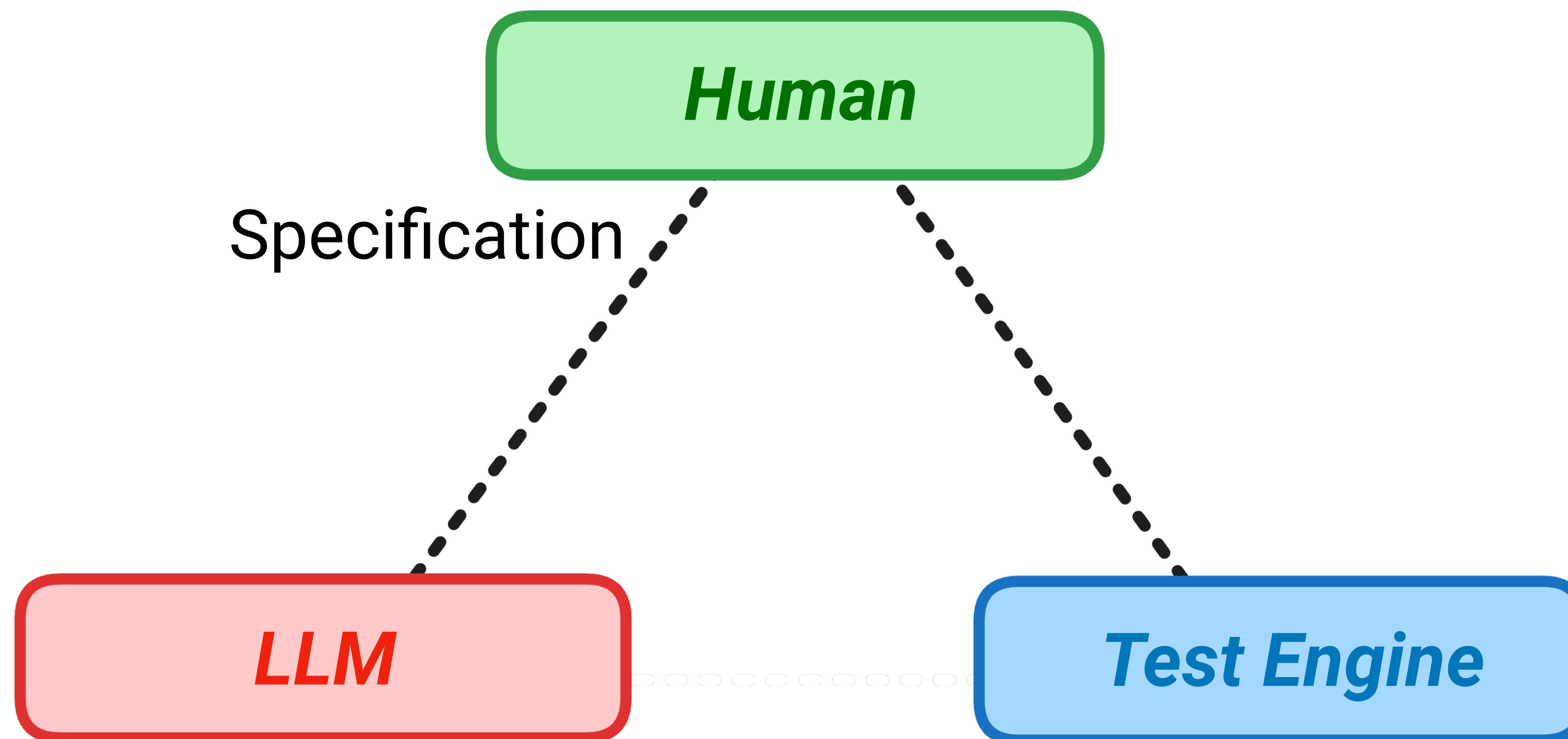
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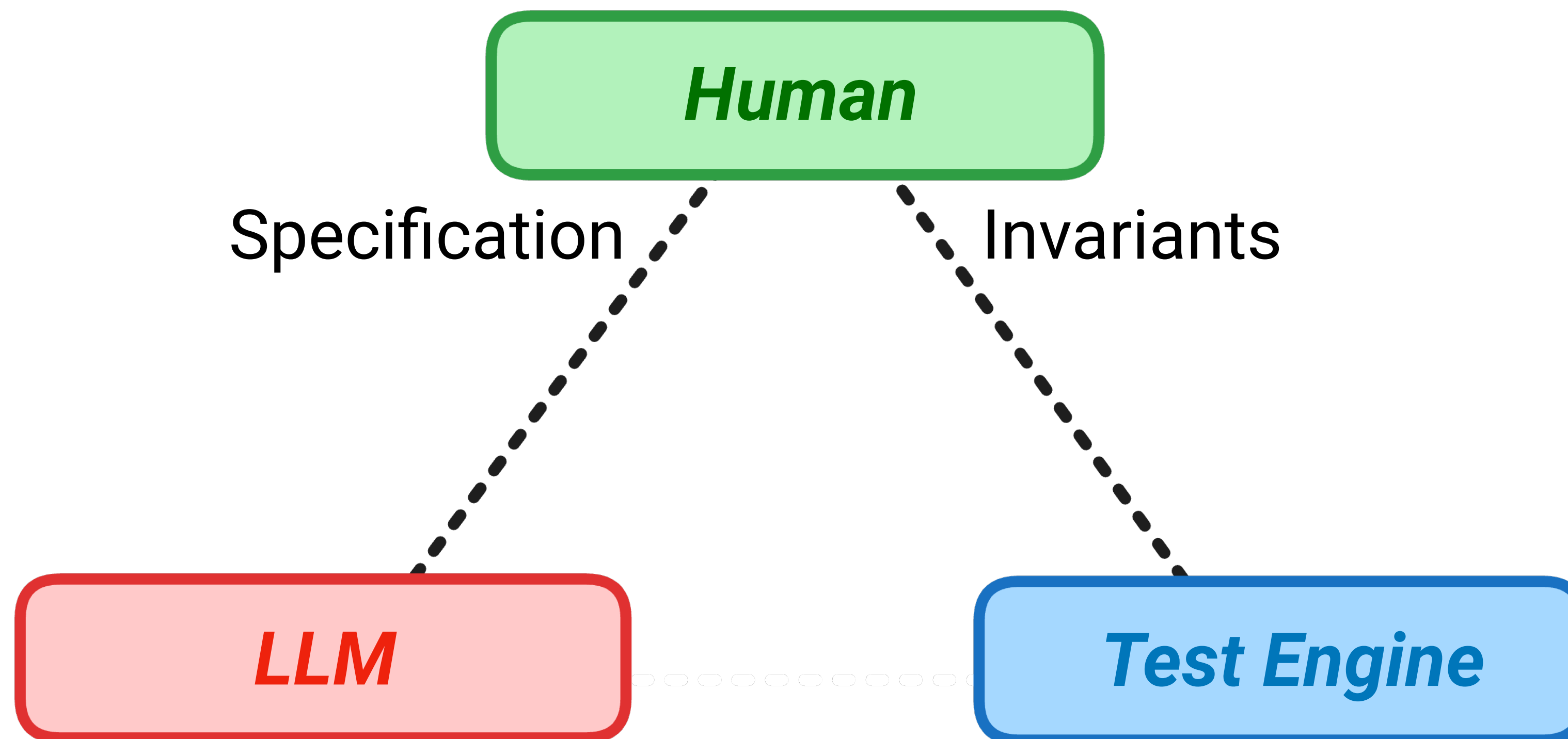
Division of Cognitive Labour



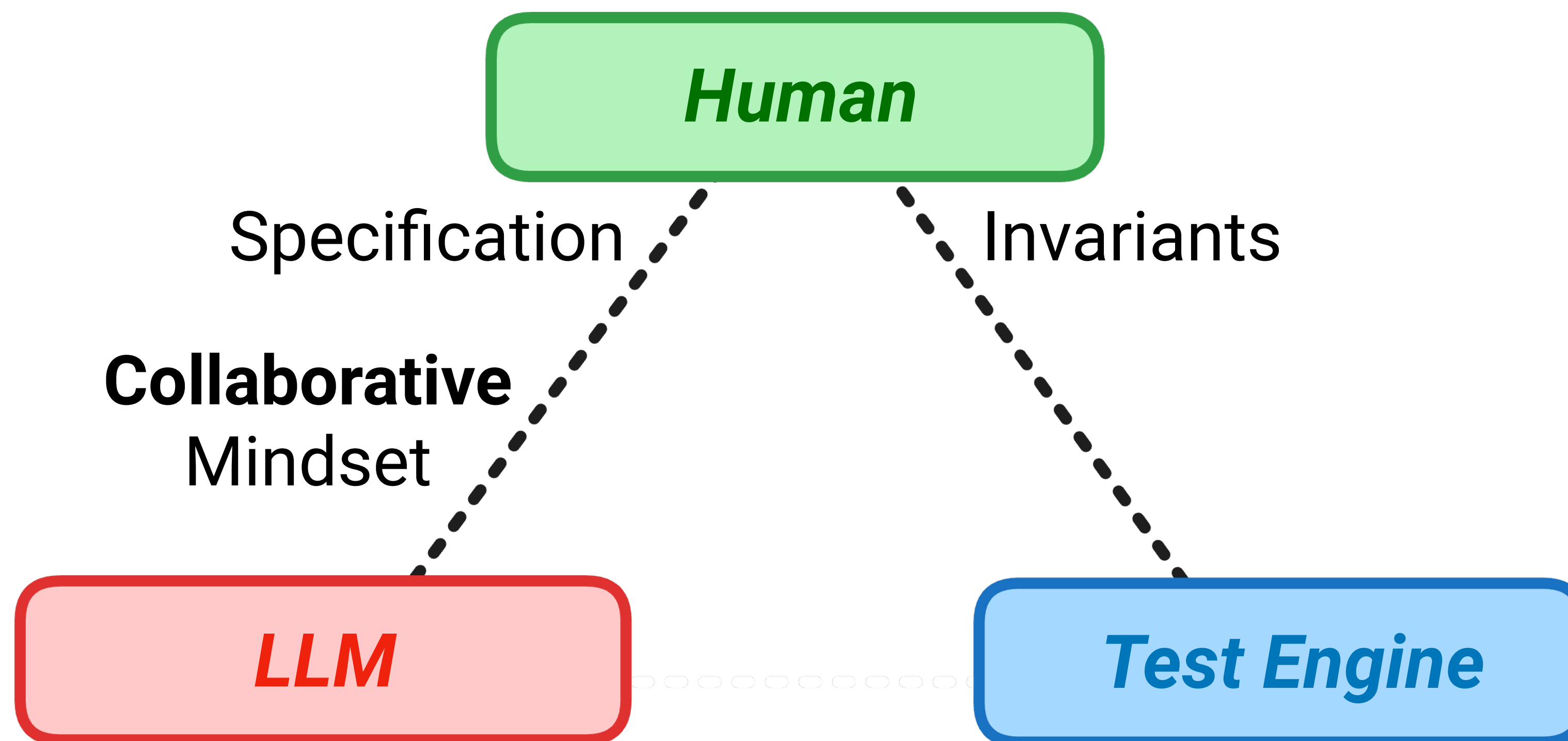
Division of Cognitive Labour



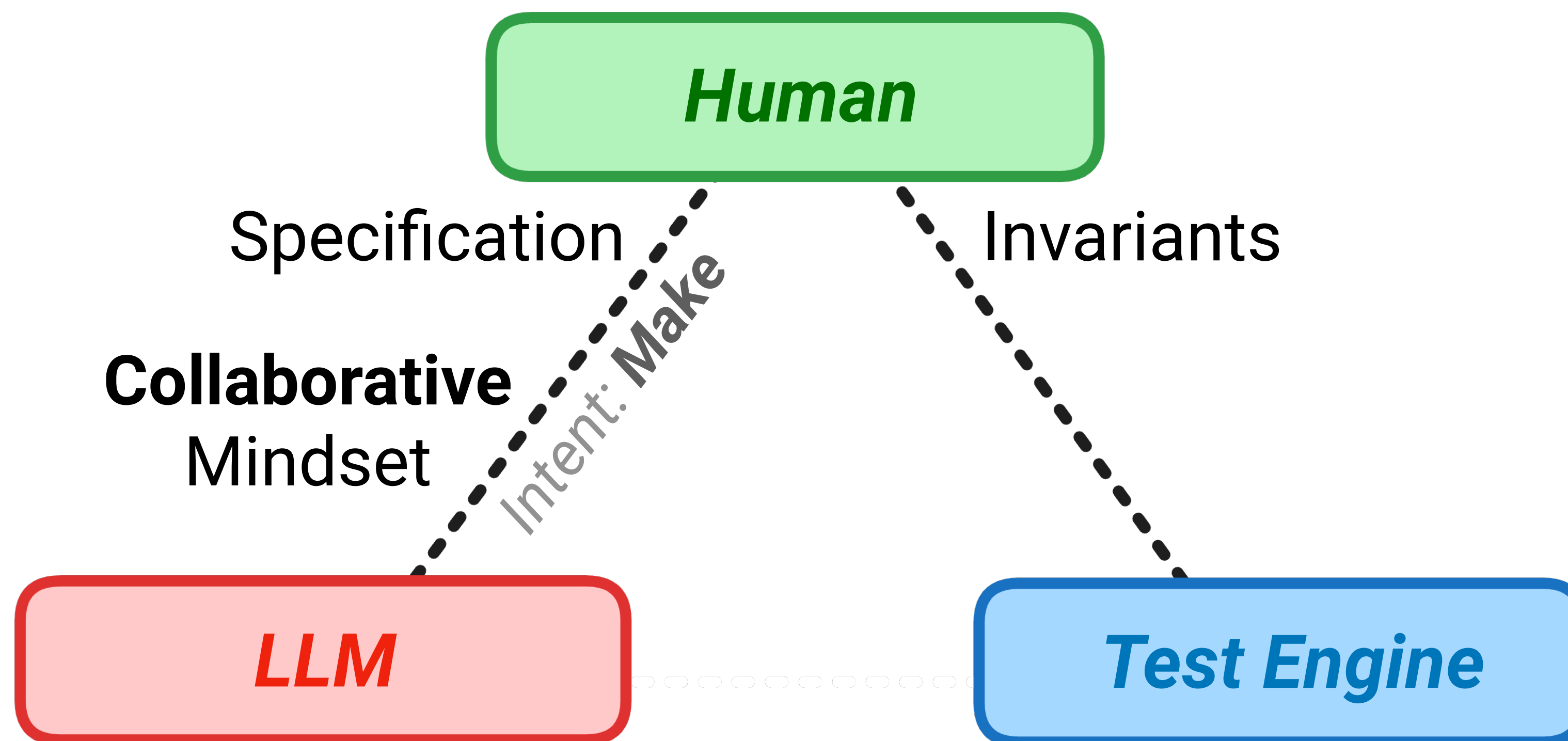
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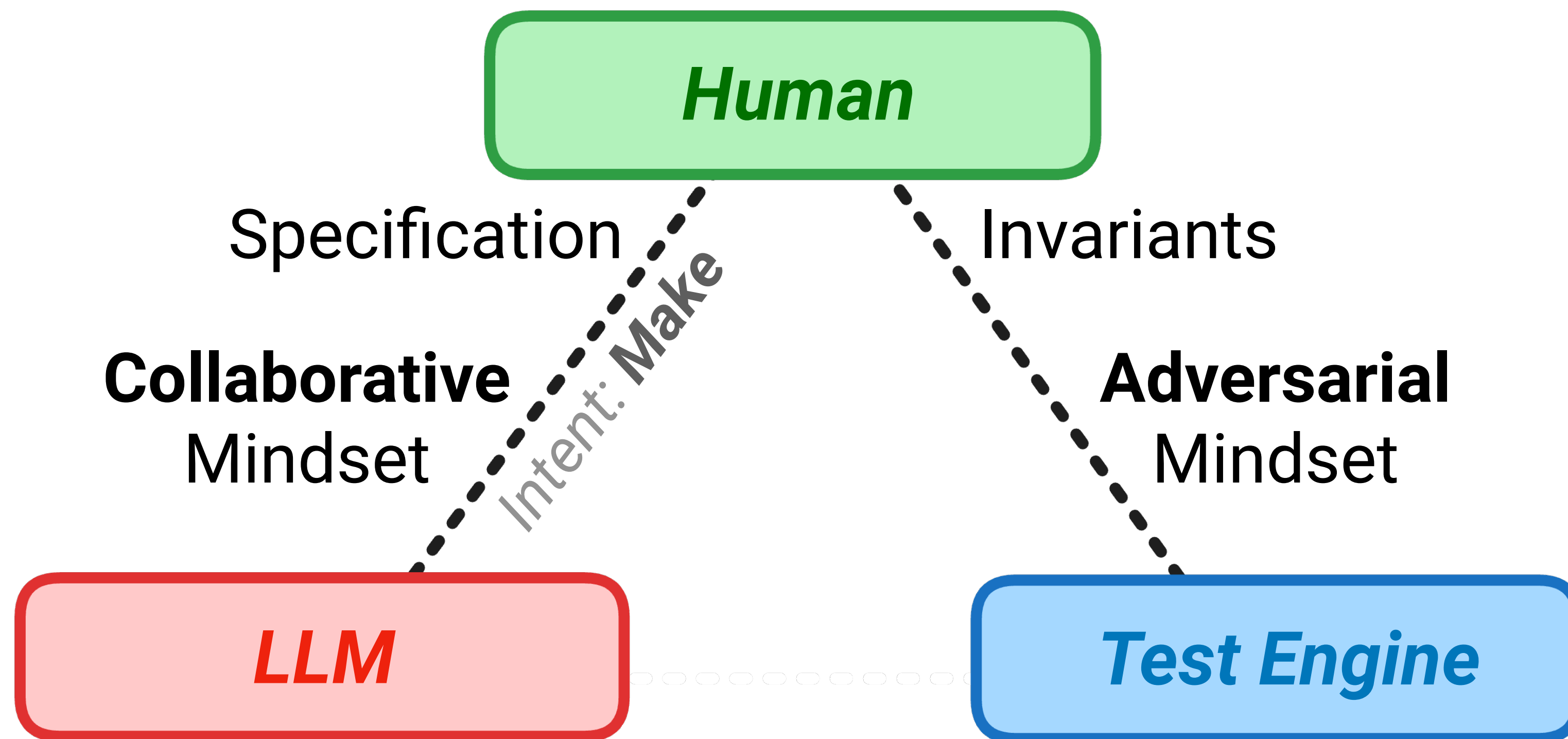
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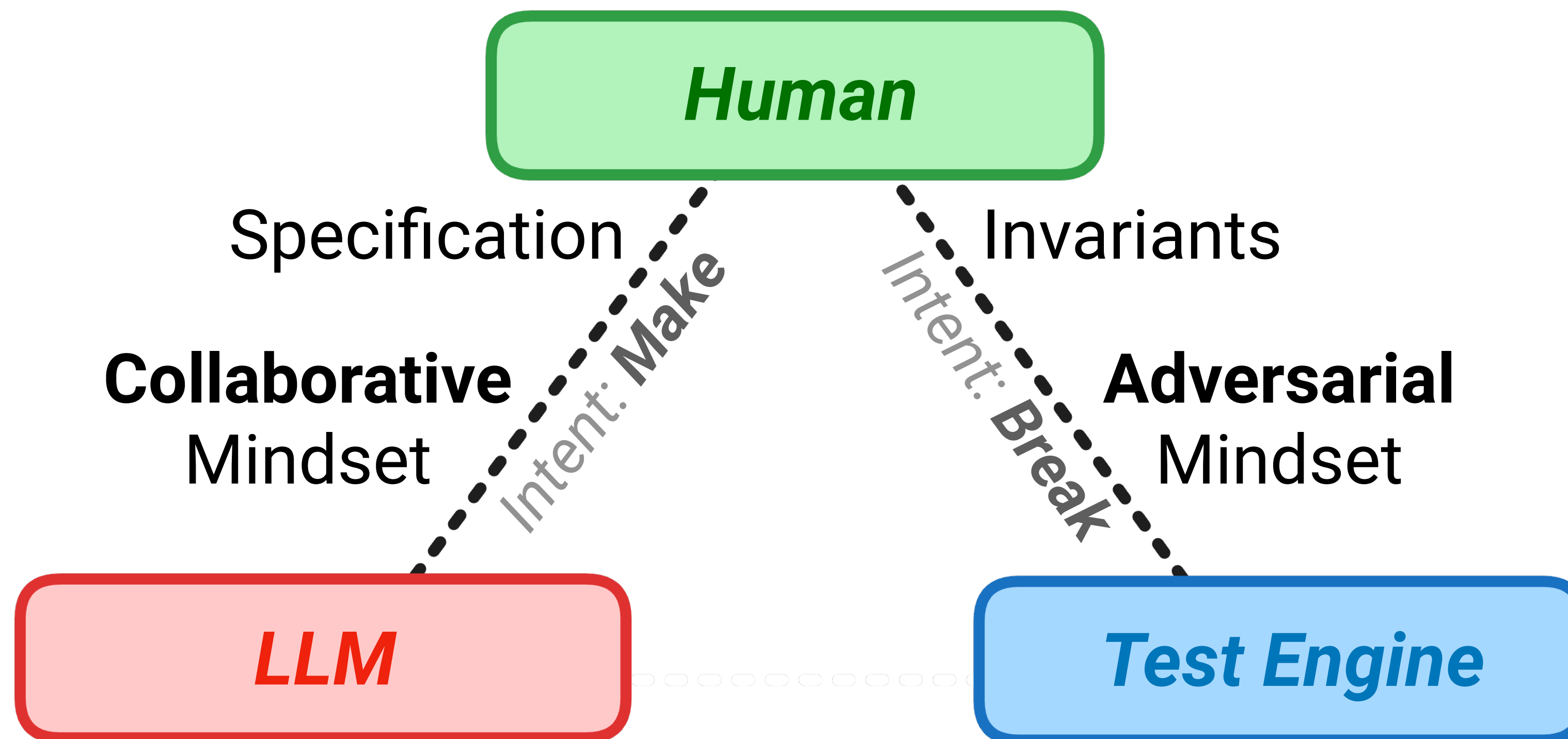
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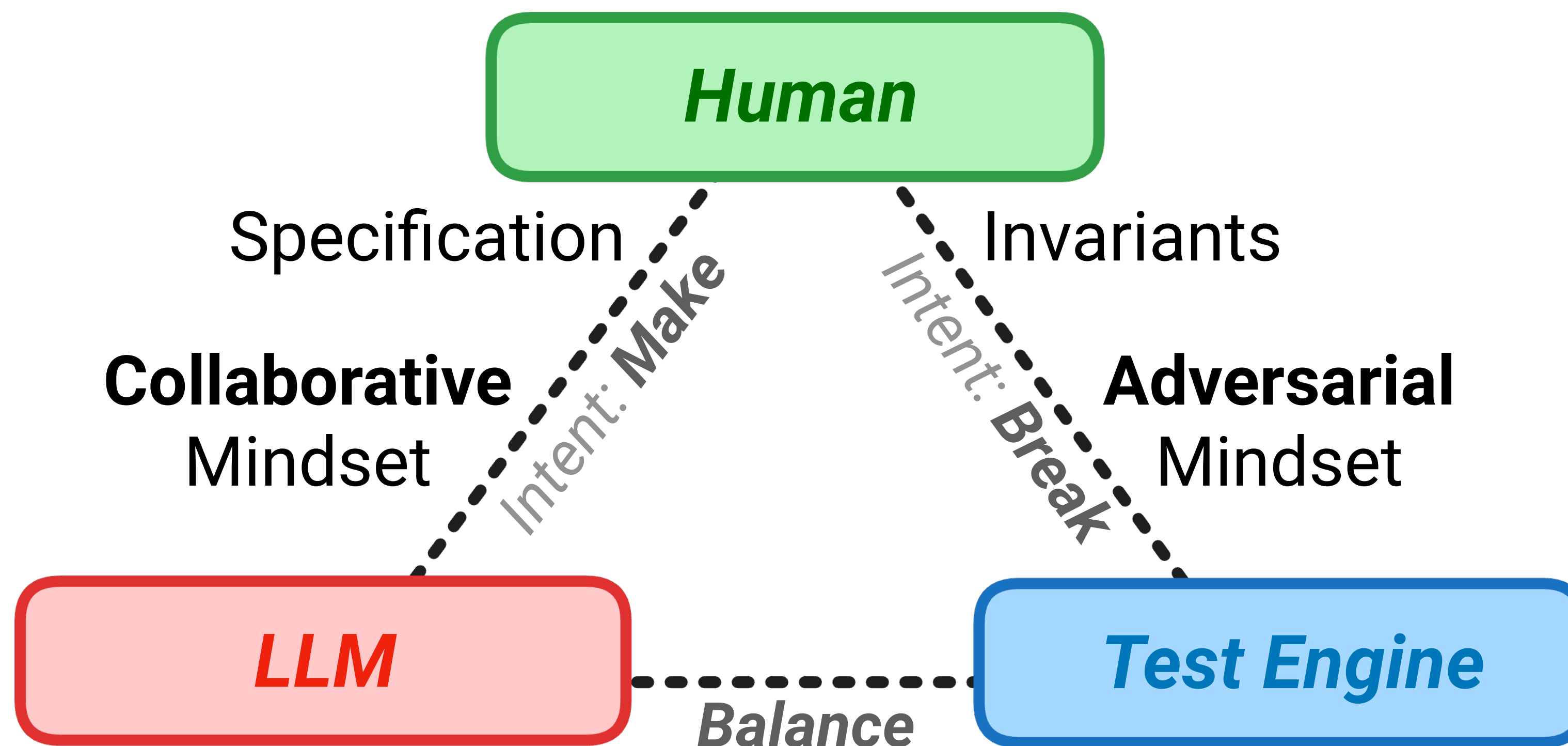
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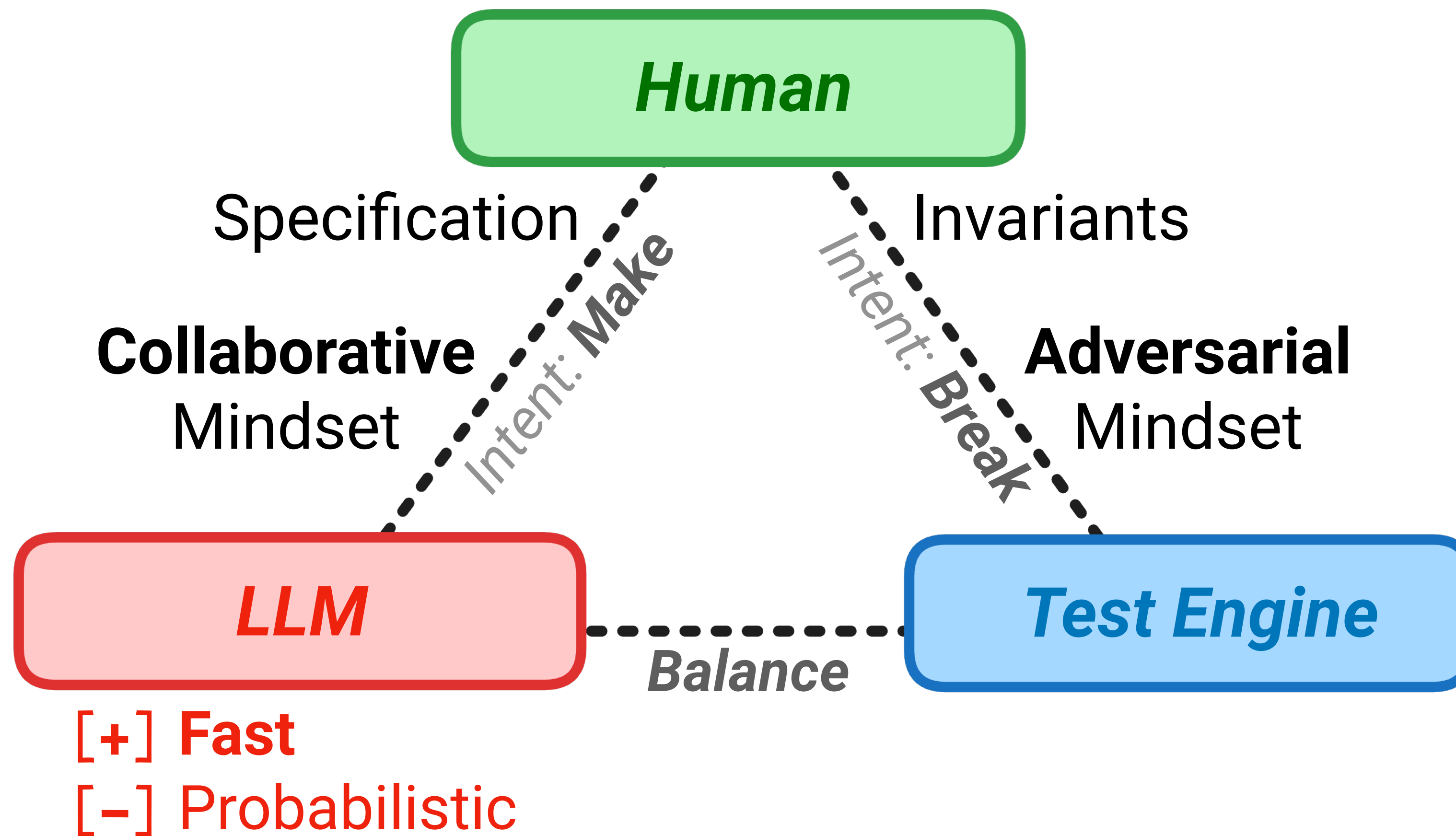
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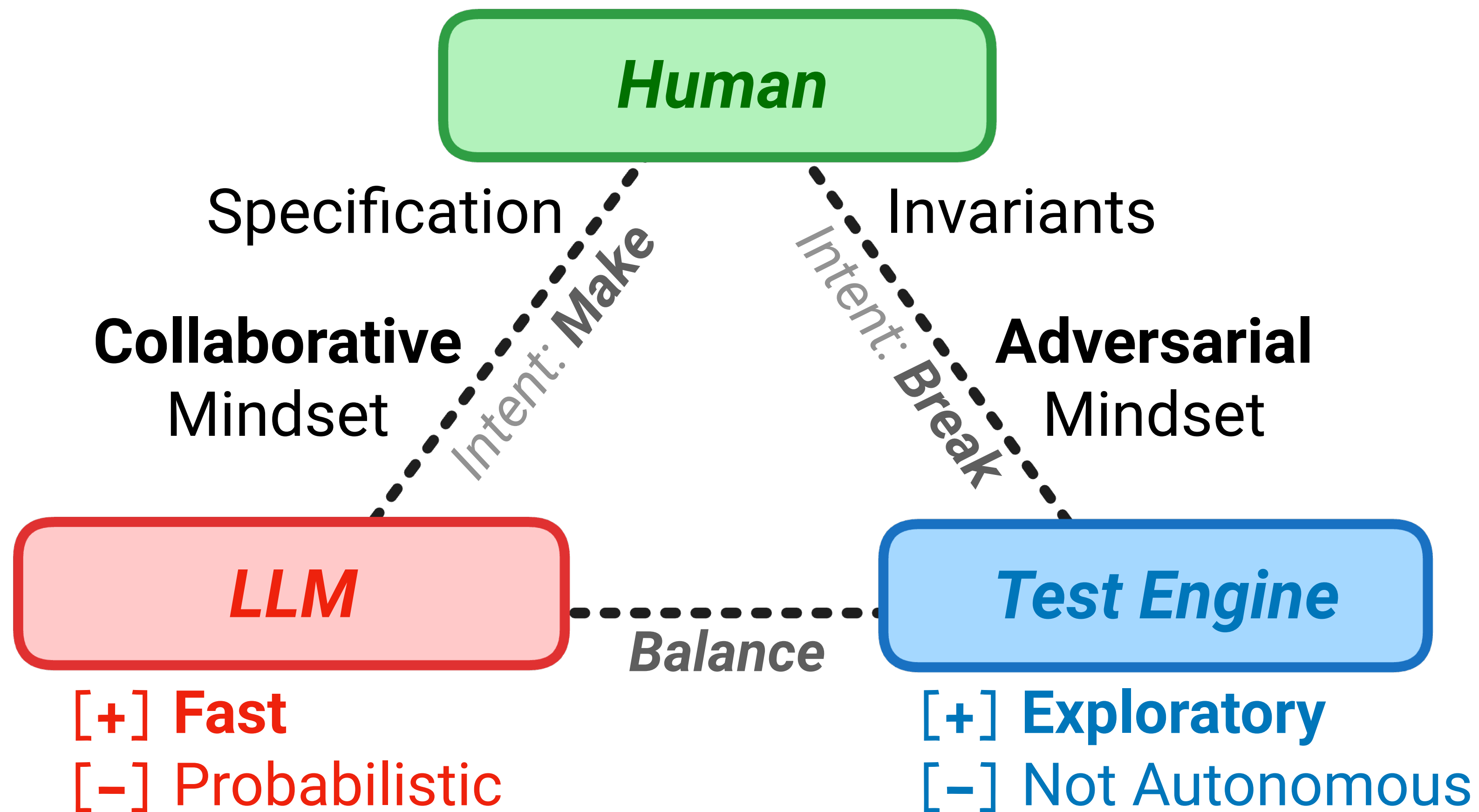
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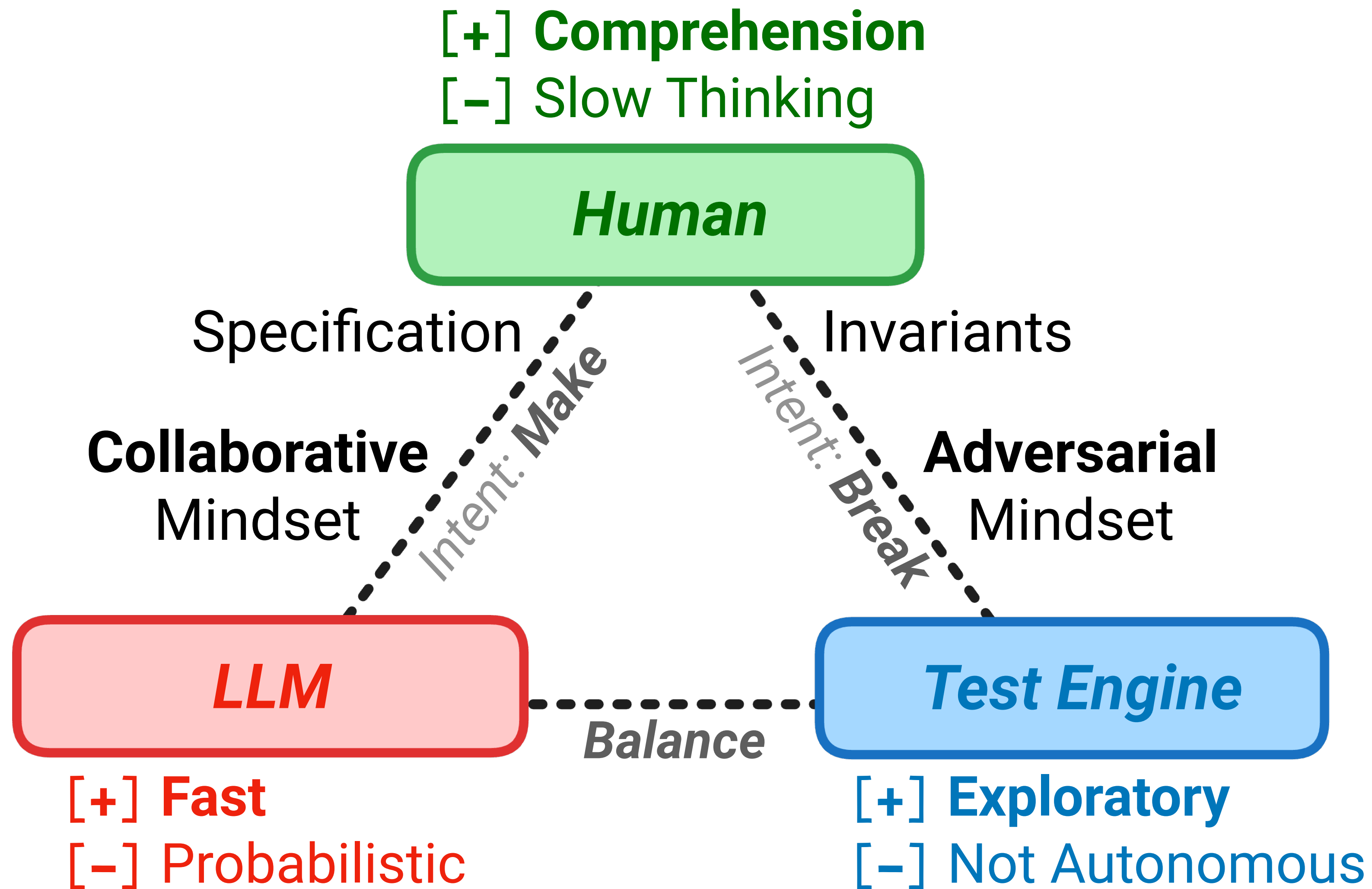
Division of Cognitive Labour



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Thank You!

- mourjo.me
- github.com/mourjo/prompt-meetings
- mourjo.me/slides/2026-09-11-signals.pdf



linkedin.com/in/mourjo